Instruments Apart: Monetary Policy and Consumption Inequality across U.S. Regions and Income Groups*

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Abstract

This paper examines how three monetary policy instruments—the federal funds rate, forward guidance, and large-scale asset purchases—differ in their distributional effects on U.S. household consumption. Using Consumer Expenditure Survey data for 1992–2019, orthogonal high-frequency shocks, and cell-level local projections, it finds no detectable difference at the national level between the consumption responses of the lowest and highest income quintiles under any instrument. This average masks two differences. In the Northeast, expansionary asset purchases raise non-housing consumption more for the lowest quintile than for the highest from the second year on, a differential statistically distinguishable from the other regions. Among lower-income households, contractionary conventional interest rate policy lowers consumption more for renters than for mortgagors, whereas contractionary asset purchase shocks do the reverse, consistent with a cash flow channel through refinancing. Forward guidance has no consistent distributional direction. The distributional consequences of monetary policy depend on the instrument, the region, and the housing contract through which policy reaches households.

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*Queensland University of Technology (PhD, 2026). Email: jiayi.yang.econ@outlook.com. This paper draws on Chapter 6 of my doctoral dissertation Yang (2026). The research question and data source are unchanged, but the empirical methodology has been substantially revised, and the results reported here therefore differ from those in the dissertation. I am grateful to Pedro Gomis-Porqueras, Jayanta Sarkar, and Ruchith Dissanayake for their guidance throughout my doctoral studies, and to Jorge Miranda-Pinto for detailed comments on this paper. I also thank Frank Mathmann and Stephen Thiele, and two anonymous examiners of my dissertation, for helpful comments on earlier versions of this work. This research was supported by QUT Postgraduate Research Award (International) and QUT HDR Tuition Fee Sponsorship. All remaining errors are my own.

1 Introduction

After the 2008 financial crisis, the Federal Reserve lowered the federal funds rate (FFR) to its effective lower bound, leaving conventional interest rate policy with no further room to cut, and turned to two unconventional instruments on a large scale: large-scale asset purchases (LSAP) and forward guidance (FG). Three rounds of quantitative easing (QE) expanded the Federal Reserve's balance sheet from under one trillion dollars before the crisis to approximately 4.5 trillion dollars by 2015. This intervention took place against a backdrop of decades of rising income and consumption inequality in the United States (Piketty and Saez, 2003), so its distributional consequences bear on the social legitimacy of central bank action. One view of quantitative easing holds that it raises equity and housing prices and primarily benefits wealthy households that hold financial assets (Domanski et al., 2016); another emphasizes that it compresses long-term interest rates and supports investment and employment, favoring lower-income households that depend on wages (Bivens, 2015). Both positions have support, but they rest on different measures and frameworks and rarely engage each other directly. Unconventional instruments are now a standard part of the central bank toolkit, and if different instruments have different distributional consequences, the choice of instrument itself carries distributional implications.

A substantial body of literature addresses this question and can be grouped into several strands. One strand examines the distributional consequences of monetary policy empirically, finding at both cross-country and single-country levels that contractionary policy generally widens inequality (Furceri et al., 2018; Sen and Sensarma, 2025), and documenting differences along dimensions such as race (Albert and Gómez-Fernández, 2024) and the composition of household income and wealth (Ampudia et al., 2018). A second strand identifies monetary policy shocks using high-frequency data, extending from two-factor to multi-factor decompositions and providing the basis for separating distinct instruments (Gürkaynak et al., 2005; Gertler and Karadi, 2015; Nakamura and Steinsson, 2018b). On the theoretical side, Heterogeneous Agent New Keynesian (HANK) models show that the distributional effects of monetary policy in general equilibrium operate primarily through labor income and aggregate demand, mattering especially for liquidity-constrained lower-income households, while direct asset price effects fall more on wealthy households (Kaplan et al., 2018; Auclert, 2019). Recent empirical work also provides evidence on the role of these channels. Del Canto et al. (2025) measure the responses of U.S. households to oil and monetary policy shocks in terms of monetized welfare changes and find that expansionary monetary policy is progressive along education and life-cycle dimensions: rate cuts raise equity, housing, and bond prices, which hurts middle-aged, highly educated households that are still accumulating these assets, while less educated households are compensated by higher labor income, as the employment channel implies. By focusing on differences across regions, this paper also builds on work examining the regional

effects of monetary policy (Carlino and DeFina, 1998; Beraja et al., 2019).

Most of these studies, however, combine the full range of Federal Reserve actions into a single composite shock. Coibion et al. (2017), using Consumer Expenditure Survey (CEX) data since 1980, find that contractionary monetary policy persistently widens consumption and income inequality. The single Romer and Romer (2004) shock they use, however, cannot distinguish the separate roles of LSAP, FG, and conventional interest rate policy, whose transmission channels differ considerably: LSAP primarily affects long-term asset prices and borrowing costs, while FG operates on expectations of future interest rates, both distinct from adjustments to the overnight rate. Beyond the choice of instrument, U.S. Census regions differ markedly in industrial structure, housing markets, and labor markets, which should produce different transmission across regions; few distributional studies examine this systematically, and consumption-side evidence covering the full quantitative easing cycle remains limited.

This paper addresses these gaps using CEX Interview survey data from 1992 to 2019. The sample period spans the pre-crisis years, the zero interest rate period, and the subsequent policy normalization, covering the full quantitative easing cycle. The CEX is a standard data source for studying consumption inequality in the United States (Attanasio and Pistaferri, 2016; Aguiar and Bils, 2015). I construct two grouped panels. The first crosses four Census regions with five income quintiles, yielding 20 cells, and is used to identify where distributional effects appear and whether they are general. Pooling the four regions gives five national income groups, used to estimate the national average. The second crosses three housing tenure types with five income quintiles, yielding 15 cells, and is used to identify the transmission channel. The dependent variable is non-housing consumption, which is total consumption net of housing expenditure. This excludes the direct effects of interest rates on housing expenditure through rents and refinancing-induced changes in monthly payments, thereby avoiding confounding changes in consumption demand with changes in housing costs. Total consumption and housing consumption are reported for comparison, and the food expenditure ratio provides supplementary evidence on the composition of expenditure.

Identifying the shocks is central, and measuring unconventional policy has long been difficult. The narrative shocks of Romer and Romer (2004) barely vary during the zero interest rate period; the shadow rate of Wu and Xia (2016) compresses the overall policy stance into a single number that masks differences across instruments; event studies of individual announcements (Krishnamurthy and Vissing-Jorgensen, 2011) do not yield a continuous series. I use the high-frequency identification data of Swanson (2021), which extracts three mutually orthogonal factors from a high-frequency window around Federal Open Market Committee (FOMC) announcements, corresponding to FFR, FG, and LSAP. The high-frequency window ensures that these shocks reflect the policy news itself rather than other information released at the same time, and the orthogonality allows consumption responses to be attributed to specific instruments. High-frequency surprises may

nevertheless be predictable from information already available before the announcement (Bauer and Swanson, 2023). Section 6 re-estimates the model using shocks orthogonalized with respect to such information. For estimation, I apply the local projections method of Jordà (2005) to the time series formed by each cell separately; this method is well suited to externally identified shock variables and facilitates comparison across cells. The shocks enter the regression directly as explanatory variables, and the reported responses are expressed per unit of the factor rather than rescaled into interest-rate units, because no usable first-stage relationship is available for the asset purchase factor at the quarterly frequency.

The paper has three findings, at the national, regional, and housing contract levels.

First, at the national level, none of the three instruments produces a detectable change in the consumption gap between high- and low-income households. With the four regions pooled, the difference between the non-housing consumption responses of the lowest income quintile ($Q1$) and the highest ($Q5$) is close to zero under all three instruments. Under asset purchases, for example, the estimated difference is -0.02 percentage points, with a standard error of 0.8. This near-zero average leaves open whether the distributional effect is absent or whether effects in opposite directions offset one another. The next two findings support the latter.

Second, the national average masks differences across regions. In the Northeast, expansionary asset purchases raise non-housing consumption more for $Q1$ than for $Q5$. I refer to this direction, in which lower-income groups benefit more, as progressive. The difference is close to zero in the first year after the shock, rises to approximately 5 percentage points in the second year, and remains at around 4 percentage points in the third. The hypothesis that the difference is the same in the four regions is rejected in the second year. The Northeast's estimated difference exceeds the average for the other three regions by 4 percentage points ($p = 0.010$), while the point estimates for the other regions run in the opposite direction, offsetting the Northeast's effect when the regions are pooled.[1] The food expenditure ratio moves in the same direction: following expansionary asset purchases, it falls more for $Q1$ than for $Q5$ in the Northeast at most horizons, which by Engel's Law points to an improvement in expenditure composition. A single federal monetary policy therefore has different distributional consequences across regions.

Third, the three instruments reach households through different housing contracts, which helps explain the first two findings. Short-term interest rates do not change the monthly payments on existing fixed-rate mortgages but do pass through to rents, whereas asset purchases can open or close the refinancing channel for existing borrowers by compressing long-term interest rates and mortgage spreads. Within the lowest income quintile, a contractionary conventional interest

[1] This difference remains significant at the 5% level after adjusting for multiple comparisons across the four regions (adjusted $p = 0.036$), so the conclusion holds even if the Northeast is treated as having been selected from the four regions after the fact.

rate shock produces a consumption decline approximately 2 percentage points larger for renters than for mortgagors at the same income level. Under a contractionary asset purchase shock, the direction reverses: the decline is approximately 3 percentage points larger for mortgagors than for renters. The difference between the two instruments, about 5 percentage points, is significant at the 1% level and remains so under alternative shock series and specifications. No corresponding difference between the two groups is found under forward guidance. The income gradient under asset purchases appears within the mortgagor group but not within the renter group. Under this identification strategy, the renter group provides the only evidence that can identify the employment channel; the absence of an income gradient among renters therefore points instead to a cash flow channel operating through refinancing. Opazo (2023) likewise finds that mortgagor households adjust expenditure more than outright owners. The gradient is concentrated among low-income borrowers, who face the greatest difficulty in meeting refinancing thresholds. It is close to zero in the first year, widens in each subsequent year, and is largest in the third year, consistent with the time required to complete refinancing and for its budgetary effects to accumulate.

These two findings answer the question left open by the first. National averages by income group combine households affected by a given instrument with those largely unaffected by it. Mortgagors, whose cash flow is affected by asset purchases through refinancing, account for only about one-ninth of the lowest income quintile, so their response is diluted in the national average. The Northeast does not have a higher share of mortgagor households; its prominence is instead consistent with housing carrying the greatest weight in local household budgets there, although this explanation is inferred from the two separate splits by region and tenure. Other studies emphasize the income effects of reduced unemployment (Chang and Schorfheide, 2024; Lee, 2026); this paper cannot rule out that channel, but finds no evidence on the consumption side that it is dominant.

These results are robust to changes in shock measurement, regression specification, and income classification. When the shocks are orthogonalized with respect to pre-announcement information, the LSAP factor is set to zero before 2009, shock lags are omitted, interest rate controls are added, or households are grouped by national income rank, the second-year difference for the Northeast remains statistically significant at the 5% level. Identification comes from the quarters in which these instruments were used on a large scale. After excluding recession quarters, the second-year difference between the Northeast and the other three regions remains significant at the 10% level, whereas the difference for the Northeast itself and the differences across housing tenure groups are no longer statistically distinguishable from zero. The findings therefore characterize the distributional effects of large-scale asset purchases; whether smaller operations have proportionate effects cannot be settled with this sample and is left for future work.

The paper makes three contributions. First, drawing on the orthogonal decomposition of Swanson (2021), this paper compares the distributional effects of conventional interest rate policy,

forward guidance, and asset purchases separately using a unified consumption panel, rather than combining these policies into a single shock as most of the literature does (Coibion et al., 2017). Even when the sample covers the zero lower bound (ZLB) and quantitative easing periods, evidence based on a single shock cannot answer which instrument the distributional consequences come from. Chang and Schorfheide (2024) find that expansionary shocks raise consumption inequality slightly, but the credible interval spans both directions. Del Canto et al. (2025) find that monetary expansion is progressive in welfare terms, but their shock is measured by unexpected movements in the interest rate path, which cannot distinguish which instrument the progressivity arises from. The decomposition in this paper offers one possible explanation: in the consumption data, none of the three instruments generates a detectable income gradient in the national average. Taken apart, the progressive direction appears in the asset purchase component, in the Northeast; the conventional rate component produces no income gradient in non-housing consumption, and forward guidance has no consistent direction. Within the same income group, conventional interest rate policy and asset purchases affect different households in opposite directions, and a direct test confirms that this difference is statistically significant. Combined into a single shock, these differences dilute or offset one another; since central banks often use several instruments at once, aggregate analysis obscures differences that matter for policy. Model-based work has already addressed this issue at the instrument level: Lee (2026) decomposes quantitative easing and forward guidance in an estimated heterogeneous-agent New Keynesian model and notes that empirical conclusions on the distributional effects of unconventional policy are not consistent. This paper provides instrument-level empirical evidence from consumption data that differs from that study in two respects: forward guidance has no consistent distributional direction here, and the gains to lower-income households operate mainly through refinancing cash flow, with no indication that the employment channel is dominant.

Second, most of these studies summarize the entire distribution using a statistic such as the Gini coefficient, a quantile ratio, or the top 10% share. Instead, this paper directly compares impulse responses across region-by-income and tenure-by-income cells. Combining the four measures of non-housing consumption, total consumption, housing consumption, and the food expenditure ratio with income quintiles defined by within-region relative income, this design reveals both which households are affected and through which type of consumption, while separating nominal changes from changes in expenditure composition.

Third, by splitting the sample by housing contract, this paper shows that the differences in distributional effects across the three instruments match the types of contracts through which they reach households, and provides evidence that asset purchases operate through the cash flow of mortgagor households. Di Maggio et al. (2017) use the reset of adjustable-rate mortgages to show that lower monthly payments substantially increase durable goods purchases, with a higher

marginal propensity to consume among low-income borrowers with low housing wealth. Cloyne et al. (2020) find that the aggregate consumption response to conventional policy is driven mainly by mortgagor households. This paper finds evidence of a cash flow channel operating through mortgagor households under large-scale asset purchases. Under conventional interest rate policy, however, comparing households at the same income level within the lower income groups shows a larger response among renters, which differs from the finding in Cloyne et al. (2020). This paper also provides an explanation for the regional differences in the distributional effects.

The rest of the paper proceeds as follows. Section 2 introduces the data sources and panel construction. Section 3 describes the empirical strategy. Section 4 reports the main results, first for the national aggregate and then by region. Section 5 uses the split by housing contract to explain why the distributional effects of the three instruments differ, why the Northeast stands out, and why the national average is close to zero. Section 6 presents the robustness checks. Section 7 concludes and discusses policy implications and future directions.

2 Data

2.1 CEX Interview Survey

The Bureau of Labor Statistics (BLS) provides detailed household expenditure data through the Consumer Expenditure Survey (CEX) (U.S. Bureau of Labor Statistics, 2025; Paulin and Krishnamurty, 2019). I use the quarterly Interview Survey as the source for household consumption, income, and related characteristics. The Interview Survey is conducted at a lower frequency than FOMC meetings, but it covers a broad range of consumption categories, including total expenditure, durable goods such as appliances and vehicles, and services, and has a relatively large sample size.

The consumption data cover 1992Q1 to 2019Q4, a total of 112 quarters. The four dependent variables are total consumption expenditure, housing consumption, non-housing consumption, and the food expenditure ratio. Among these, non-housing consumption serves as the primary outcome variable, with the food expenditure ratio providing supplementary analysis, and total consumption and housing consumption used for comparison. Non-housing consumption is constructed as the difference between total consumption expenditure and housing expenditure in the raw dataset. I use this variable as the primary outcome because central bank interest rate policy directly affects lending rates and thus monthly mortgage payments. This creates the possibility that the response of housing consumption may reflect both the direct effect on mortgage payments and the distributional effect of policy operating through consumption demand.

After winsorizing total expenditure and income at the 99.9th percentile within each calendar year, I assign all households to region-income cells defined by Census region and within-region

income quintile based on pre-tax household income, as recorded in the BLS interview questionnaire. From 2004 onward, pre-tax income is measured using the BLS-imputed `FINCBTXM`. For 1992–2003, no imputed values are available, so I use the respondent-reported `FINCBTAX` and drop households with incomplete income reports in those years, since their income is zero, missing, or understated, and retaining them would concentrate them in the lowest income quintile. Households with negative income are dropped in all years. This yields 20 groups: four Census regions by five income quintiles. Within each cell, I compute quarterly weighted averages using the sampling weights provided by the CEX, producing a representative value for each cell-quarter observation.

I construct a parallel set of cells stratified by housing tenure to provide complementary evidence on the mechanisms through which monetary policy shocks operate. Following Cloyne et al. (2020), who show that households in different tenure categories—renters, mortgagors, and outright owners—respond differently to monetary policy shocks, I form 15 groups: five income quintiles by three tenure types. In the region-by-income panel, the median number of households per cell per quarter is approximately 220, compared with approximately 270 in the tenure-by-income panel. The standard errors of the cell means are therefore much smaller than the dispersion across individual households. Sample sizes for the region-by-income cells are reported in Appendix A.2.

2.2 Monetary Policy Shocks

Swanson (2021) analyzes changes in asset prices in a 30-minute window around Federal Reserve FOMC announcements from July 1991 to June 2019, applying principal component analysis to the joint responses of federal funds futures, Eurodollar futures, and 2-, 5-, and 10-year Treasury yields. This procedure extracts three mutually orthogonal policy factors: an FFR factor capturing unexpected changes in the current policy rate; an FG factor capturing revisions to the expected path of future policy rates; and an LSAP factor capturing unexpected changes in balance sheet policy. The orthogonality of these three series is central to the analysis, as it allows me to attribute any heterogeneity in the estimated household consumption responses to specific policy instruments. Here, orthogonality means that the three factors are orthogonal to one another, not that they are orthogonal to information already available before the announcement (Bauer and Swanson, 2023). In addition, the LSAP factor is not zero before 2009, when no asset purchases were taking place. Section 6 reports results after addressing each of these two points separately. I estimate the effects of the unanticipated policy information contained in FOMC announcements, rather than the full effect of any particular policy program.

For the FFR and FG factors, negative values indicate expansion, while the LSAP factor has the opposite orientation. Following Swanson (2021), I reverse the sign of the LSAP factor so that negative values indicate expansion and positive values indicate contraction for all three series. The

shock series are summed within each quarter, aggregating them from meeting-date frequency to quarterly frequency to match the consumption data (Appendix A.4).

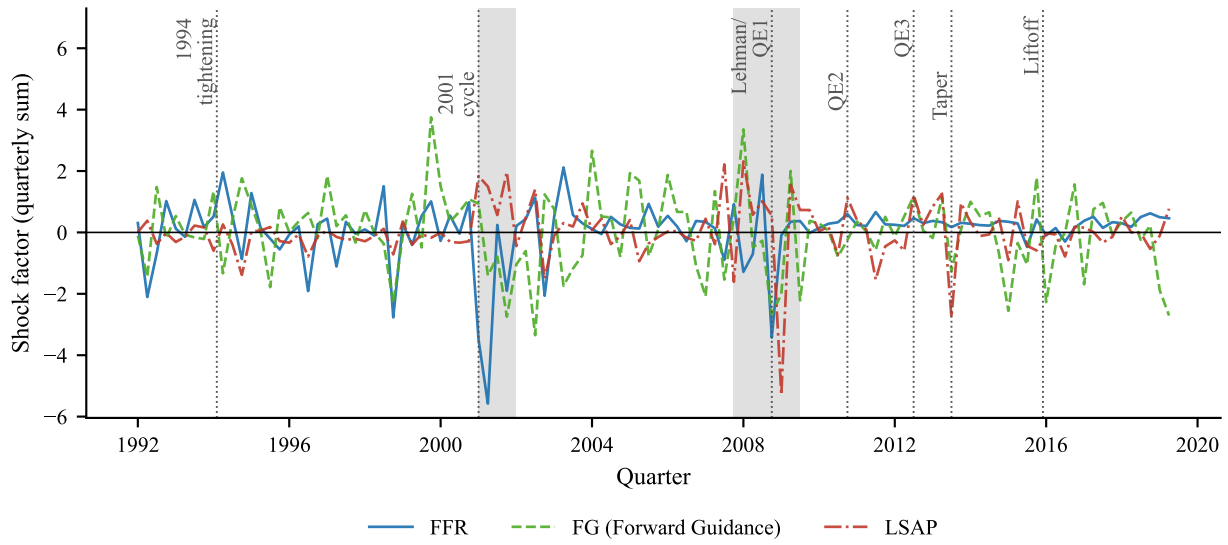


Figure 1: Monetary Policy Shock Factors, 1992Q1–2019Q2

Notes: Quarterly aggregates of FOMC-meeting-level FFR, FG, and LSAP shock factors. All three factors are signed so that positive values denote monetary tightening and negative values denote easing (including QE expansions). Shaded bars indicate National Bureau of Economic Research (NBER) recession quarters.

Figure 1 plots the three shock series. The FFR factor fluctuates within a narrow band around zero during 2009–2015, reflecting the constraint imposed by the zero lower bound, while the LSAP factor is concentrated mainly after 2008. The Federal Reserve did not conduct asset purchases before 2008, and movements in the LSAP factor during this period reflect unexpected changes in long-term interest rates that are orthogonal to the other two factors. The standard deviations of the three series are 1.03, 1.29, and 0.91, respectively, so a one-unit shock corresponds roughly to one standard deviation. Across the full sample, 15 quarters record shocks with absolute values exceeding two standard deviations, concentrated mainly around 2001 and at the peak of the 2008–2009 financial crisis; these extreme values may affect the precision of the estimates, and Section 6 returns to this issue.

2.3 Macro Control Variables

Household consumption decisions respond to a broad range of macroeconomic conditions, not only to unexpected monetary policy shocks. To prevent the model from attributing variation driven by domestic or global economic conditions to monetary policy, I include a set of macroeconomic control variables covering consumption, employment, output growth, housing prices, exchange rates, and financial asset prices. I draw ten representative series from the Federal Reserve Economic

Data (FRED) database (Federal Reserve Bank of St. Louis, 2026); variable definitions are provided in Table A.1 of Appendix A. In selecting these controls, I include the NASDAQ Composite Index to capture movements in financial markets, but exclude financial indicators with direct links to Federal Reserve policy instruments—such as the FFR, Treasury yields, and mortgage rates—as their prices move closely with the policy instruments themselves, and including them risks absorbing part of the variation attributable to monetary policy shocks.

These ten series capture distinct yet interrelated dimensions of macroeconomic conditions and exhibit substantial correlation with one another, making their direct inclusion in the model prone to multicollinearity. To address this and reduce model complexity, I apply principal component analysis to the ten series as a pre-processing step (Bernanke et al., 2005). The first three principal components explain 68.2 percent of the total variance (see Figure A.2 in Appendix A). As these variables serve only as controls, this level of explanatory power is sufficient here. The three factors correspond to an economic activity factor (PC1), a price and inflation factor (PC2), and a labor market pressure factor (PC3).

Merging the CEX panel, the monetary policy shock series, and the macroeconomic factor series yields the estimation panel, covering 1992Q1–2019Q2, a total of 110 quarters. The shock variables and macroeconomic controls are common across all cells at any given point in time, while consumption variables, income variables, and household size—used to construct per-capita consumption and income—vary across panel units.

3 Empirical Strategy

I use two panels to address two questions: the region-by-income panel identifies geographic heterogeneity in the distributional effects of monetary policy, while the tenure-by-income panel identifies the transmission channels of the shocks (Cloyne et al., 2020). A panel crossing region, income, and tenure cannot be reliably estimated at the CEX sample size: many cells have too few observations in a non-trivial share of quarters, which amplifies the influence of individual households and increases residual variance. The tenure-by-income panel is therefore a compromise given this constraint, and its estimates should be interpreted as national-level results. In addition, for each income group, the four regional cells in the regional panel are aggregated using survey weights to construct five national income groups, which are then used to estimate the average effect at the national level.

The primary estimation framework is the local projections method of Jordà (2005). Swanson's monetary policy shock series are identified from high-frequency data and enter the model as exogenous regressors, which fits the local projections setting and reflects current practice in the literature (Plagborg-Møller and Wolf, 2021). This choice is also supported on inferential grounds.

Montiel Olea et al. (2026) show that, in a locally misspecified vector autoregression setting, confidence intervals based on local projections retain correct coverage, whereas those based on vector autoregressions with short to medium lag orders can substantially undercover. Since the core conclusions of this paper hinge on assessing the significance of gaps between quantiles, reliable inference under model misspecification is particularly important for the choice of estimator. To improve the interpretability of the impulse response functions, I complement the standard local projections specification with a smoothed variant.

3.1 Local Projection Specification and Identification

The local projections specification used to identify the effects of monetary policy shocks on consumption is given in Equation (1). The dependent variable is specified as $y_{g,t+h} - y_{g,t-1}$, following Jordà (2005), so that the estimated coefficients at each horizon h can be interpreted as cumulative effects, and the resulting impulse response functions are cumulative impulse response functions. Rather than pooling observations into a single panel estimator—which would prioritize identifying common effects across cells—I estimate the model separately for each cell by OLS. This approach is better suited to the objective of documenting heterogeneity across cells in response to common shocks, and prevents cross-cell dependencies from confounding the cell-level estimates that form the basis of the subsequent comparisons.

$$\begin{aligned} y_{g,t+h} - y_{g,t-1} = \alpha_g^h + \beta_g^h \cdot s_t + \sum_{j=1}^{p_s} \delta_j^h \cdot s_{t-j} + \sum_{j=1}^{p_y} \phi_j^h \cdot \Delta y_{g,t-j} \\ + \sum_{j=1}^{p_x} \mathbf{\Gamma}_j^h \cdot \mathbf{PC}_{t-j} + \sum_{q=1}^{3} \gamma_q^h D_{q,t} + \varepsilon_{g,t+h} \end{aligned} \tag{1}$$

where g indexes panel cells (region × income quintile in the regional estimation, housing tenure × income quintile in the tenure estimation), t indexes calendar quarters, and $h = 1, \ldots, 12$ denotes the forecast horizon. α_g^h is the regression intercept; s_t is one of the three Swanson (2021) monetary policy shock series (FFR, FG, or LSAP), and β_g^h is the coefficient of interest, tracing out the impulse response of cell g's consumption to the shock at horizon h. The $p_s = 2$ lags of the shock (δ_j^h), $p_y = 3$ lags of consumption growth (ϕ_j^h), and $p_x = 1$ lags of the three macroeconomic principal components ($\mathbf{\Gamma}_j^h, \mathbf{PC}_{t-j}$) absorb the dynamic feedback between the shock, macroeconomic conditions, and household consumption. $D_{q,t}$, $q = 1, 2, 3$, are quarter dummies (Q1–Q3, with Q4 as the reference quarter) with coefficients γ_q^h that absorb seasonality in the CEX data, and $\varepsilon_{g,t+h}$ is the regression error. Equation (1) is estimated separately for each cell g and each horizon h by OLS on cell g's own quarterly time series ($T_g \approx 108$ observations); all coefficients are therefore

cell-specific, although the subscript g is suppressed on δ_j^h, ϕ_j^h, $\mathbf{\Gamma}_j^h$, and γ_q^h for notational economy, and α_g^h is simply that regression's intercept rather than a fixed effect estimated jointly across a pooled panel. Standard errors and hypothesis tests are based on the bootstrap procedure described in Section 3.3.

The maximum horizon is $H = 12$ (three years). The lag orders p_y and p_x are determined by information criteria, while p_s is determined by autocorrelation diagnostics on the shock series and the sensitivity of the point estimates, yielding $(p_s, p_y, p_x) = (2, 3, 1)$. Under the baseline specification, each regression uses 96 anchors and estimates 13 parameters. The selection procedure is described in Appendix B.1.

Consumption variables are measured in nominal per-capita dollars. Price-level movements are broadly similar across cells, so CPI deflation has little effect on the $Q1$–$Q5$ difference, the regional comparisons, or the tenure comparisons. The level of each group's impulse response should be interpreted as nominal consumption growth. The deflated results are reported in Section 6, while the assumption that all income groups face the same price index is discussed in Appendix F.4.

Swanson (2021) extracts monetary policy shocks from asset price movements in a window extending from 10 minutes before to 20 minutes after each FOMC announcement. The identifying assumption is that price movements within this window are driven mainly by monetary policy information and that the effects of other information arriving at the same time are negligible. Households cannot adjust their consumption decisions within such a short window, so these shocks can be treated as exogenous with respect to consumption decisions. A potential concern is the Fed information effect associated with FOMC announcements, whereby market reactions may partly reflect information about the economic outlook conveyed by the Federal Reserve rather than changes in the policy stance itself (Nakamura and Steinsson, 2018a; Jarociński and Karadi, 2020). The factor decomposition mitigates but does not fully eliminate this concern, so Section 6 re-estimates the model after excluding the periods most likely to be affected, namely NBER recession quarters and extreme shock quarters.

The shocks enter Equation (1) directly as explanatory variables rather than as instruments for an interest rate. This paper compares responses across cells to the same shock, and rescaling the shock into interest-rate units amounts to dividing the responses of all cells to that shock by the same first-stage coefficient (Stock and Watson, 2018). This transformation changes neither the sign of the responses nor the direction of the differences across cells, and a response is zero exactly when its reduced-form counterpart is zero. Moreover, no usable first-stage relationship is available for the asset purchase factor at the quarterly frequency: its F-statistics for the first-stage regressions on quarterly changes in the 10-year Treasury yield and the 30-year mortgage rate are both below 3 (Table B.2 of the appendix). The paper therefore reports responses per unit of the factor, and comparisons across instruments concern the direction of the responses rather than their size.

3.2 Smooth LP

Standard local projections are susceptible to sawtooth impulse response functions due to correlated residuals across horizon-specific regressions, making the pattern of lagged dynamics harder to discern. Following Barnichon and Brownlees (2019), I parameterize the impulse response function using a B-spline basis with K functions, imposing smoothness and continuity across the forecast horizon without changing the identifying assumptions.

I parameterize the impulse response vector using K cubic B-spline basis functions, as follows:

$$\beta_g^h = \sum_{k=1}^{K} \theta_k^g \cdot B_k(h), \quad h = 1, \ldots, H \tag{2}$$

where θ_k^g, $k = 1, \ldots, K$, are cell-specific B-spline coefficients to be estimated and $B_k(h)$ are cubic B-spline basis functions evaluated over the forecast horizon $h = 1, \ldots, H$. The number of basis functions K in Equation (2) is selected from $\{3, 4, 5\}$ by generalized cross-validation (GCV) separately for each cell g.

In practice, all cells select the upper bound $K = 5$, which I treat as a fixed regularization choice rather than an unconstrained optimum in the GCV sense (Appendix B.2). The standard errors of the smoothed local projections rely on a diagonal covariance approximation and may underestimate the true standard errors, so the smoothed specification is used only for graphical display; all tests and significance annotations in the figures are based on the standard local projections. The confidence bands in the figures are obtained by smoothing the bootstrap paths from the standard local projections using the same B-spline projection (Appendix B.2), and a comparison of point estimates between the two specifications is provided in Appendix C.

3.3 Measuring Distributional Effects

The central question of this paper is how the effects of monetary policy shocks on consumption differ across groups. Conventional inequality measures such as the Gini coefficient summarize the entire distribution with a single statistic, which facilitates computation and allows straightforward tracking of inequality over time, but does not reveal which population groups or consumption categories are driving the inequality. I therefore adopt a disaggregated approach, examining each consumption variable separately by income quintile, and characterize inequality through visual comparisons and simple algebraic contrasts across income groups within the same shock-consumption variable combination. This approach preserves more information about the distribution of policy effects and avoids reducing economy-wide inequality to a single index. I examine the distributional effects using four tools: fan charts, adjacent-gap charts, gap-path plots, and a test of the difference between

$Q1$ and $Q5$.

Fan charts place the impulse responses for $Q1$ through $Q5$ in a single figure, with one panel per region, so that both the within-region income gradient and differences across regions can be assessed from the same figure. $Q1$ and $Q5$ are shown with 68% and 90% confidence bands, while the three middle quintiles show point estimates only, following Cloyne et al. (2020) and Coibion et al. (2017).

The adjacent-gap charts plot the differences in impulse responses between each pair of neighboring income quintiles, indicating where in the income distribution the gap is concentrated; the definition is provided in Appendix D.4.

Gap-path plots show the path of the difference between $Q1$ and $Q5$ itself over time, together with a confidence band for that difference, and mark the boundaries of the three yearly windows defined below and the test result for each window in the figure. Like all inference in this paper, these plots use the standard local projections without smoothing; their confidence bands are obtained from the same bootstrap procedure.

The three graphical tools allow differences in impulse responses to be assessed visually, but do not support formal statistical inference. I therefore test the difference between $Q1$ and $Q5$, and restrict the test to these two groups for two reasons. First, testing all pairs of income quintiles would yield up to ten pairwise test statistics, making it difficult to draw clear conclusions. Second, the estimates reveal substantial within-group heterogeneity among $Q2$–$Q4$, suggesting that consumption differences among the middle 60 percent of the income distribution reflect a range of factors that the model does not fully capture. The comparison of $Q1$ and $Q5$ targets the most salient dimension of consumption inequality: the gap between the bottom and top of the income distribution.

For each cell combination (region or tenure type × consumption variable × shock), define $\delta^h \equiv \hat{\beta}^h_{g,Q5} - \hat{\beta}^h_{g,Q1}$, where the two coefficients are obtained from Equation (1) estimated for the corresponding cells. The analysis reports the average of δ^h over a horizon window:

$$\bar{\delta}_w = \frac{1}{|w|} \sum_{h \in w} \delta^h, \tag{3}$$

where w denotes either the full set of 12 horizons (h = 1–12) or one of three yearly windows (h = 1–4, 5–8, and 9–12). The full-horizon average captures the mean difference over the three-year period, while the yearly averages show when the difference emerges. The three windows are defined in advance by year after the shock and do not depend on inspection of the estimation results.

The average $\bar{\delta}_w$ combines regression coefficients from two cells over 4 to 12 horizons. These estimates are correlated across horizons and across cells, so the standard error and p-value of $\bar{\delta}_w$ are obtained from a residual moving block bootstrap (Künsch, 1989). The procedure uses a block length of L = 12 and B = 5000 replications, with the same block indices applied to $Q1$, $Q5$, and

all horizons within each bootstrap draw. Before resampling, the residuals are scaled by $\sqrt{n/(n-k)}$ so that the variance of the resampled errors matches the unbiased estimate of the error variance (Davidson and MacKinnon, 2006). The null hypothesis is $\bar{\delta}_w = 0$, and the p-value is the share of bootstrap draws whose deviation from the estimate is at least as large in absolute value as the estimate itself. Comparisons across regions, policy instruments, and housing tenure types use the same procedure: the relevant differences are averaged over the specified window, and standard errors and p-values are calculated from the same set of bootstrap draws. In a simulation calibrated to the data in which the true effect is zero, the test's rejection rate at the nominal 10% level ranges from 9% to 16%. The conclusions are also insensitive to the choice of block length (Appendix B.3).

Throughout, I treat an effect as present if it is significant at the 10% level; tables mark significance at the 10%, 5%, and 1% levels using *, **, and ***, respectively, and report each estimate with its bootstrap standard error.

4 Main Results

All impulse responses in this section are cumulative consumption responses to a +1 unit contractionary shock, and the response to an expansionary shock is equal in magnitude and opposite in sign. A larger consumption decline for $Q1$ than for $Q5$ ($Q1$ lying below $Q5$ in the figures) therefore implies that $Q1$ gains more under an expansionary shock, that is, a progressive effect. The paths in the figures are estimated using smoothed local projections, while the differences, test statistics, and p-values cited in the text all come from the standard local projections; the qualitative conclusions are the same under both specifications (Appendix C). Table 1 summarizes the main findings in the order in which they are discussed in this and the following section: the $Q1$–$Q5$ differences under the three policy instruments at the national level; the corresponding difference in the Northeast under asset purchases and its comparison with those in the other regions; and the difference between renters and mortgagors within the lowest income group under asset purchases and conventional interest rate policy.

4.1 National Aggregate

At the national level, the four regional cells are aggregated into five national income groups using survey weights. The resulting $Q1$–$Q5$ differences in non-housing consumption are close to zero under all three instruments: 0.6 percentage points under conventional interest rate policy (bootstrap standard error 0.6), −0.2 percentage points under forward guidance (0.5), and −0.02 percentage points under asset purchases (0.8). The same pattern holds for total consumption and the food expenditure ratio (Table D.3 of the appendix). By year, only the first-year difference

Table 1: Main Results: Log Non-Housing Consumption per Capita

	Horizons	Estimate (pp)[a]	Boot. SE	p[b]
Panel A: National aggregate, $Q1$–$Q5$ gap				
FFR	1–12	+0.60	0.62	0.317
FG	1–12	−0.17	0.51	0.734
LSAP	1–12	−0.02	0.80	0.972
Panel B: Regional estimation, LSAP				
Northeast, $Q1$–$Q5$ gap	1–12	+3.29	1.52	0.031
Northeast, $Q1$–$Q5$ gap, second year	5–8	+5.02	1.75	0.005
Northeast, $Q1$–$Q5$ gap, third year	9–12	+3.83	1.82	0.033
Northeast gap minus the mean gap of the other three regions	1–12	+4.21	1.63	0.010
Panel C: Tenure estimation, lowest income quintile, renter minus mortgagor				
LSAP	1–12	+3.20	1.12	0.005
FFR	1–12	−1.90	0.95	0.044
LSAP minus FFR	1–12	+5.10	1.71	0.002

[a] Average over the stated horizons, in percentage points per unit contractionary shock. Panels A and B: $\beta^h_{Q5} - \beta^h_{Q1}$; a positive value means that $Q1$ loses more under a contraction and gains more under an expansion. Panel C: $\beta^h_{\text{Renter}} - \beta^h_{\text{Mortgaged}}$; a positive value means that mortgagor households lose more than renters with the same income.

[b] Bootstrap test that the average in the row is zero (residual moving block bootstrap, $B = 5000$, $L = 12$). 'Boot. SE' is the bootstrap standard deviation of the estimate.

Notes: the national aggregate pools the four regions (Table D.3 of the appendix). The results for every region, instrument, and outcome are reported in Table D.1 of the appendix, and those for every income quintile and tenure group in Tables 2 and E.5.

under conventional interest rate policy is significant at the 10% level (1.3 percentage points, $p = 0.081$), and it does not persist. None of the three instruments consistently widens or narrows the consumption gap between high- and low-income households at the national level.

A zero average has two possible readings: monetary policy has no distributional effect on consumption, or effects in opposite directions offset one another in the national aggregate. The precision of the estimated national difference helps distinguish between these possibilities. Under asset purchases, the standard error is 0.8 percentage points, smaller than that in any individual region. This precision is sufficient to rule out a difference exceeding 3 percentage points if that difference were uniform across the country, but it cannot rule out regional differences in opposite directions. The analysis therefore turns first to regional estimates and then to differences by housing tenure.

4.2 LSAP: Progressive Distributional Effects

The literature offers two competing predictions about the distributional effects of unconventional monetary policy. If LSAP operates primarily through the asset price channel (Domanski et al., 2016)—compressing long-term interest rates and pushing up equity and housing valuations—higher-income households, which hold more financial assets, should benefit more, producing a

regressive distributional outcome ($Q5 > Q1$). If instead the dominant channel is employment and labor income (Kaplan et al., 2018)—LSAP expanding aggregate demand and boosting job creation—lower-income households should gain more, as labor income accounts for a larger share of their total income and they are more sensitive to employment conditions, producing a progressive outcome ($Q1 > Q5$). Distinguishing between these two predictions is the starting point for the regional analysis.

Figure 2 plots the cumulative impulse responses of non-housing consumption to a contractionary LSAP shock across the four Census regions. In the Northeast, the decline is larger for $Q1$ than for $Q5$: the $Q1$–$Q5$ difference averages 3.3 percentage points over the three-year horizon (bootstrap standard error 1.5, $p = 0.031$) and peaks at close to 7 percentage points. This difference emerges with a clear lag. It averages approximately 1 percentage point in the first year and is not statistically distinguishable from zero, increases to approximately 5 percentage points in the second year ($p = 0.005$), and remains at approximately 4 percentage points in the third year ($p = 0.033$). Total consumption shows the same timing, with differences of 3.8 and 2.6 percentage points in the second and third years, respectively ($p = 0.002$ and 0.034). This is not a mechanical artifact of cumulation: between the first and second years, the difference increases almost fivefold, while its bootstrap standard error remains nearly unchanged. In the other three regions, the point estimates are around -1 percentage point, opposite in sign to the Northeast, and none is distinguishable from zero. Regional estimates are reported in Table D.1 of the appendix.

Figure 3 plots the path of the difference δ^h directly, with dotted lines marking the boundaries of the three windows and panel titles reporting the test results for each window.

The adjacent-quintile gap charts in Appendix D show that the progressive effect in the Northeast is not distributed evenly across the income distribution. Of the four adjacent gaps, only those at the bottom (between $Q2$ and $Q1$) and at the top (between $Q5$ and $Q4$) are clearly positive, while the two gaps in the middle of the distribution are both close to zero, with one slightly negative. No test is conducted on the adjacent gaps themselves, so they are used here only to indicate where in the distribution the gap is concentrated, not as effect sizes. The overall gap between $Q1$ and $Q5$ therefore arises mainly from the widening of the two tails relative to their neighboring quintiles, rather than from a stepwise progression across the distribution.

By Engel's Law, the food expenditure ratio can serve as an indirect indicator of changes in consumption composition. In the Northeast, the increase in the food expenditure ratio for $Q1$ following a contractionary LSAP shock is significantly larger than that for $Q5$, with this difference emerging in the second year; the difference is negative at most horizons and reaches its largest magnitude between the end of the second year and the middle of the third. This runs in the same direction as the non-housing consumption results: $Q1$ not only shows a larger decline in nominal consumption but also a more pronounced deterioration in expenditure composition. The food

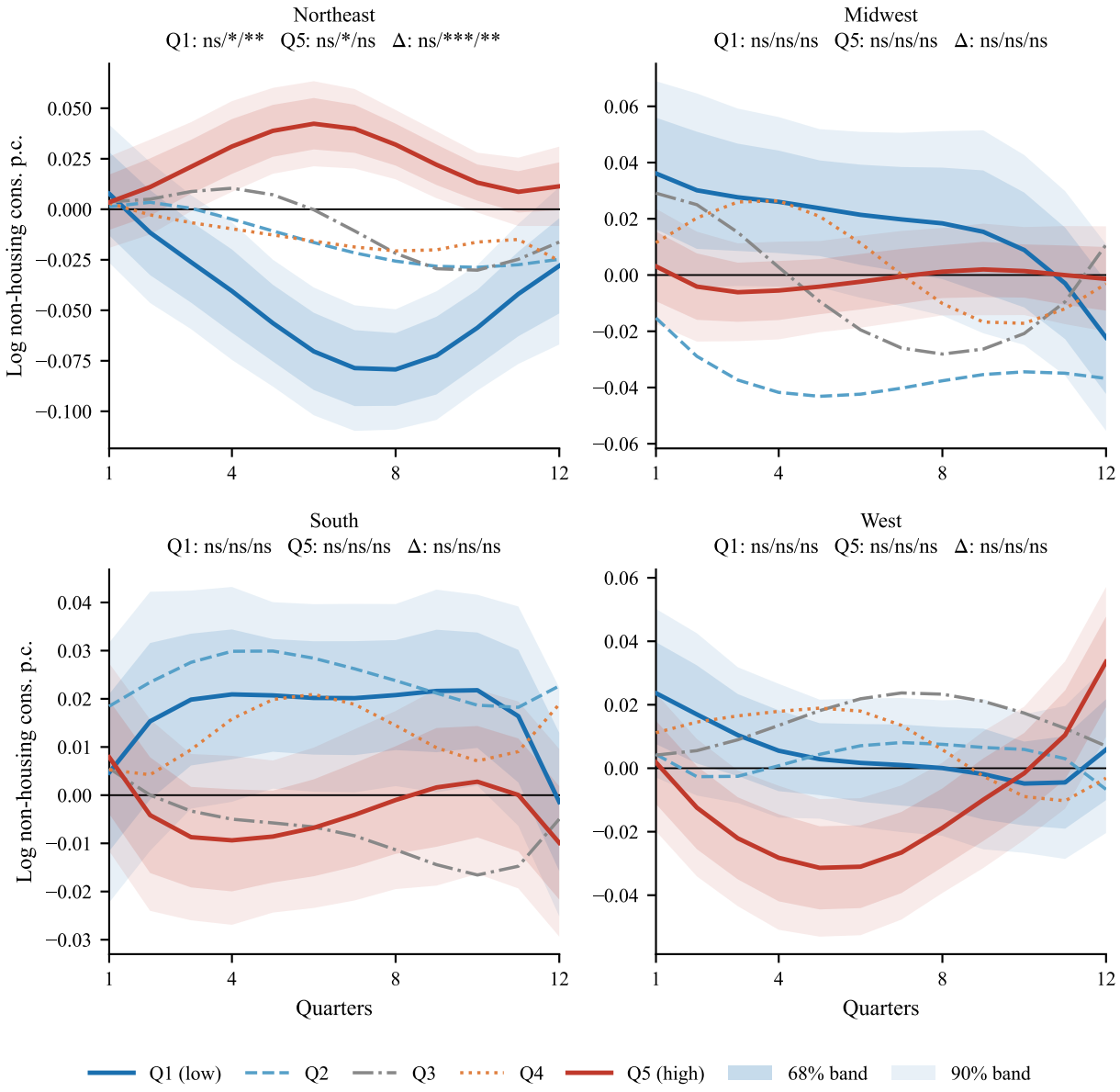


Figure 2: IRF: Log Non-Housing Consumption per Capita — LSAP Shock

Notes: (1) Non-housing consumption is total consumption expenditure minus housing costs (rent and mortgage payments), in log per-capita nominal dollars. (2) Responses to a one-unit contractionary LSAP shock; the response to an expansionary shock of equal magnitude is the mirror image (see Section 2.2). (3) IRF paths are Smooth LP estimates ($K = 5$, GCV-selected; Section 3.2). Confidence bands and significance annotations come from the residual moving block bootstrap on the Standard LP estimates ($B = 5000$, $L = 12$; Section 3.1); no Newey–West band is plotted. (4) Shaded bands: 68% CI (inner, darker) and 90% CI (outer, lighter) for Q1 and Q5; Q2–Q4 show point estimates only. (5) Panel titles report the bootstrap tests that the window average is zero, in the order h = 1–4 / 5–8 / 9–12, for $Q1$, for $Q5$, and for the difference $\Delta \equiv \delta^h$; ns denotes $p \geq 0.10$, * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The tests of the difference are tabulated in Table D.1 of the appendix. (6) Analysis sample: 1992 Q1–2019 Q2 (110 quarters).

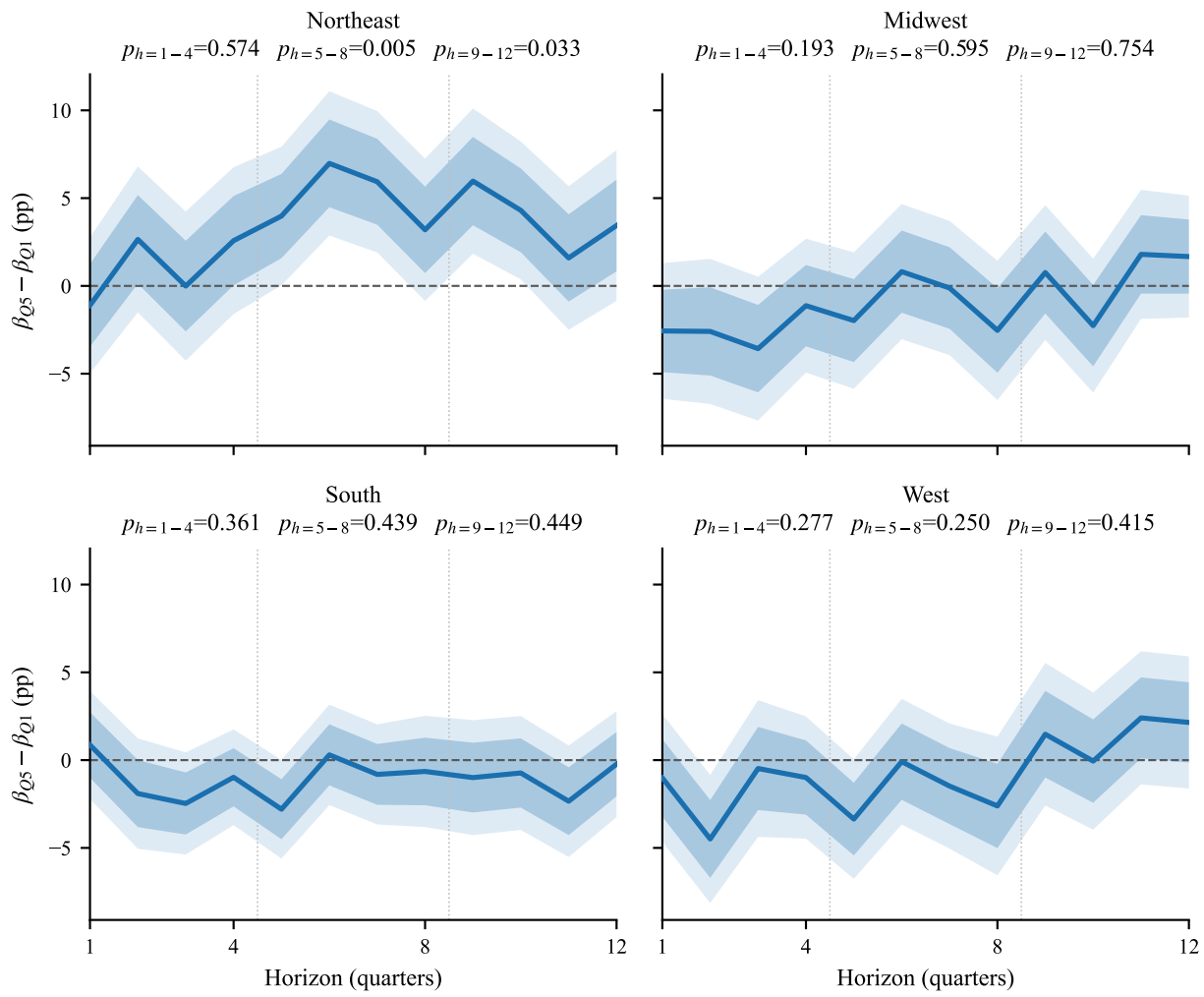


Figure 3: The $Q1$–$Q5$ Gap in Log Non-Housing Consumption per Capita — LSAP Shock

Notes: (1) Each panel plots the distributional gap itself, $\delta^h \equiv \beta^h_{Q5} - \beta^h_{Q1}$, rather than the two quintiles' separate responses as in Figure 2. A positive value means that $Q1$'s cumulative response lies below $Q5$'s, that is, $Q1$ loses more under a contractionary shock. (2) Estimates are Standard LP, matching the basis for all inference in this paper; no smoothing is applied. (3) Shaded bands are ±1 and ±1.645 bootstrap standard deviations (68% and 90%), obtained from the residual moving block bootstrap ($B = 5000$, $L = 12$). (4) The three p-values in each panel title are from the bootstrap tests that the average over $h = 1$–4, 5–8, and 9–12 is zero (Section 3.3); dotted vertical lines mark the window boundaries. (5) Analysis sample: 1992 Q1–2019 Q2 (110 quarters).

expenditure ratio provides no robust evidence in the remaining three regions or under FFR and FG shocks (Appendix D).

4.3 Forward Guidance

Compared with LSAP, FG produces no consistent regional pattern. The full-horizon average difference is not statistically distinguishable from zero in any region for any consumption measure. Across the yearly windows, four differences are statistically significant at the 10% level: non-

housing consumption (−2.3 percentage points, $p = 0.082$) and the food expenditure ratio in the Northeast in the second year, non-housing consumption in the South in the first year, and housing consumption in the South in the second year. These occur in different periods and do not point in a consistent direction, and the second-year difference in the Northeast has the opposite sign to that under LSAP. FG therefore transmits differently from LSAP; its own distributional direction cannot be determined from these results.

The differences across regions may be related to factors such as industrial structure, but this paper does not incorporate region-by-industry data to test this possibility. Full impulse responses for each region are reported in Appendix D.

In the Northeast and West, non-housing consumption for $Q1$ is higher than that for $Q5$ following a contractionary FG shock, in the opposite direction to LSAP, while this pattern does not hold in the Midwest and South. The evidence from the food expenditure ratio is directionally consistent but weak: the $Q1$–$Q5$ difference for the Northeast passes the test only in the second-year window and is insignificant over the full horizon, while the direction in the Midwest is not consistent.

The adjacent-gap charts in Figure D.6 of the appendix show that the FG differences are close to zero and insignificant in most regions. Taking the two unconventional instruments together, $Q1$ declines systematically more than $Q5$ under LSAP in the Northeast, while the $Q1$–$Q5$ ordering under FG varies across regions in both direction and statistical significance. The distributional consequences of quantitative easing and forward guidance therefore cannot be treated as interchangeable forms of "monetary easing".

4.4 Conventional Monetary Policy (FFR)

The distributional effects of conventional interest rate policy appear in housing consumption. Following a contractionary shock, the decline in housing consumption is smaller for $Q1$ than for $Q5$: the $Q1$–$Q5$ difference is −2.7 percentage points in the Northeast in the first year ($p = 0.009$) and −2.2 percentage points in the South in the third year ($p = 0.009$). Total consumption in the South shows a difference in the same direction in the third year (−1.5 percentage points, $p = 0.083$). For non-housing consumption and the food expenditure ratio, no statistically detectable income gradient is found in any of the four regions. The housing consumption response for $Q1$ is positive in the Northeast, Midwest, and South, but negative in the West. This pattern is consistent with rents rising following a contractionary shock while monthly payments on existing fixed-rate mortgages remain unchanged in response to short-term interest rate movements. Conventional interest rate policy thus affects housing costs and not consumption demand, which is also why I use non-housing consumption as the primary outcome variable; Section 5 examines this by housing contract type. The full regional results, including fan charts and adjacent-gap charts, are reported in Appendix D.

4.5 Differences across Regions and Instruments

Table D.1 of the appendix reports the estimates, standard errors, and test results for each region, policy instrument, and consumption measure. This subsection asks two questions: whether the differences across regions are significant, and whether the differences across instruments are significant.

A statistically significant result in one region and a non-significant result in another does not, by itself, establish a significant difference between the two (Gelman and Stern, 2006). I therefore test directly whether the $Q1$–$Q5$ differences are equal across the four regions. Under asset purchases, this hypothesis is rejected in the second year for both non-housing consumption ($p = 0.017$) and total consumption ($p = 0.008$), while the corresponding full-horizon p-values are 0.076 and 0.104, respectively. The difference lies between the Northeast and the other three regions. The $Q1$–$Q5$ difference in the Northeast exceeds the average across the other three regions by 4.2 percentage points for non-housing consumption ($p = 0.010$) and 3.2 percentage points for total consumption ($p = 0.013$) over the full horizon.[2] In the second year, these differences increase to 6.3 and 5.0 percentage points, respectively ($p = 0.002$ and $p = 0.003$). Pairwise comparisons of the Northeast with the Midwest, South, and West yield full-horizon p-values of 0.026, 0.015, and 0.045 for non-housing consumption, and 0.029, 0.050, and 0.031 for total consumption, respectively. No statistically detectable differences are found among the remaining three regions. The national average is therefore zero because regional differences cancel in the aggregate. The same expansionary asset purchases narrow the consumption gap between low- and high-income households in the Northeast and not in the other regions. Under conventional interest rate policy and forward guidance, the hypothesis of equal $Q1$–$Q5$ differences across the four regions is not rejected.

Instruments can also be compared pairwise within the same region and consumption measure (Table D.2 of the appendix). In the Northeast, the $Q1$–$Q5$ difference under asset purchases exceeds that under forward guidance by 7.4 percentage points in the second year ($p < 0.01$). It exceeds that under conventional interest rate policy by 4.8 percentage points ($p = 0.072$), and by 3.9 percentage points for total consumption ($p = 0.034$). Outside the Northeast, the only full-horizon result under asset purchases that is statistically significant is for housing consumption in the South (1.6 percentage points, $p = 0.042$). Housing consumption is directly affected by interest rates, and this paper does not treat it as evidence on consumption demand; within the same region, however, it is

[2]The differences remain significant at the 5% level after adjusting for multiple comparisons across the four regions. Treating each region's difference relative to the average of the remaining regions as a family of tests and adjusting the p-values using the bootstrap distribution of the maximum statistic within that family yields p-values of 0.036 for non-housing consumption and 0.048 for total consumption in the Northeast. The Northeast's differences therefore remain statistically significant even when accounting for the possibility that this region was selected post hoc from among the four regions.

the outcome for which the difference between asset purchases and conventional interest rate policy is most pronounced: in the third-year window, the differences under LSAP and FFR are +2.52 and −2.23 percentage points, respectively, and the gap between the two, +4.76 percentage points, is significant at the 1% level.

Among the three instruments, the direction is clear under asset purchases; under forward guidance and conventional interest rate policy it varies by region and consumption measure and cannot be classified simply as progressive or regressive. The differences in distributional effects across the three instruments reflect differences in their transmission channels, rather than differences in the strength of a common "monetary easing" effect.

5 Mechanism Analysis

Section 4 raises four questions: why the national average is close to zero; why only asset purchases produce a robust income gradient, and why it appears in the Northeast; why this gradient takes shape in the second year; and why conventional interest rate policy shows up in housing consumption while forward guidance has no consistent direction.

The regional split alone cannot answer these questions, because monetary policy reaches households through their contracts: it changes the cash payments currently due under a given type of contract or the terms on which that contract can be renegotiated. This section therefore reslices the same sample by housing contract type, crossed with income quintile. This yields fifteen cells, with the estimation and inference procedures unchanged, and the results are at the national level.

5.1 Housing Contracts and Testable Predictions

Mortgage contracts in the United States are predominantly long-term and fixed-rate, unlike those in most advanced economies (Badarinza et al., 2016). As a result, the monthly payments of households that have already entered into such contracts do not move with market interest rates over the life of the loan. The contract also embeds a prepayment option, which is the only route through which monetary policy can affect the cash flow of existing borrowers, and whether it is worth exercising depends on long-term mortgage rates rather than the federal funds rate. Leases, by contrast, are short and reprice frequently, and both landlords' holding costs and the supply of rental housing are sensitive to the short end. The housing cash flow of renter households is therefore floating, while that of mortgagor households is fixed, and this contractual difference explains why the same shock has different consequences for the two groups.

Conventional interest rate policy moves the short end. It does not change the monthly payment on an existing loan, but it passes through to rents and the price of new credit, so its impact falls on the

renter side. Asset purchases move the long end and compress mortgage-backed securities spreads through direct purchases; these are the two prices that determine the value of the prepayment option. LSAP is therefore the only instrument that can open or close the refinancing channel for existing borrowers, and its impact falls on the mortgagor side. Forward guidance moves the expected path of the short end without changing current rents or mortgage payments, and expectations alone are not sufficient to trigger refinancing.

Comparing mortgagors and renters within the same income group therefore implies a larger response for renters under conventional interest rate policy, a larger response for mortgagors under asset purchases, and no difference between the two under forward guidance. These three predictions point in different directions, making it difficult for a single alternative explanation to account for all three simultaneously. They also imply two mutually exclusive criteria: the cash flow channel requires the income gradient to be concentrated within the mortgagor group, whereas the employment channel requires a gradient to appear within all three groups simultaneously. Renter households carry no refinanceable debt, so a gradient within the renter group provides the only source of identification for the employment channel.

5.2 Results from the Tenure Estimation

Tenure and income are highly correlated, so comparisons are made within a given income quintile. Mortgagors account for roughly one-ninth of the lowest quintile, making this cell thinner than the other two cells in the same quintile.

The pattern under all three instruments is consistent with the predictions in the previous subsection. Under conventional interest rate policy, the decline in non-housing consumption among renters in the lowest income group is 1.9 percentage points larger than that among mortgagors in the same income group ($p = 0.044$). The difference for $Q3$ has the same sign, at 1.3 percentage points ($p = 0.062$), while the difference for $Q2$ is in the same direction but is not statistically significant (Table 2). For $Q4$, the difference reverses direction and is smaller in magnitude (0.9 percentage points, $p = 0.091$). The difference in the lowest income group is no longer statistically significant when orthogonalized shocks are used or when all three shocks enter the regression simultaneously, but its difference from the corresponding gap under asset purchases remains statistically significant in all these specifications (Panel C of Table 3 and Table F.3 of the appendix). No income gradient in non-housing consumption is evident within any of the three groups. Conventional policy therefore generates differences across tenure rather than across income. This pattern differs from the finding in Cloyne et al. (2020), who report that under conventional policy, the consumption response of mortgagor households is larger than that of renters. One possible explanation is that this paper compares households within the same income group, whereas mortgagors are concentrated at the

high-income end of the distribution; without conditioning on income, differences across tenure are confounded with differences across income.

The direction reverses under asset purchases. In the lowest income group, the decline in non-housing consumption is 3.2 percentage points larger among mortgagors than among renters ($p = 0.005$). This reversal can be tested directly. Within the lowest income group, the difference in consumption responses between renters and mortgagors under asset purchases differs from the corresponding difference under conventional interest rate policy by 5.1 percentage points. This contrast is significant at the 1% level over the full horizon and is also significant in each of the three yearly windows ($p = 0.061$, 0.021, and 0.006); relative to forward guidance, the contrast is significant at the 5% level over the full horizon (Table E.5 of the appendix). The income gradient also shifts to the mortgagor group and widens over time (Figure 4). No income gradient appears within the renter group under any measure or in any window, except for the food expenditure ratio in the first year ($p = 0.073$). Neither pattern is observed under forward guidance. Among outright owners, the only significant gradient is in housing consumption, which reflects a direct change in holding costs rather than consumption demand (Appendix E); the gradient in non-housing consumption is not significant.

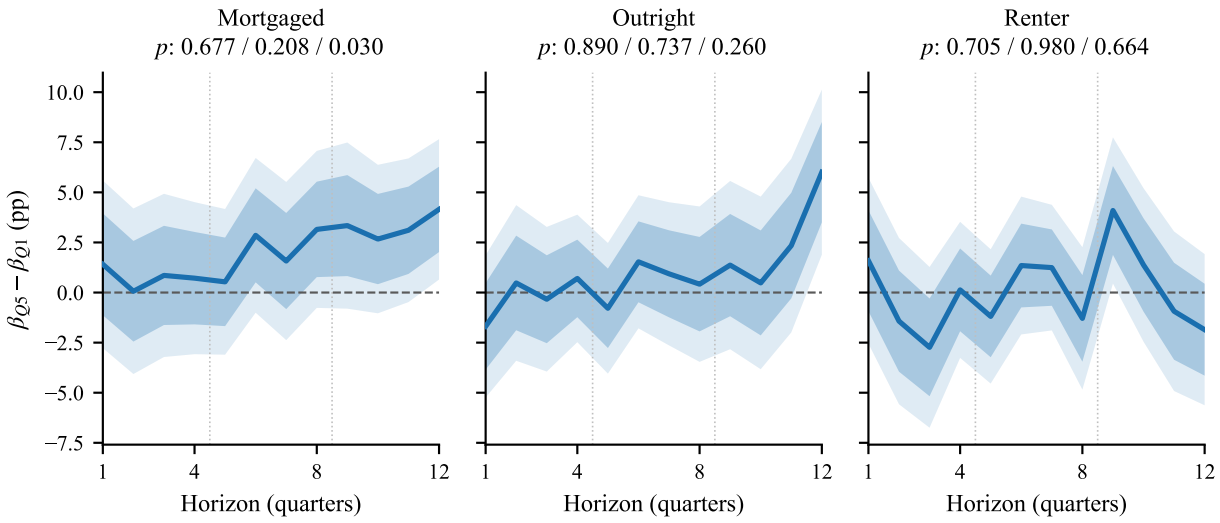


Figure 4: The $Q1$–$Q5$ Gap in Log Non-Housing Consumption per Capita — LSAP Shock, by Housing Tenure

Notes: (1) Each panel plots the distributional gap itself, $\delta^h \equiv \beta^h_{Q5} - \beta^h_{Q1}$, for one tenure group, rather than the five quintile responses separately as in Figure E.1 of the appendix. A positive value means that $Q1$'s cumulative response lies below $Q5$'s, that is, $Q1$ loses more under a contractionary shock. (2) Estimates are Standard LP, matching the basis for all inference in this paper; no smoothing is applied. (3) Shaded bands are ±1 and ±1.645 bootstrap standard deviations (68% and 90%), obtained from the residual moving block bootstrap ($B = 5000$, $L = 12$). (4) The three p-values in each panel title are from the bootstrap tests that the average over $h = 1$–4, 5–8, and 9–12 is zero (Section 3.3); dotted vertical lines mark the window boundaries. The same window tests for all outcomes are reported in Table E.3. (5) Analysis sample: 1992 Q1–2019 Q2 (110 quarters).

The income gradient in non-housing consumption among mortgagors is close to zero in the first year but widens in each subsequent year, reaching 3.3 percentage points in the third year ($p = 0.030$). The three-year average is 2.0 percentage points ($p = 0.124$). This gradual accumulation matches the lag predicted by the refinancing channel and differs in character from the forward guidance window results in Section 4.3, which are scattered across regions and periods and mostly reverse or decay thereafter. The expenditure composition within the same group points in the same direction: following a contractionary asset purchase shock, the food expenditure ratio for low-income mortgagor households rises more than that for high-income mortgagor households, and this gradient is significant in both the second- and third-year windows and in the full-horizon test ($p = 0.008$), in the same direction as the gradient in non-housing consumption, with low-income borrowers facing greater budgetary pressure (Table E.3). For conventional interest rate policy, the expenditure composition is likewise consistent with the contract explanation: following a contractionary shock, the food expenditure ratio for low-income households within the renter group rises more than that for high-income households, with this difference significant in both the first- and third-year windows, suggesting that rent increases place greater pressure on low-income renters; this does not alter the conclusion regarding non-housing consumption set out above.

In short, no gradient appears within the renter group, so this design finds no evidence for the employment channel. The gradient is concentrated within the mortgagor group, and the condition for the cash flow channel holds in the baseline. Full cell-by-cell test results for all consumption measures are provided in Table E.3 of the appendix.

Forward guidance does not reject the null in any pairwise comparison within income groups, consistent with the contractual interpretation: it generates no current-period cash flow and therefore produces no systematic differences across tenure. The two unconventional instruments are not interchangeable: asset purchases operate through mortgage contracts, whereas forward guidance operates through no contractual cash flow channel.

Contracts explain which households are reached but not why those reached differ among themselves. The key point is that every borrower holds a prepayment option, but exercising it is subject to several thresholds: closing costs, verifiable income, sufficient equity, and a credit assessment that must be passed (Beraja et al., 2019). Low-income borrowers are the group for whom these conditions bind most tightly, with monthly payments accounting for the largest share of disposable income and the thinnest financial buffers; higher-income borrowers, whose payments account for a much smaller share of income, are less dependent on this route for relief (Di Maggio et al., 2017). When a contractionary asset purchase shock raises long-term rates and spreads together, refinancing becomes less attractive for all borrowers, but those with the most constrained budgets are the households that relied most on this option for relief.

This also explains the timing. Refinancing must be initiated by the borrower, and application,

Table 2: Renter versus Mortgagor Households within the Same Income Quintile: Log Non-Housing Consumption per Capita

	FFR				FG				LSAP			
Quintile	h=1–12	1–4	5–8	9–12	h=1–12	1–4	5–8	9–12	h=1–12	1–4	5–8	9–12
$Q1$	-1.90^{**}	-1.68	-2.09^{*}	-1.93	-0.34	-0.89	-0.03	-0.12	$+3.20^{***}$	$+2.81^{*}$	$+2.88^{**}$	$+3.91^{***}$
	(0.95)	(1.25)	(1.15)	(1.19)	(0.81)	(1.00)	(1.05)	(0.98)	(1.12)	(1.58)	(1.39)	(1.37)
$Q2$	-1.25	-1.58	-1.11	-1.04	$+0.85$	$+0.19$	$+1.01$	$+1.34$	$+0.02$	$+1.87$	-0.61	-1.20
	(0.98)	(0.96)	(1.22)	(1.23)	(0.85)	(0.85)	(1.07)	(1.10)	(1.24)	(1.30)	(1.57)	(1.51)
$Q3$	-1.32^{*}	-1.09	-1.97^{**}	-0.91	$+0.32$	$+0.10$	$+0.45$	$+0.40$	$+0.05$	$+0.10$	-0.18	$+0.24$
	(0.71)	(0.84)	(0.85)	(0.83)	(0.61)	(0.68)	(0.78)	(0.70)	(0.92)	(1.10)	(1.11)	(1.06)
$Q4$	$+0.89^{*}$	$+0.65$	$+1.25$	$+0.78$	$+0.77$	$+1.13$	$+0.73$	$+0.44$	-0.82	$+0.00$	-1.76^{*}	-0.69
	(0.52)	(0.71)	(0.77)	(0.63)	(0.55)	(0.68)	(0.76)	(0.60)	(0.72)	(1.00)	(0.97)	(0.80)
$Q5$	-0.96	-0.78	-0.45	-1.65	-0.59	-1.52	-0.41	$+0.17$	$+1.19$	$+1.44$	$+0.88$	$+1.26$
	(1.00)	(1.26)	(1.01)	(1.30)	(0.78)	(0.95)	(0.90)	(1.00)	(1.30)	(1.74)	(1.26)	(1.67)

Notes: each entry is the average of $d^h \equiv \beta^h_{\text{Renter}} - \beta^h_{\text{Mortgaged}}$ within the stated income quintile over the stated horizons, in percentage points per unit contractionary shock, with its bootstrap standard error in parentheses. Stars refer to the bootstrap test that the average is zero (residual moving block bootstrap, $B = 5000$, $L = 12$): $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. A positive value means that the consumption of mortgagor households falls more than that of renter households with the same income; a negative value means that renters' consumption falls more. Results for the other outcomes are reported in Table E.4 of the appendix.

appraisal, approval, and closing each take time; what matters is the cumulative budgetary effect of refinancing remaining out of reach. If the distributional effect were driven mainly by asset prices and wealth effects, it should emerge soon after the announcement. The observed pattern is the reverse: the gradient within the mortgagor group is close to zero in the first year, widens from one window to the next thereafter, and is largest in the third year. The timing provides evidence independent of the tenure decomposition.

5.3 The Northeast and the National Average

A natural explanation for the Northeast's prominence is that it has a higher share of mortgagor households, but the data do not support this. Tenure composition is very similar across the four regions, with the gap in the mortgagor share between the highest and lowest income quintiles at around 60 percentage points in each region (Table E.2 of the appendix). A pure composition difference would, moreover, predict the opposite sign: mortgagors are concentrated at the high-income end, so if asset purchases affected only mortgagor households, the decline should be larger for the high-income group. One possible explanation is that housing expenditure accounts for a slightly larger share of household budgets in the Northeast, so the same change in housing cash flow crowds out more non-housing consumption. The directions of the three measures in the Northeast are also consistent with this explanation, with non-housing consumption declining relatively more and housing consumption and the food expenditure ratio rising relatively more. The data do not

allow a direct test of this explanation.

The housing contract channel also helps explain why the national average is close to zero. The income gradient under asset purchases emerges among mortgagors, who account for only about one-ninth of the lowest income quintile (Table E.1 of the appendix). When consumption responses are averaged by income group at the national level, the responses of these households are combined with those of renters and outright owners, who constitute the majority and are not directly exposed to this refinancing channel. Conventional interest rate policy, by contrast, affects renters through a different channel and produces effects in the opposite direction. The national average by income group can therefore be close to zero even when consumption responses differ significantly across housing contract types.

The two estimations provide two ways of slicing the same sample rather than a joint estimation; the number of cells does not permit the data to be partitioned along all three dimensions simultaneously. The claim that the effect in the Northeast is driven by low-income borrowers is therefore an inference: the tenure estimation does not distinguish regions, while the regional estimation does not distinguish contracts. The employment channel (Kaplan et al., 2018) is likewise not ruled out, but this design cannot identify its contribution separately, and no gradient appears within the renter group in this sample. The employment channel therefore cannot be regarded as dominant.

After excluding either recession quarters or extreme shock quarters, the gradient within the mortgagor group is no longer statistically significant. After excluding extreme shock quarters, a significant gradient emerges within the renter group in the third year, which is the pattern required by the employment channel criterion. The excluded quarters are those in which the unconventional instruments were deployed on a large scale. The distinctions along the tenure dimension, including the absence of a gradient within the renter group, are therefore identified from crisis-period variation, as is the main result (Appendix F).

6 Robustness

This section reports five sets of robustness checks to assess the sensitivity of the main findings to the estimation method, income classification, sample selection, the shock series, and the regression specification. Table 3 summarizes how the three main results perform under each check, and full results for each check are provided in Appendix F.

Consistency between standard and smoothed local projections. In the regional estimation, the relative ordering of $Q1$ and $Q5$ is the same under both specifications across all regions and consumption variables, and the significance judgments are identical. In the tenure estimation, the paths for some tenure groups under the LSAP shock show visible magnitude shifts, reflecting bias introduced by the B-spline parameterization under weak-signal conditions, which is why Figure 4

Table 3: Robustness of the Regional and Mechanism Results

Specification	$\bar{\delta}$ (pp)[a]	Boot. SE	p (1–12)	p (1–4)	p (5–8)	p (9–12)
Panel A: Northeast, $Q1$–$Q5$ gap, LSAP (regional estimation)						
Baseline	+3.29	1.52	0.031	0.574	0.005	0.033
INC_RANK income classification	+4.05	1.67	0.013	0.331	0.009	0.015
Excluding extreme-shock quarters	+4.39	2.82	0.124	0.997	0.058	0.033
Excluding recession quarters	+2.29	2.60	0.380	0.879	0.320	0.273
LSAP factor set to zero before 2009	+4.58	2.26	0.049	0.329	0.013	0.100
Shocks orthogonalized to pre-announcement information	+3.22	1.79	0.071	0.758	0.012	0.071
Excluding 2009Q1	+2.68	1.57	0.087	0.910	0.030	0.041
Excluding 2013Q3	+3.91	1.59	0.014	0.536	0.002	0.009
No lags of the shock ($p_s = 0$)	+2.63	1.24	0.033	0.599	0.012	0.027
Lagged interest rates added	+3.21	1.51	0.034	0.440	0.005	0.087
Three shocks entered jointly	+3.99	1.77	0.024	0.293	0.012	0.021
Panel B: Mortgagors, $Q1$–$Q5$ gap, LSAP (tenure estimation)						
Baseline	+2.04	1.32	0.124	0.677	0.208	0.030
INC_RANK income classification	+2.24	1.21	0.064	0.376	0.110	0.042
Excluding extreme-shock quarters	+1.59	2.16	0.488	0.562	0.613	0.517
Excluding recession quarters	+1.45	2.16	0.511	0.846	0.825	0.200
LSAP factor set to zero before 2009	+2.23	2.12	0.300	0.479	0.712	0.137
Shocks orthogonalized to pre-announcement information	+3.25	1.50	0.030	0.308	0.064	0.011
Excluding 2009Q1	+2.19	1.40	0.115	0.735	0.215	0.017
Excluding 2013Q3	+1.59	1.36	0.244	0.809	0.316	0.081
No lags of the shock ($p_s = 0$)	+1.38	1.08	0.198	0.695	0.340	0.069
Lagged interest rates added	+2.51	1.27	0.047	0.509	0.086	0.010
Three shocks entered jointly	+2.13	1.55	0.174	0.657	0.380	0.031
Panel C: Renters minus mortgagors in $Q1$, LSAP minus FFR (tenure estimation)						
Baseline	+5.10	1.71	0.002	0.061	0.021	0.006
INC_RANK income classification	+4.79	1.40	0.001	0.028	0.006	0.015
Excluding extreme-shock quarters	+1.81	2.69	0.511	0.727	0.310	0.780
Excluding recession quarters	+2.95	2.63	0.268	0.515	0.494	0.181
LSAP factor set to zero before 2009	+5.64	2.24	0.010	0.060	0.084	0.020
Shocks orthogonalized to pre-announcement information	+5.47	1.80	0.001	0.027	0.009	0.029
Excluding 2009Q1	+5.15	1.71	0.003	0.068	0.015	0.007
Excluding 2013Q3	+4.59	1.69	0.006	0.094	0.036	0.012
No lags of the shock ($p_s = 0$)	+3.99	1.56	0.007	0.097	0.025	0.024
Lagged interest rates added	+5.74	1.74	<0.001	0.044	0.008	0.002
Three shocks entered jointly	+3.53	1.41	0.012	0.146	0.048	0.019

[a] In Panels A and B, $\bar{\delta}$ is the gap $\beta^h_{Q5} - \beta^h_{Q1}$ averaged over the 12 horizons, in percentage points per unit contractionary shock. In Panel C it is the difference $\beta^h_{\text{Renter}} - \beta^h_{\text{Mortgaged}}$ within the lowest income quintile under LSAP minus the same difference under FFR, averaged in the same way. 'Boot. SE' is its bootstrap standard deviation.

Notes: the p-values are from the bootstrap test that the average of the row's quantity over the stated horizons is zero (residual moving block bootstrap, $B = 5000$, $L = 12$; $B = 2000$ for INC_RANK). The second to fourth rows of each panel change the income classification or drop quarters from the sample; the last seven rows change the shock series, remove a single quarter, or change the specification. All rows are defined in the notes to Table F.2 in the appendix, which reports the same checks for the other main results.

uses the standard local projections for display. Full comparisons are provided in Appendix C.

Alternative income classification variable. The baseline specification assigns households to quintiles based on their relative position within the annual regional income distribution; as an alternative, I use the BLS `INC_RANK` variable, which gives each household's percentile in the national income distribution. $Q1$ in the baseline consists of households that are relatively poorest within each region, while $Q1$ under `INC_RANK` consists of households that are poorest nationally; households in high-cost-of-living regions that are relatively poor locally but rank in the middle nationally are assigned to different groups under the two schemes. The two classifications use exactly the same sample, so their significance levels can be compared directly. The results show that the Northeast is insignificant in the first year and significant in the second and third years under both classifications, with a slightly larger difference under `INC_RANK`, while the remaining three regions are insignificant under both (Table 3 and Table F.2 of the appendix). The core finding of this paper therefore does not depend on the classification based on within-region relative income.

Excluding recession and extreme shock quarters. I re-estimate the model after separately dropping the recession quarters defined by the National Bureau of Economic Research (2001Q1–2001Q4 and 2007Q4–2009Q2, reducing the available anchors from 96 to 85) and the extreme quarters in which any shock series exceeds two standard deviations in absolute value (15 quarters, leaving 81 anchors). The following discussion concerns the income difference in non-housing consumption under the LSAP shock in the Northeast. The two exclusions push the point estimate in opposite directions: excluding extreme shock quarters widens the difference, whereas excluding recession quarters narrows it (Table 3). The excluded quarters are those with the largest shock variance and the main source of information for identifying the coefficient, so the standard errors rise by approximately 70% and 85%, respectively. After excluding recession quarters, all three window tests are far from significant; after excluding extreme shock quarters, the third-year window remains significant ($p = 0.033$) and the second-year window is significant at the 10% level ($p = 0.058$). Whether this effect can be statistically distinguished therefore depends on whether the sample contains policy shocks of sufficient magnitude. The Northeast's difference relative to the other three regions is less sensitive to these exclusions: the second-year difference is 6.5 percentage points when recession quarters are excluded ($p = 0.056$) and 7.5 percentage points when quarters with extreme shocks are excluded ($p = 0.049$). Details are provided in Table F.2 of the appendix.

Deflating consumption by the price level. Because the specification includes each cell's own lagged dependent variable, deflation does not affect the estimates for the two quintiles identically, so this check is not mechanical. After deflating non-housing consumption, total consumption, and housing expenditure using CPIAUCSL from FRED and re-estimating the models, the conclusions for the Northeast under the LSAP shock and for the South under the FFR shock are unchanged, indicating that the distributional effects reported in this paper are not driven by price-level movements

but reflect differences in real consumption (Table F.2 of the appendix).

Shock series, individual quarters, and regression specification. Four further sets of checks address the measurement of the shocks and the specification in Equation (1), with results reported in the last seven rows of each panel of Table 3 and in Table F.2 of the appendix. First, the three factors are predictable from information already available before the announcement (Table F.1 of the appendix); after extracting announcement-level residuals following Bauer and Swanson (2023), the second-year difference for the Northeast increases from 5.0 to 5.2 percentage points and remains statistically significant at the 5% level. Second, the LSAP factor is not zero before 2009, and 41% of its sum of squares within the estimation sample comes from this period; setting these pre-2009 values to zero leaves the second-year difference for the Northeast statistically significant at the 5% level. Third, 2009Q1 alone accounts for 32% of the sum of squares of the LSAP factor; excluding this quarter leaves the second-year difference for the Northeast significant at the 5% level. Fourth, omitting the shock lags, adding lagged interest rates, or entering the three shocks simultaneously leaves the second-year difference significant at the 5% level in each specification. Across these four sets of checks, the full-horizon test for the Northeast is significant at the 5% or 10% level and becomes insignificant only when orthogonalization and zeroing are applied together. The comparison between asset purchases and conventional interest rate policy in the lowest income group (Panel C) remains significant at the 10% level or below under all four sets of checks, but is no longer significant after excluding recession quarters or extreme shock quarters. The gradient within the mortgagor group (Panel B) is not robust: it becomes insignificant once the LSAP factor is set to zero before 2009, while the gradient in the food expenditure ratio within the same group appears only in the pre-2009 observations.

7 Conclusion

7.1 Main Findings

This paper uses the three-factor monetary policy shock series of Swanson (2021) and CEX consumption data to estimate the distributional effects of FFR, FG, and LSAP at the national, regional, and housing contract levels. At the national level, the difference between the consumption responses of high- and low-income households is close to zero under all three instruments. This average masks two differences. The first is regional: in the Northeast, expansionary asset purchases raise consumption more for the lowest income quintile than for the highest; this difference appears in the second year, persists through the third, and is statistically distinguishable from the other three regions. The second is across housing contract types: among lower-income households, conventional interest rate policy acts on renters and asset purchases on mortgagors, in opposite directions.

Among the three instruments, asset purchases have the clearest direction and the most complete chain of evidence. Forward guidance shows no consistent distributional direction, while the effects of conventional interest rate policy appear in housing consumption and cannot be interpreted directly as an improvement in household welfare.

The food expenditure ratio helps distinguish nominal consumption changes from changes in expenditure composition. Under the FFR shock, the $Q1$–$Q5$ difference in the food expenditure ratio is insignificant in every window across all four regions; the smaller declines in nominal housing consumption and total consumption for $Q1$ partly reflect rent increases driven by higher interest rates, consistent with the result in Section 5.2 that renters show a larger consumption decline within the same income group. Under the LSAP shock, the difference in the food expenditure ratio in the Northeast is significant at the 10% level in the second-year window ($p = 0.050$), indicating that the progressive effect in non-housing consumption in this region reflects a genuine change in expenditure composition rather than a purely nominal or price effect.

The estimation split by housing contract shows that the three instruments reach households through different types of contracts. Within the same income group, renters show a larger consumption decline under conventional interest rate policy, mortgagors decline more under asset purchases, and there is no difference between the two under forward guidance; in the third-year window, the consumption decline for low-income mortgagors is approximately 4 percentage points larger than that for renters at the same income level. This reversal between asset purchases and conventional interest rate policy is confirmed by a direct test in the lowest income group. The income gradient under asset purchases appears only within the mortgagor group, which is inconsistent with an explanation in which the employment channel operates alone. The gradient is concentrated among low-income borrowers, who face the greatest difficulty in meeting the refinancing thresholds, and appears only after the second year, consistent with the time required for refinancing, whereas asset price effects should appear immediately. The prominence of the Northeast is consistent with housing carrying the greatest weight in local household budgets rather than with a higher share of mortgagor households. This conclusion does not conflict with the HANK framework of Kaplan et al. (2018), in which cash flow effects on borrowers are also present.

This progressive effect is robust to alternative income classifications, shock measures, and regression specifications. It is identified from the quarters in which asset purchases were conducted on a large scale. Excluding quarters with extreme shocks increases the point estimate, whereas excluding recession quarters reduces it by approximately 30%. Under the latter restriction, the second-year difference between the Northeast and the other three regions remains significant at the 10% level, but the difference for the Northeast itself is no longer statistically distinguishable from zero. The tenure-related results are identified from the same set of quarters. The findings therefore describe asset purchases on the scale seen after 2008; whether they extend to smaller-scale

operations remains an open question for future research, as discussed below.

7.2 Policy Implications

These results provide direct consumption-side evidence for the debate on the distributional consequences of QE. Wealth-based critiques such as Domanski et al. (2016) argue that QE raises asset prices and benefits wealthy households that hold financial assets. This paper does not dispute the existence of the wealth channel, but the regressive effect it predicts does not show up in consumption. At the national level, the consumption gap between high- and low-income households is unchanged; in the Northeast, where a gap does open, expansionary LSAP raises consumption more for low-income households, an effect that appears to operate mainly through borrowers' cash flow and that holds under both income classifications. The distributional consequences of monetary policy therefore depend on which side of the balance sheet is considered: wealth effects on the asset side favor wealthier households, whereas mortgage cash flow on the liability side favors low-income borrowers. At the same time, the difference in distributional outcomes across the three instruments is itself a policy-relevant fact: in an environment where the ZLB renders the FFR ineffective, LSAP and FG are not interchangeable forms of "easing" and differ materially in their capacity to improve outcomes for lower-income households. The evidence for the three instruments is unevenly distributed geographically, a pattern visible only when the estimation is grouped by region: the national aggregate shows no gradient for any instrument. Because the distributional effect of LSAP in the Northeast differs significantly from that in the remaining three regions, a uniform federal monetary policy can produce different distributional consequences across regions.

7.3 Limitations and Future Research

Several limitations apply. The rotating panel structure of the CEX tracks individual households for at most five quarters; I therefore construct time series from grouped cell means, bypassing the short-panel constraint at the cost of being unable to observe consumption dynamics at the household level. The three-factor decomposition of Swanson (2021) mitigates but does not eliminate the Fed information effect: the signal about the economic outlook embedded in FOMC announcements cannot be fully separated from the monetary policy stance itself, and the robustness checks excluding recession and extreme shock quarters reduce but do not resolve this concern.

The evidence from the mechanism analysis is consistent with the mortgage cash flow channel and inconsistent with an explanation in which the employment channel operates alone, but the design of this paper cannot separately quantify the contributions of the two channels, so a substantive role for the employment channel cannot be ruled out. The three limitations discussed in Section 5 also apply: the explanation that regional differences stem from the weight of housing in household

budgets is inferred separately from the two splits, the distinctions along the tenure dimension depend on crisis-period shock variation, and the low-income mortgagor cell is small, so its estimates are imprecise. The difference between the Northeast and the other regions is supported by direct tests, but the sample length is insufficient to determine whether the remaining three regions also differ from one another.

These limitations point to several directions for future research. Incorporating quarterly labor income microdata such as the Survey of Income and Program Participation (SIPP) would allow the employment channel to be tested directly. Similarly, combining loan-level refinancing data would allow the cash flow channel to be tested directly, as well as whether regional differences are indeed driven by the weight of housing in household budgets, without relying on indirect inference from the two separate slices. Whether the distributional effect varies with shock size could be tested with a formal nonlinear specification, such as a threshold or breakpoint regression, rather than the sample-split approach used here. Extending the same framework to other advanced economies that have implemented asset purchase programs, such as the European Central Bank's Public Sector Purchase Programme (PSPP), would allow the generalizability of these findings to be assessed and clarify the meaning of the within-region relative income measure in a cross-country context.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used ChatGPT (OpenAI) and Claude (Anthropic) in order to assist in writing and debugging code, obtain suggestions for code improvements, format chart styles and layouts to improve readability, polish academic language, translate academic texts, obtain suggestions for revising drafts, check cross-references and citations, and format and adjust LaTeX documents. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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A Data Description and Diagnostics

A.1 CEX Interview Survey: Variable Construction

Household pre-tax income is taken from FINCBTXM, the BLS multiply imputed total family income before taxes; for 1992–2003, before this variable is available, the reported (non-imputed) measure FINCBTAX is used instead. For 1992–2003, households with incomplete income reports (RESPSTAT = 2) are dropped. FINCBTAX is not imputed in those years, and the income of such households is either zero or missing (13.9% of records in this period) or positive but understated (a further 5.8%). Retaining them would mechanically assign them to lower income quintiles, with the first group falling entirely into the lowest quintile. From 2004 onward, FINCBTXM imputes incomplete reports, so all households are retained in those years. Households with negative income are dropped in all years. Income quintiles are constructed within each calendar year $\times$ Census region cell, ranking households by their position in that year-region income distribution; this within-year, within-region ranking is the baseline income classification used throughout the paper, in contrast to the national ranking (INC_RANK) used in the robustness check of Appendix F.1.

Consumption variables are constructed from CEX Interview Survey expenditure totals. Total consumption is TOTEXPCQ; non-housing consumption subtracts housing-related expenditure, HOUSCQ (which includes rent and mortgage payments), from TOTEXPCQ, so that interest-rate-driven changes in monthly housing payments do not mechanically enter the measure of consumption demand. All consumption variables are converted to a per-capita basis by dividing by household size (FAM_SIZE) and are expressed in nominal dollars (see the note to Table A.2 below). Household-level survey weights (FINLWT21) are applied throughout when computing cell means, so that the resulting group averages are nationally representative rather than artifacts of the CEX's rotating sample design. Total expenditure and income are winsorized at the 99.9th percentile within each calendar year to limit the influence of extreme outliers on cell-level averages.

Housing tenure status is coded from CUTENURE into three categories: mortgaged owners (households with an outstanding mortgage), outright owners (households who own their home with no mortgage), and renters. This three-way classification, crossed with the same five income quintiles, forms the tenure-by-income panel used in Section 5.

A.2 Sample Coverage and Adequacy

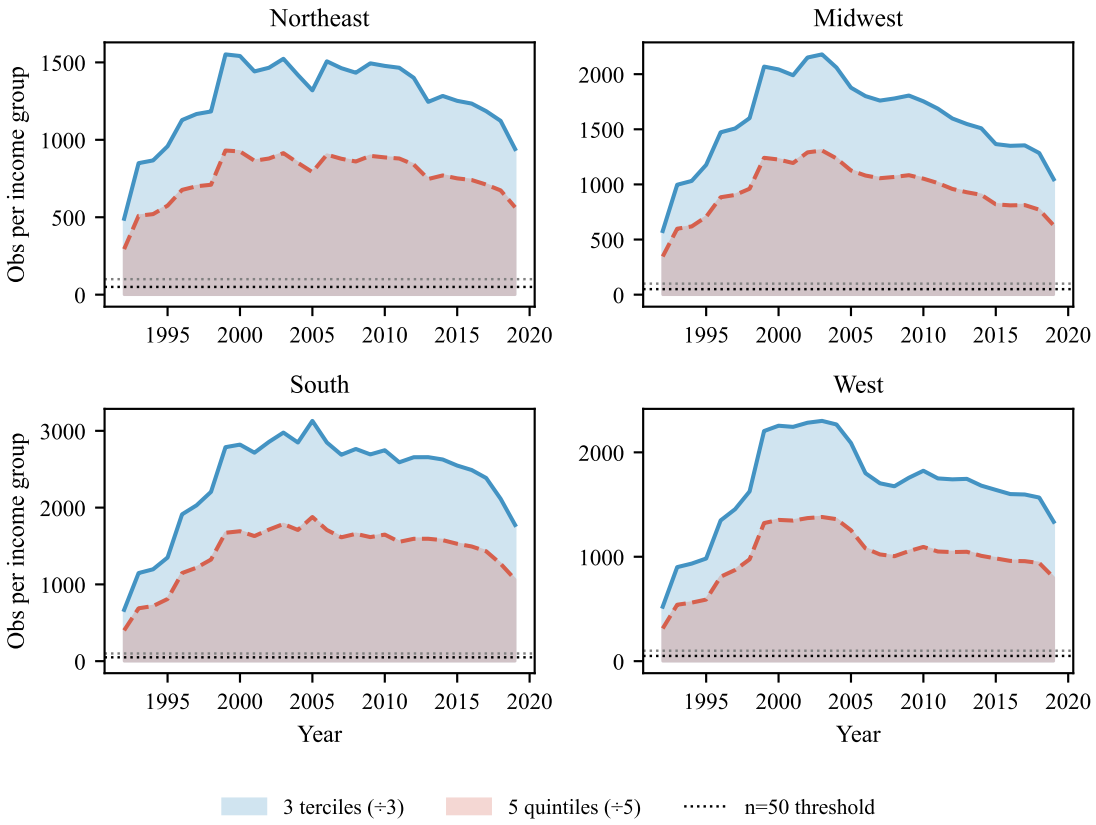


Figure A.1: Annual Observations per Income Group within Region

Notes: For each Census region, the panel plots the annual number of CEX Interview Survey household observations per income group (summed across the four quarters of each calendar year), under two alternative partitions: terciles (3 income groups, shaded blue) and quintiles (5 income groups, shaded red, the partition used in the main analysis). Dotted lines mark reference thresholds of $n = 50$ and $n = 100$ observations. The counts plotted are raw CEX Interview Survey observations, before the sample restrictions described in Section 2.1; the estimation sample is smaller, and more so before 2004, where the income-variable correction removes incomplete income reporters. The conclusion is unaffected either way: on the estimation panel itself the smallest region × quintile cell contains 240 observations in a year (median 926), so the cells used to construct the quarterly averages are not constrained by small-sample concerns.

A.3 PCA: Macro Control Variables

Table A.1 lists the ten FRED series used to construct the macroeconomic control factors described in Section 2.3. Four interest rate series — the FFR, the 3-month Treasury bill rate, the 10-year Treasury yield, and the 30-year mortgage rate — are deliberately excluded from the principal component analysis, since including them would let the resulting factors partially absorb variation in the monetary policy shocks themselves, attenuating the estimated impulse responses.

Standardizing the remaining ten variables and extracting principal components yields three factors with eigenvalues exceeding one under the Kaiser criterion (3.77, 1.82, and 1.29), jointly explaining 68.2% of the total variance (37.4%, 18.0%, and 12.8% individually). Figure A.2 plots

Table A.1: Macro Control Variables: Data Sources and Transformations

FRED Mnemonic	Variable	Source Agency	Frequency	Transformation
CPIAUCSL	Consumer Price Index for All Urban Consumers	U.S. Bureau of Labor Statistics	Monthly	100× log-difference
GDPC1	Real Gross Domestic Product	U.S. Bureau of Economic Analysis	Quarterly	100× log-difference
UNRATE	Unemployment Rate	U.S. Bureau of Labor Statistics	Monthly	Level
INDPRO	Industrial Production: Total Index	Board of Governors of the Federal Reserve System	Monthly	100× log-difference
PCEPI	Personal Consumption Expenditures: Chain-type Price Index	U.S. Bureau of Economic Analysis	Monthly	100× log-difference
PAYEMS	All Employees, Total Nonfarm	U.S. Bureau of Labor Statistics	Monthly	100× log-difference
HOUST	New Privately-Owned Housing Units Started	U.S. Census Bureau / U.S. Dept. of Housing and Urban Development	Monthly	100× log-difference
CSUSHPISA	S&P/Case-Shiller U.S. National Home Price Index	S&P Dow Jones Indices LLC	Monthly	100× log-difference
TWEXMMTH	Trade Weighted U.S. Dollar Index: Broad	Board of Governors of the Federal Reserve System	Monthly	100× log-difference
NASDAQCOM	NASDAQ Composite Index	Nasdaq OMX Group, Inc.	Daily	100× log-difference

Notes: All series are retrieved from the FRED database maintained by the Federal Reserve Bank of St. Louis; the table reports each series' FRED mnemonic and the agency that originally compiles the underlying data. All series are resampled to quarterly frequency (mean within quarter for monthly/weekly/daily series; last value for already-quarterly series), log-differenced (×100, to approximate percentage changes) where appropriate, standardized, and restricted to 1992Q1–2019Q2 before entering the principal component analysis described in the text.

the full eigenvalue spectrum. The first factor loads most heavily on real activity measures (real GDP growth, industrial production, nonfarm payrolls) and is interpreted as an economic activity factor; the second loads on price measures (CPI and PCE inflation) and is interpreted as a price and inflation factor; the third loads on labor market slack (the unemployment rate) and is interpreted as a labor market pressure factor. These three factors enter the local projection specification as $\mathbf{PC}_t$ in Equation (1).

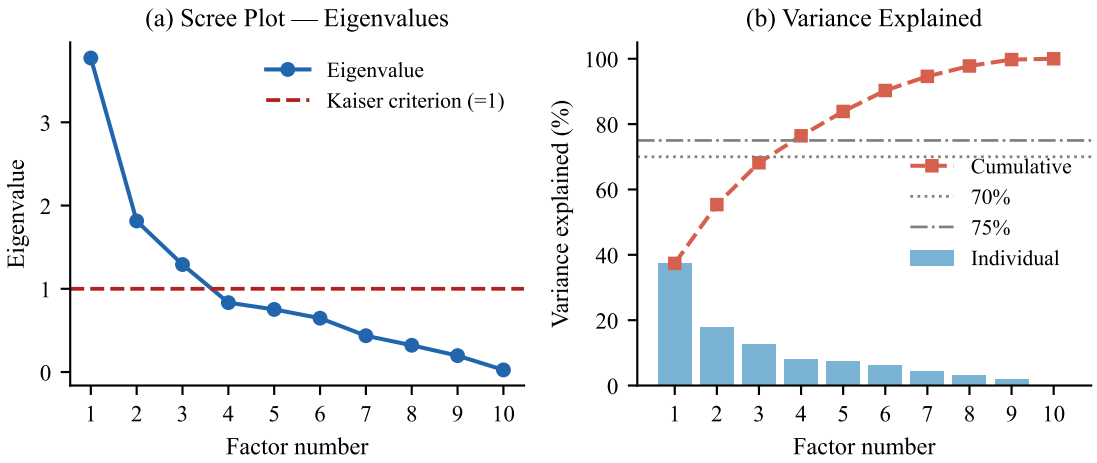


Figure A.2: PCA Scree Analysis — Macro Control Variables

Notes: Eigenvalues from the principal component analysis of the ten macroeconomic control variables in Table A.1. The dashed line marks the Kaiser threshold (eigenvalue = 1); the first three components lie above this threshold and are retained, jointly explaining 68.2% of total variance.

A.4 Summary Statistics

Table A.2 reports summary statistics for the region-by-income panel underlying the regional estimation in Section 4: mean observations per quarter, mean consumption, and food share for each of the twenty region × income-quintile cells. Consumption figures are in nominal per-capita dollars (see Section A.1); since the distributional comparisons throughout the paper are differences between quintiles within the same quarter, deflation affects them very little, as the deflation check in Appendix F.4 confirms. Table A.3 reports descriptive statistics for the three Swanson (2021) shock series — mean, standard deviation, and extreme observations — over the full sample period, complementing the discussion of shock timing in Section 2. The shock series are summed within each quarter, aggregating them from meeting-date frequency to quarterly frequency. This ensures consistency with the consumption dataset and avoids the mechanical zeros that would otherwise arise in roughly one-third of monthly observations due to the irregular spacing of FOMC meetings.

Table A.2: Summary Statistics: CEX Interview Survey Panel (1992–2019)

Region	Income	Quarters with data	Mean obs per quarter	Non-housing cons. ($/quarter, pc)	Total cons. ($/quarter, pc)	Food share (%)
Northeast						
	Low (Q1)	112	173	$965	$1,663	19.4
	Q2	112	173	$1,255	$2,083	17.4
	Q3 (Mid)	112	172	$1,606	$2,524	15.9
	Q4	112	172	$1,884	$2,848	14.1
	High (Q5)	112	172	$2,742	$4,109	12.1
Midwest						
	Low (Q1)	112	216	$1,051	$1,657	18.0
	Q2	112	216	$1,334	$2,038	16.2
	Q3 (Mid)	112	215	$1,604	$2,365	14.8
	Q4	112	216	$1,827	$2,625	13.7
	High (Q5)	112	215	$2,502	$3,537	11.9
South						
	Low (Q1)	112	329	$936	$1,512	20.1
	Q2	112	329	$1,195	$1,821	18.1
	Q3 (Mid)	112	329	$1,460	$2,166	16.4
	Q4	112	328	$1,749	$2,529	14.9
	High (Q5)	112	328	$2,508	$3,577	12.4
West						
	Low (Q1)	112	234	$1,072	$1,790	18.4
	Q2	112	234	$1,307	$2,070	16.9
	Q3 (Mid)	112	234	$1,602	$2,470	15.3
	Q4	112	233	$1,966	$2,977	13.9
	High (Q5)	112	234	$2,860	$4,261	11.9

Notes: Averages over 1992Q1–2019Q4 of quarterly cell statistics; quarterly cell means use CEX survey weights (`FINLWT21`). Consumption columns report the exponential of the average quarterly log; observations and food share are simple averages. Income quintiles (Q1–Q5) are computed within each calendar year × Census region using before-tax income: `FINCBTXM` (BLS imputed) from 2004 and the reported `FINCBTAX` for 1992–2003, excluding incomplete income reporters in the latter period. Consumption variables are expressed in nominal dollars per capita. "Non-housing consumption" excludes `HOUSCQ`. Sample: CEX Interview Survey, 4 Census regions, 5 income quintiles, 112 quarters.

Table A.3: Descriptive Statistics: Swanson (2021) Monetary Policy Shocks

Shock factor	N	Mean	SD	Min	Median	Max
FFR (Fed Funds Rate)	110	0.053	1.033	-5.576	0.255	2.117
FG (Forward Guidance)	110	0.003	1.291	-3.349	0.046	3.751
LSAP (Asset Purchases)	110	0.015	0.913	-5.298	-0.018	2.312

Notes: Quarterly aggregates (sum within quarter) of FOMC-meeting-level Swanson (2021) shocks, 1992Q1–2019Q2.

B Estimation and Inference Details

This appendix reports the details of the estimation and inference in Section 3, including the selection of lag orders, the number of basis functions for the smoothed local projections and the construction of the confidence bands, and the size simulation and settings for the bootstrap test.

B.1 Lag Selection

The parameter settings are determined by information criteria and the relevant literature (Coibion et al., 2017; Cloyne et al., 2020). The maximum horizon is $H = 12$ (three years), with a Newey–West bandwidth of $h + 2$. Among the lag orders, p_y and p_x are determined by a grid search over 1–8 using four information criteria: AIC, BIC, HQIC, and AICc. All four criteria agree on $p_x = 1$; the four criteria for p_y range from 3 to 6, and I use the BIC result of $p_y = 3$, the most parsimonious choice. p_s is not determined by information criteria, since these measure the trade-off between goodness of fit and parameter parsimony, while the role of p_s is to control for serial correlation in the shock series itself to ensure identification; some adjustment based on the properties of the variable itself and its economic meaning is therefore needed. In the grid, the BIC value decreases monotonically in p_s with no interior optimum, which is consistent with this distinction. I therefore rely on two diagnostics instead. The Ljung-Box white noise test (Ljung and Box, 1978) on the three shock series shows that all three are close to white noise, with only the LSAP series exhibiting a significant autocorrelation at lag 1. A sensitivity analysis on the IRF point estimates shows that the standardized shift in the point estimate when p_s increases from 1 to 2 is only 0.03 standard errors, a shift that increases gradually thereafter and becomes substantial beyond lag 5. I therefore set $p_s = 2$, since this controls for the only significant lag-1 autocorrelation, and controlling for one additional lag has a negligible effect on the estimation results. The final parameters are $(p_s, p_y, p_x) = (2, 3, 1)$. This set of lag orders, together with the 12-horizon lead, determines the anchors available for estimation. Each cell's series spans 1992Q1–2019Q4, a total of 112 quarters. This is two quarters longer than the 110-quarter period stated in Section 2, because the shocks and control variables end in 2019Q2, whereas the consumption data extend through 2019Q4. The 12-horizon lead from the final anchor, 2016Q3, therefore extends to 2019Q3 in the consumption data. The earliest 3 quarters are reserved for the lag terms and the final 13 quarters for the lead, leaving anchors from 1992Q4 to 2016Q3, or 96 in total. The same set of anchors is used at all horizons, and each regression estimates 13 parameters.

B.2 Smooth LP: Basis Selection and Confidence Bands

The candidate set for K is $\{3, 4, 5\}$, with the upper bound set at $5 = \lfloor H/2 \rfloor - 1$ to prevent overfitting when K is too large. K is selected independently for each cell via GCV, and in practice, $K = 5$, the upper bound, is selected for all region-by-income-group-by-consumption-variable combinations. To check whether this upper bound is too restrictive, I expand the candidate set to $\{2, \ldots, 10\}$ and rerun GCV; in the cells examined, the new upper bound of $K = 10$ is selected in every case, rather than the selection converging to some interior value. GCV repeatedly selecting the boundary value after the candidate set is expanded indicates a tendency to overfit, indicating that the upper bound $K_{\max} = \lfloor H/2 \rfloor - 1$ is itself a necessary regularization constraint. I therefore treat $K = 5$ as a fixed regularization choice, rather than claiming it to be an unconstrained optimum in the GCV sense.

The 68% and 90% confidence bands for $Q1$ and $Q5$ in the figures do not come directly from the standard errors of the smoothed specification; instead, they are constructed as follows. Each bootstrap resample path from the standard local projections is smoothed using the same B-spline projection matrix $P = B(B^\top B)^{-1}B^\top$ applied to the point estimates, and the standard deviation of these smoothed paths at each horizon is taken as the width of the confidence band, centered on the point estimate from the smoothed local projections. The confidence bands therefore have the same degree of smoothness as the point estimate curve; their width comes entirely from the bootstrap, while the point estimate itself still comes from the smoothed local projections.

B.3 Size of the Bootstrap Test

Table B.1 reports the rejection rates of the test used in the main text under the null hypothesis. Across the six comparisons, rejection rates range from 9.0% to 15.6% at the nominal 10% level and from 5.4% to 9.2% at the nominal 5% level. The national aggregate under LSAP deviates most from the nominal level.

In the bootstrap used for inference, the same set of block indices is applied across all 12 horizons, so that each resample corresponds to the same set of anchors at every horizon. The conclusions are insensitive to the choice of block length: 82 of 84 cells yield the same significance judgments across $L \in \{4, 8, 12, 16\}$.

Table B.1: Rejection Rates of the Bootstrap Test under a True Null (%)

	h = 1–12			Yearly windows, nominal 10%		
Nominal level	10%	5%	1%	h = 1–4	5–8	9–12
Northeast, $Q1$–$Q5$ gap, LSAP	12.8	5.6	1.0	9.4	12.2	12.8
Northeast gap minus the other three regions, LSAP	12.4	7.4	1.2	9.4	10.0	14.4
$Q1$, renter minus mortgagor, LSAP	10.6	6.0	1.2	11.4	10.6	10.4
$Q1$, renter minus mortgagor, LSAP minus FFR	11.6	5.6	1.6	10.2	12.4	11.6
National $Q1$–$Q5$ gap, LSAP	15.6	9.2	2.4	12.4	12.6	15.4
National $Q1$–$Q5$ gap, FFR	9.0	5.4	1.2	9.2	9.0	9.8

Notes: each entry is the share of 500 synthetic datasets in which the bootstrap test that the window average is zero rejects at the stated nominal level. A synthetic dataset is built by regressing each cell's outcome on the controls of Equation (1) alone, without the shock or its lags, and adding to the fitted values the residuals of that regression, resampled in moving blocks with one block index shared by all cells. No shock therefore has any effect on any cell, while the dependence across cells and horizons is that of the data. The procedure of the main text is then applied to each synthetic dataset, with 2,000 bootstrap replications and block length $L = 12$. With 500 datasets the simulation standard error of a rejection rate is about 1.3 percentage points at the 10% level, 1.0 at 5%, and 0.4 at 1%.

B.4 Quarterly First Stage

Table B.2 reports the first-stage results required when the quarterly shock series are used as instruments for interest rates. The F-statistic for the FFR factor with respect to the federal funds rate is above 38, whereas the F-statistics for the FG factor with respect to the 2-year Treasury yield and for the LSAP factor with respect to the 10-year Treasury yield and the 30-year mortgage rate are all below 3. Apart from the federal funds rate, the shocks explain less than 3% of the quarterly variation in the corresponding interest rates, so the main text does not rescale the shocks into interest-rate units.

Table B.2: Quarterly First Stage: Changes in Interest Rates on the Policy Factors

				F, shock only			F, with	SD of rate
Shock	Interest rate	N	Partial R^2	OLS	HC1	Newey–West	controls[a]	change (bp)
FFR	Federal funds rate	96	0.394	61.2	38.5	60.0	63.1	44
FG	2-year Treasury yield	96	0.026	2.5	1.4	2.6	0.7	53
LSAP	10-year Treasury yield	96	0.004	0.4	0.3	0.4	1.0	48
LSAP	30-year mortgage rate	96	0.011	1.0	1.8	2.0	1.4	40
LSAP, 2009Q1 on	30-year mortgage rate	31	0.196	7.1	4.4	8.7	7.0	31

[a] Adds two lags of the shock, the first lag of the three principal components, and quarter dummies; heteroskedasticity-robust (HC1) standard errors.

Notes: each row regresses the quarterly change in the stated interest rate, measured as the last observation of the quarter minus the last observation of the previous quarter, on the quarterly shock series. The sample is the 96 anchor quarters of the estimation, 1992Q4–2016Q3, except in the last row, which uses the 31 anchor quarters from 2009Q1 on. F is the squared t-statistic of the shock under homoskedastic (OLS), heteroskedasticity-robust (HC1), and Newey–West (4 lags) standard errors. The last column is the standard deviation of the quarterly change in the rate over the same sample, in basis points.

C Standard LP and Smooth LP Comparison

This appendix presents the standard and smoothed local projection estimates side by side, covering the results reported in Sections 4 and 5. The two specifications differ only in that the smoothed specification constrains the sequence $\{\hat{\beta}_g^h\}_{h=1}^H$ in Equation (1) to a B-spline basis; they are otherwise identical. The main-text figures for the regional estimation use the smoothed specification, so the corresponding standard local projection estimates are given here; the main-text figures for the tenure estimation use the standard specification, so the corresponding smoothed estimates are given here. Since all tests and significance annotations reported in the main text are already based on the standard local projections (Section 3.2), the comparisons below concern only the shape of the plotted paths and do not affect any of the reported significance conclusions.

C.1 LSAP Shock — Non-Housing Consumption (cf. Figure 2)

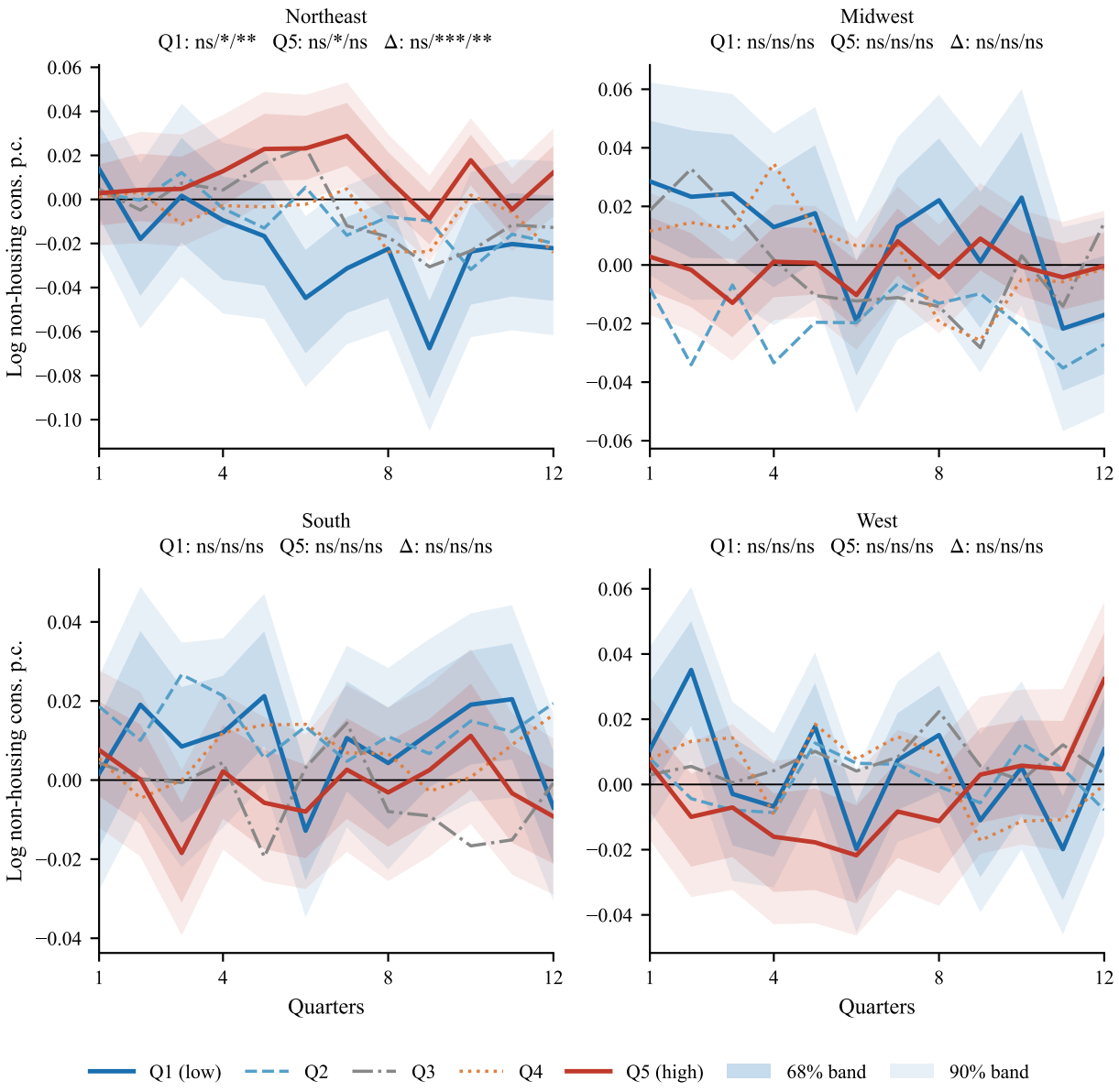


Figure C.1: Standard LP IRF: Log Non-Housing Consumption — LSAP Shock. Compare with Figure 2 (Smooth LP).

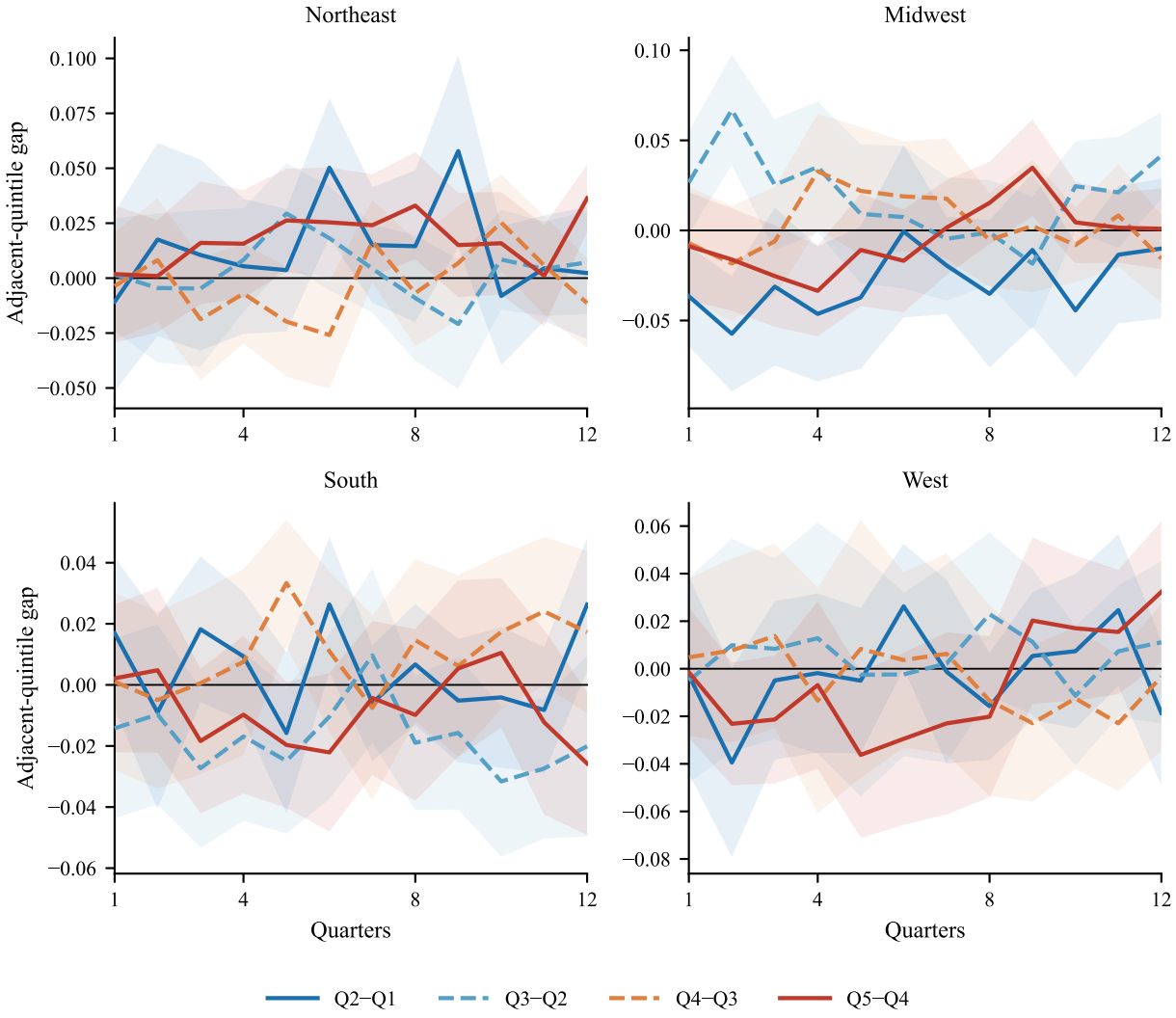


Figure C.2: Standard LP Adjacent Gap: Log Non-Housing Consumption — LSAP Shock. Compare with Figure D.4 (Smooth LP, this appendix).

C.2 FG Shock — Non-Housing Consumption (cf. Figure D.5)

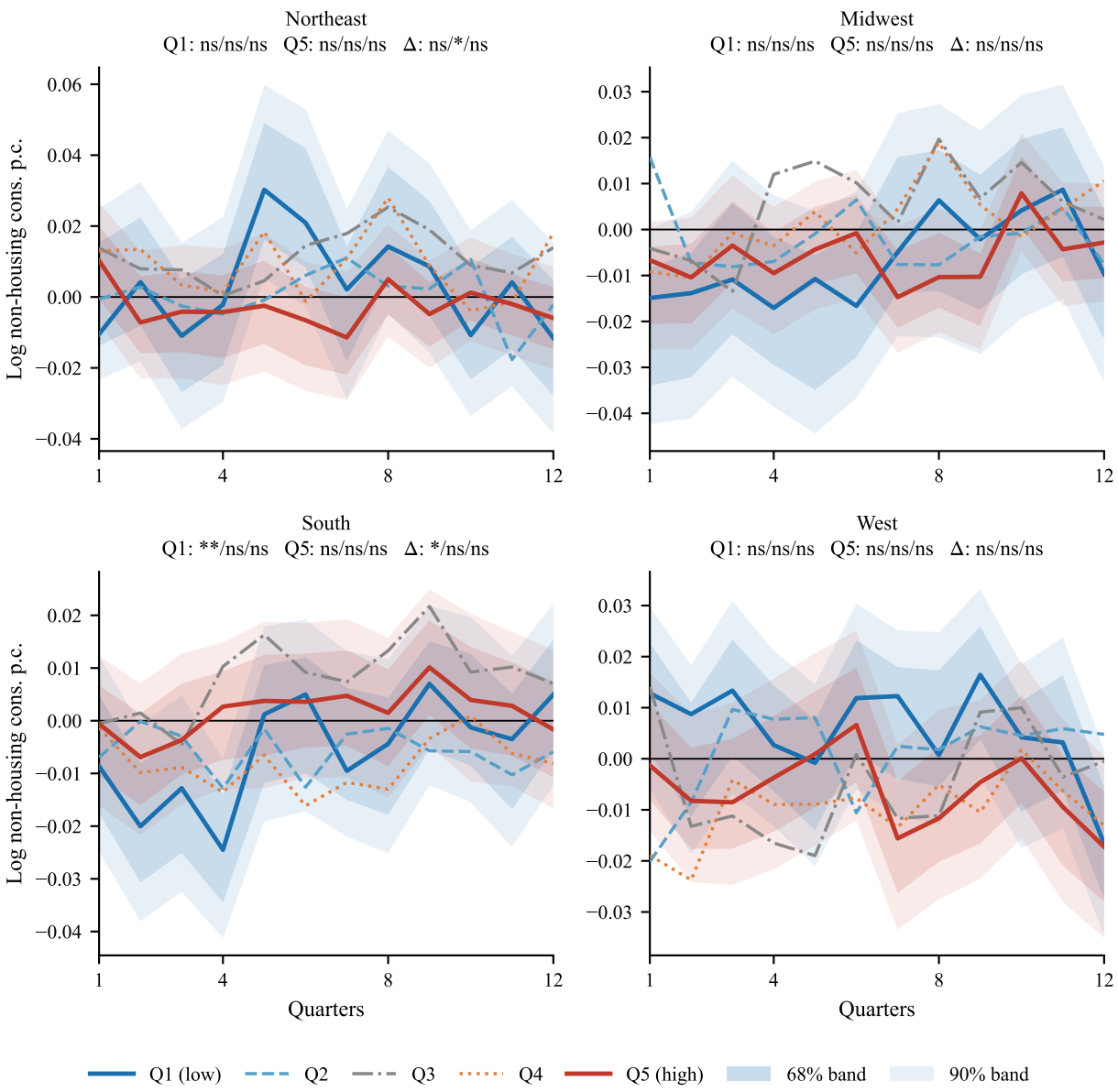


Figure C.3: Standard LP IRF: Log Non-Housing Consumption — FG Shock. Compare with Figure D.5 (Smooth LP).

C.3 FFR Shock — Non-Housing Consumption

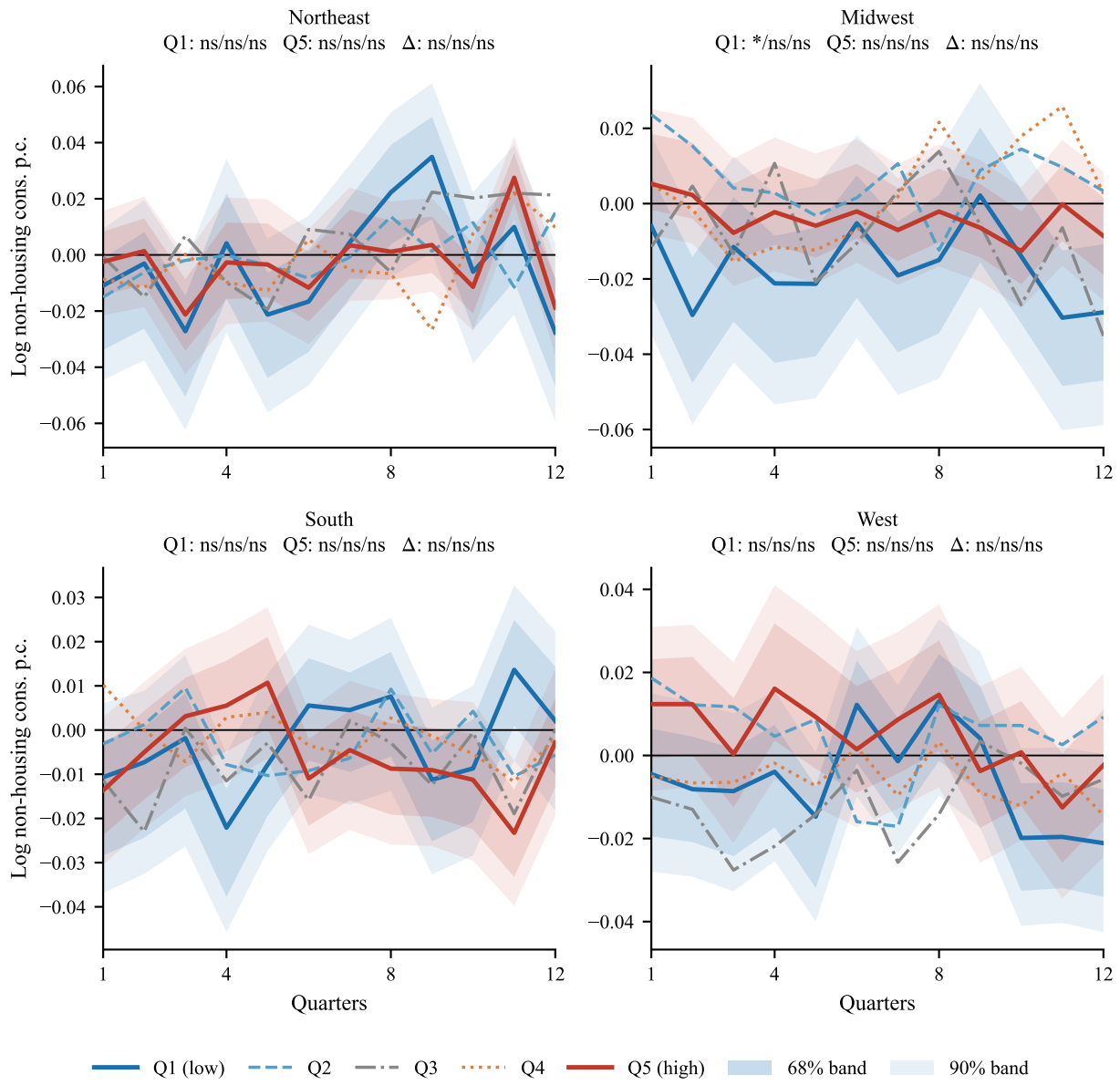


Figure C.4: Standard LP IRF: Log Non-Housing Consumption — FFR Shock

C.4 Tenure Estimation — LSAP Shock (cf. Figure E.1)

The tenure decomposition in both Figure 4 in the main text and Figure E.1 in the appendix uses standard local projections rather than smoothed local projections. Figure C.5 presents the smoothed version of the same estimates for comparison: under the tenure grouping, the sample size in each cell is smaller than under the regional grouping, so the signal is correspondingly weaker, and the B-spline parameterization produces visible magnitude shifts at several horizons, most notably for the outright owner group. The overall shape of the paths is the same under both specifications, and the relative position of $Q1$ and $Q5$ is unchanged within each group, but because the shifts

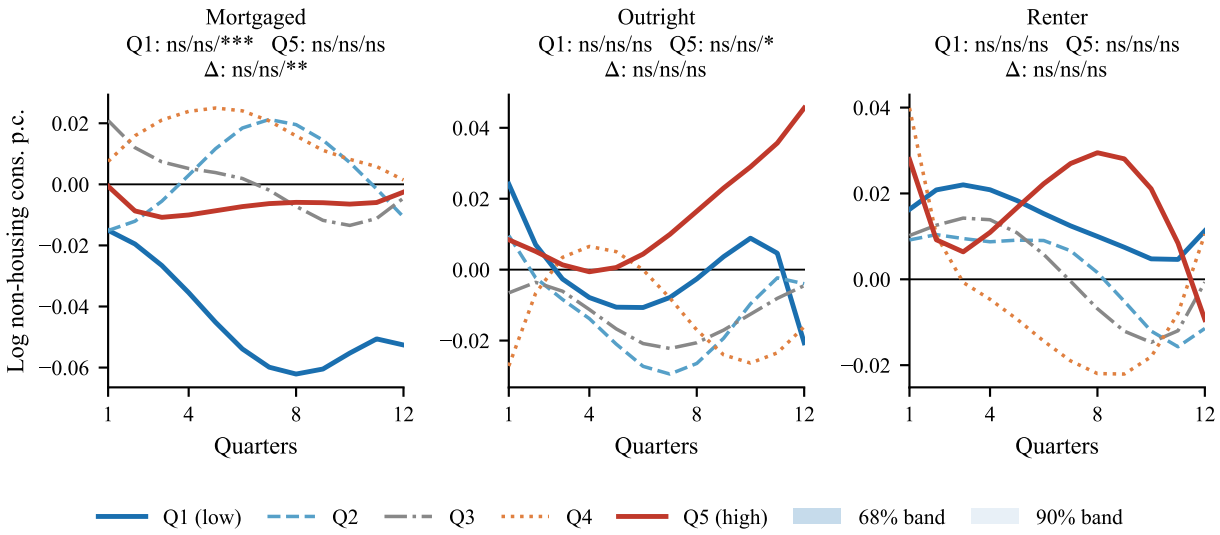


Figure C.5: Smooth LP IRF: Log Non-Housing Consumption per Capita — LSAP Shock, by Housing Tenure × Income Quintile. Compare with Figure E.1 (Standard LP).

Notes: The direction of this comparison is the reverse of the preceding subsections: Figure E.1 in Appendix E reports the Standard LP estimates for the tenure estimation, and the Smooth LP counterpart is shown here. Cell sizes are smaller under the tenure grouping than under the regional grouping, so the signal is weaker and the B-spline parameterization shifts the path visibly at several horizons. See the notes to Figure 2 for the confidence bands, significance annotations, and sample period.

introduced by the smoothed specification are not negligible here, the main text uses the standard local projections for the tenure decomposition figures.

D Additional Outcome Variables and Regional Detail

This appendix reports the smoothed local projections results for the regional estimation that are not shown in the main text. Section 4 presents only the fan chart for non-housing consumption under the LSAP shock (Figure 2). The corresponding figure for the FG shock, the adjacent quintile gap charts for both, and the fan chart for the food expenditure ratio under the LSAP shock are all reported in this appendix, along with supplementary material along two dimensions.

First, total consumption and housing expenditure are reported for shocks where the main text discusses only non-housing consumption. Second, the FFR shock and the food expenditure ratio for FG and FFR are reported here in full, since Section 4.4 discusses FFR only briefly and refers to this appendix for the underlying figures. All figures use the same Smooth LP specification, confidence bands, and significance annotations as the main-text figures; see the notes to Figure 2 for details. Adjacent gap figures are provided alongside the corresponding fan charts throughout this appendix, including the non-housing consumption gap charts for LSAP and FG referenced in Section 4 and

the food expenditure ratio gap charts for LSAP, FG, and FFR. The full set of Q1–Q5 gap estimates and tests across all four outcome variables, shocks, and regions is reported in Table D.1 below.

D.1 Regional Dynamics: LSAP Non-Housing Consumption

The magnitude and shape of the effect differ considerably across regions. The Northeast shows the largest full-horizon effect among the four regions, with the path peaking in the second year and narrowing thereafter, crossing the zero line only once over the full horizon; the gap is significant in the second year and remains significant in the third. The full-horizon means for the remaining three regions are all negative, at approximately one percentage point in absolute value, in the opposite direction to the Northeast, and none is statistically significant. The differences for the Midwest and West move from negative to positive, so their full-horizon means do not summarize their paths; the difference for the South is negative in most horizons, and its full-horizon mean is close to the means of the three windows. Region-specific values are reported in Table D.1.

D.2 Total Consumption (including housing costs)

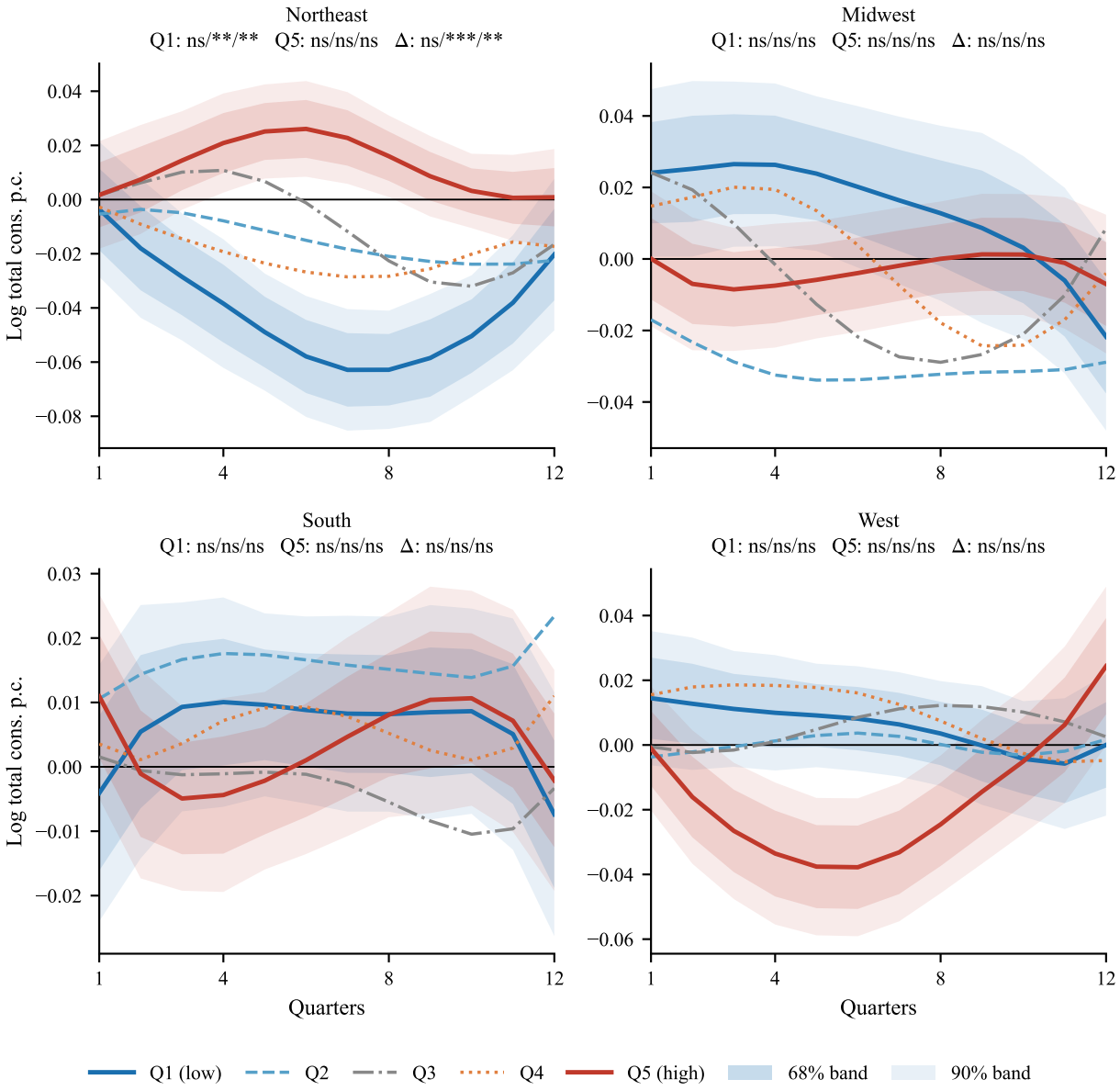


Figure D.1: Smooth LP IRF: Log Total Consumption per Capita — LSAP Shock

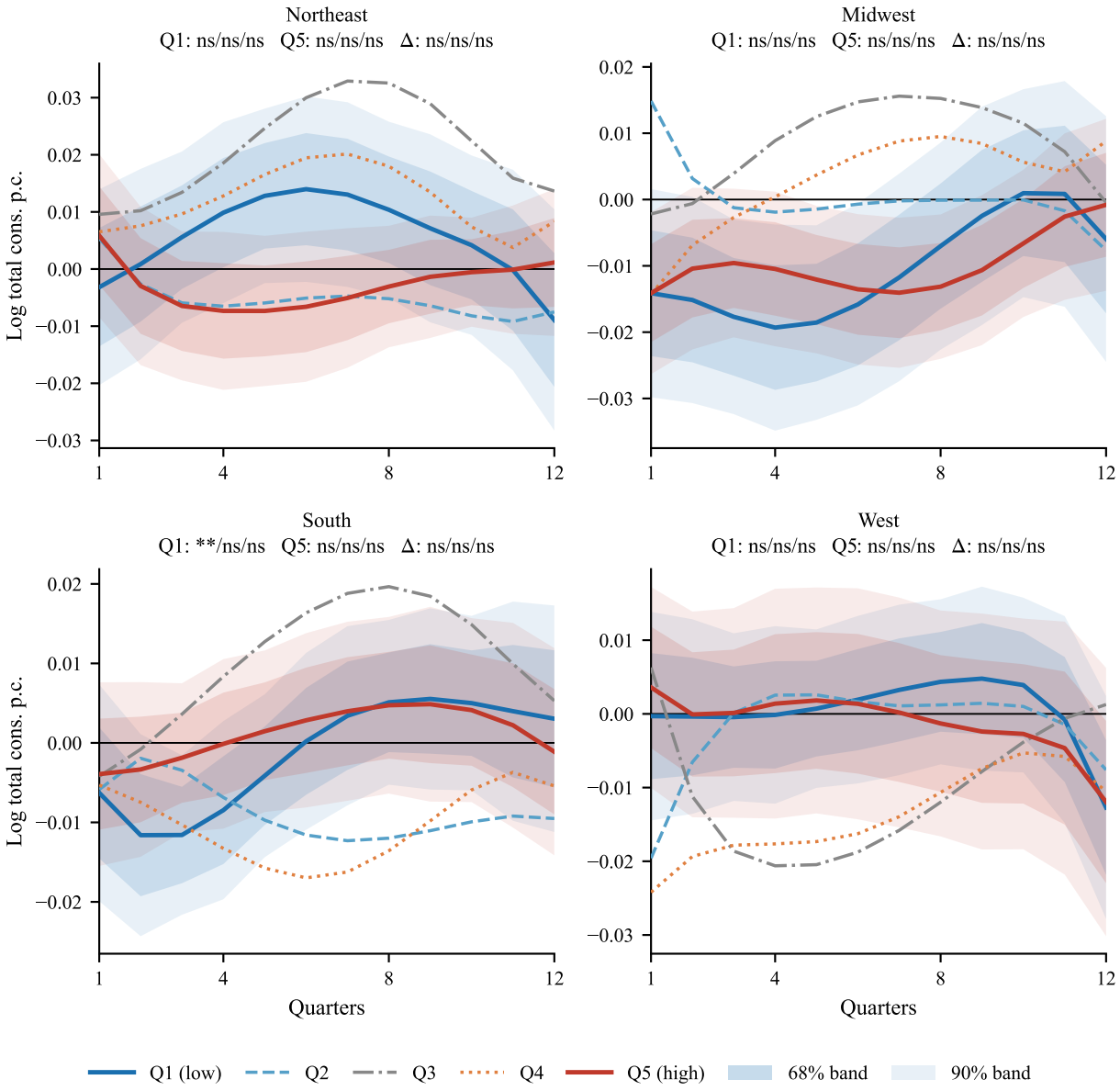


Figure D.2: Smooth LP IRF: Log Total Consumption per Capita — FG Shock

D.3 Housing Expenditure

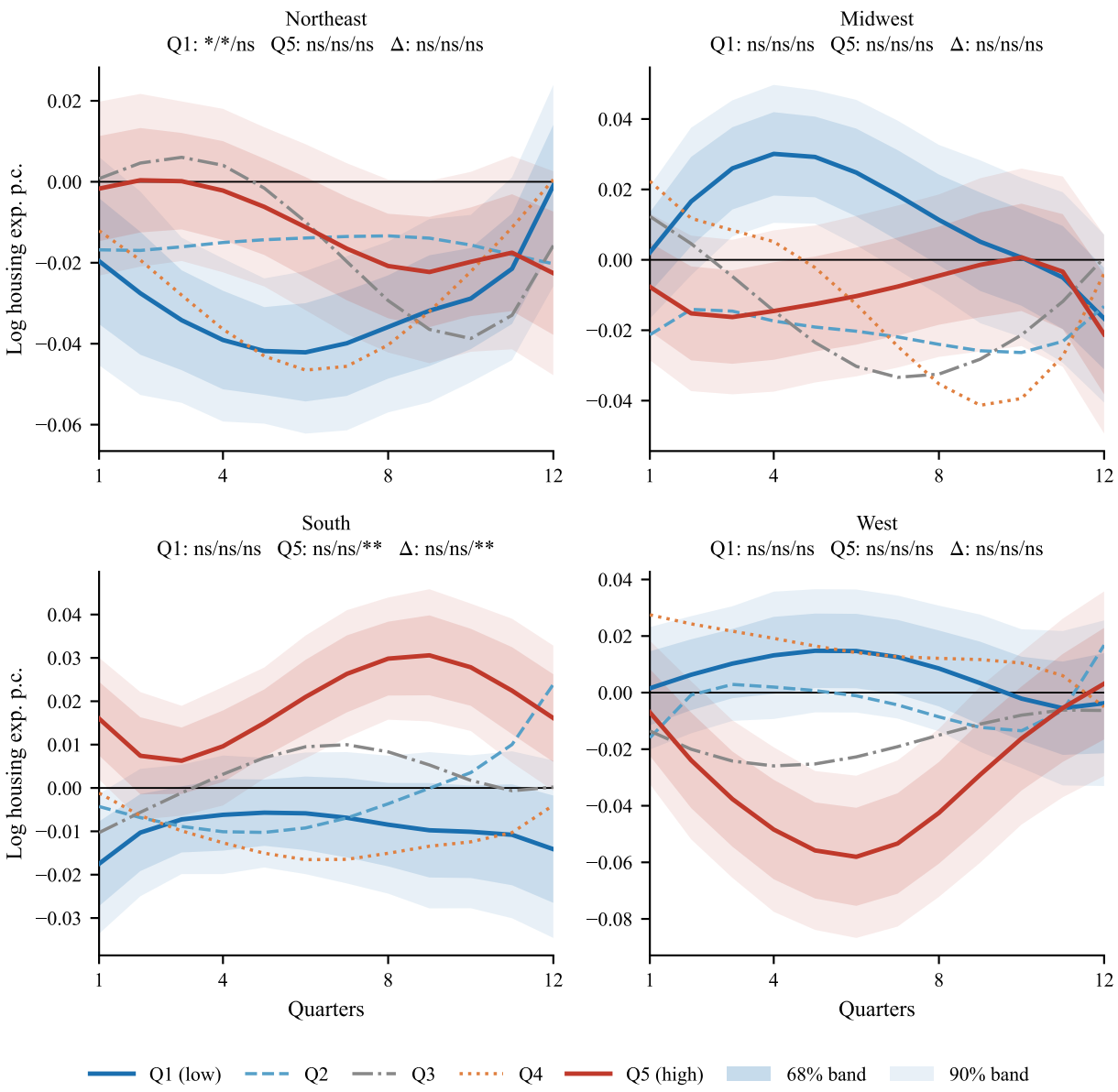


Figure D.3: Smooth LP IRF: Log Housing Expenditure per Capita — LSAP Shock

D.4 Adjacent Gap Figures — All Outcomes and Shocks

The adjacent gap chart plots the difference in impulse responses between each pair of neighboring income quintiles, indicating whether consumption inequality is concentrated among lower- or

higher-income groups. It is defined as follows:

$$G_q^h = \hat{\beta}_{g,q}^h - \hat{\beta}_{g,q-1}^h, \quad q \in \{2, 3, 4, 5\} \tag{D.1}$$

where $\hat{\beta}_{g,q}^h$ denotes the estimated impulse response (equation (1)) of income quintile q within cell-group g at horizon h, and $q \in \{2, 3, 4, 5\}$ indexes adjacent income quintiles. G_q^h thus measures the marginal difference in the consumption response between quintile q and the next-lower quintile $q - 1$ at horizon h.

This measure has direct implications for policy design. Depending on the income structure of a given society, inequality may be concentrated at the bottom of the distribution—reflecting an outsized gap between the poorest households and the rest—or between the middle class and the wealthy. These two patterns call for different policy responses. Lower-income households are more sensitive to prices of necessities such as food and to employment conditions in low-wage labor markets, whereas middle-income households are more exposed to mortgage rates, housing markets, taxation, and financial asset prices. Since different monetary policy instruments operate through different transmission channels, they affect income groups with different degrees of sensitivity, generating heterogeneous consumption responses across the distribution.

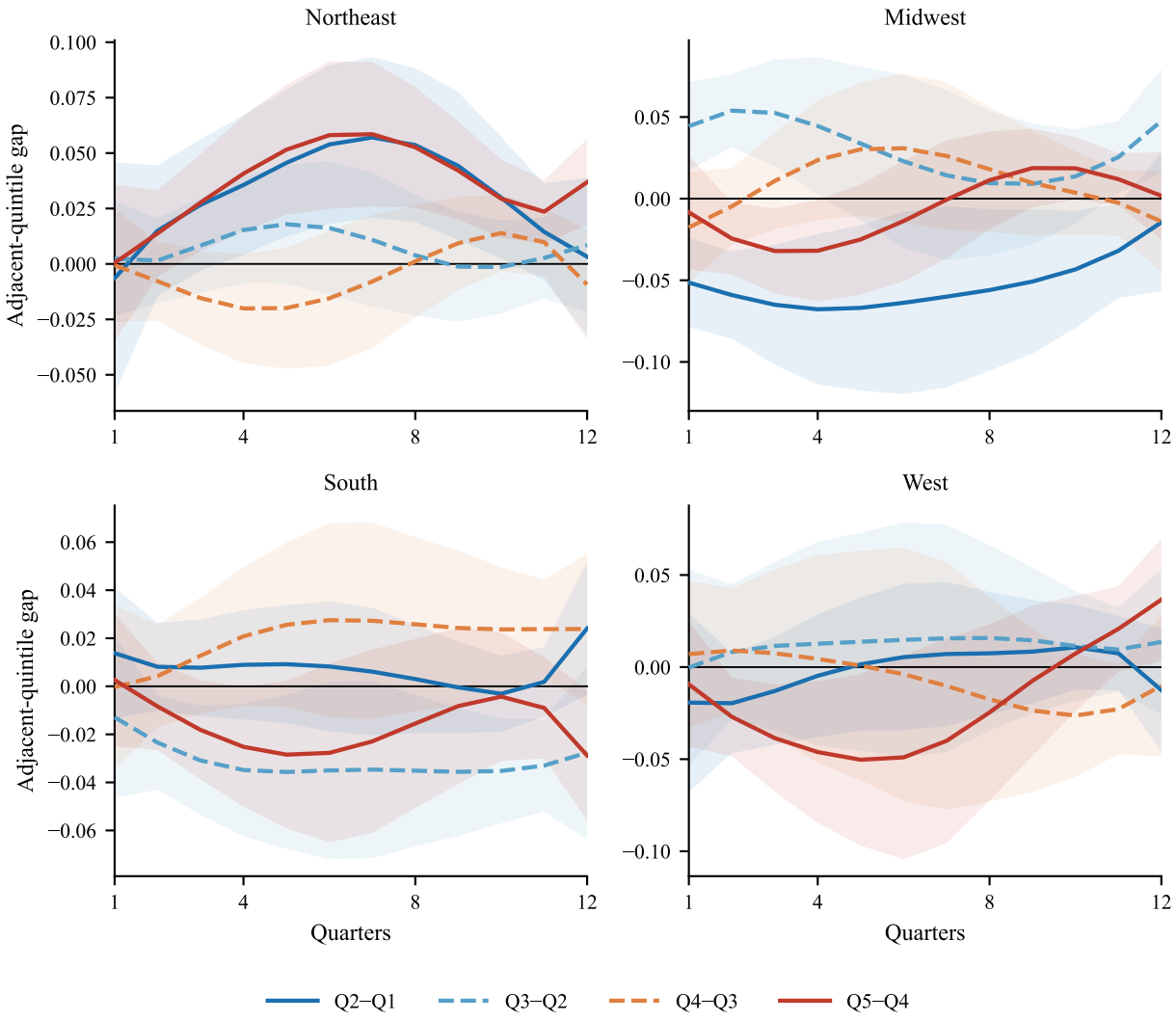


Figure D.4: Adjacent Income Gap: Log Non-Housing Consumption — LSAP Shock

Notes: (1) $G_q^h = \hat{\beta}_q^h - \hat{\beta}_{q-1}^h$, the difference in cumulative impulse responses between adjacent income quintiles. (2) Shaded bands: 68% (inner) and 90% (outer) confidence intervals from the residual moving block bootstrap ($B = 5000$, $L = 12$), for $Q1$ and $Q5$ only. (3) Contractionary LSAP shock direction; see Figure 2 notes for general specification.

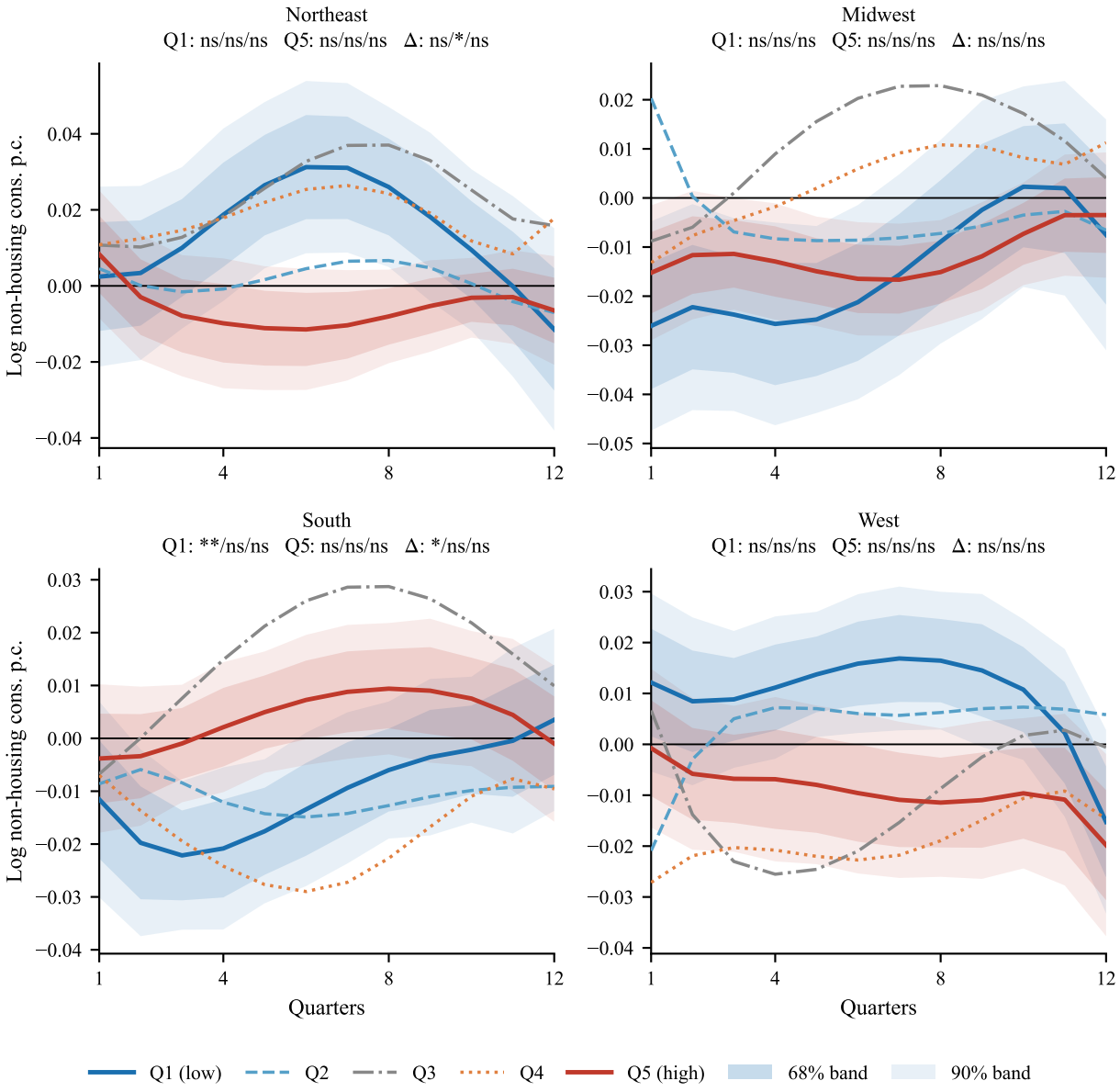


Figure D.5: IRF: Log Non-Housing Consumption per Capita — FG Shock

Notes: Smooth LP impulse responses by Census region and income quintile, as in Figure 2 but with FG as the shock variable. See the notes to Figure 2 for the confidence bands, significance annotations, and sample period.

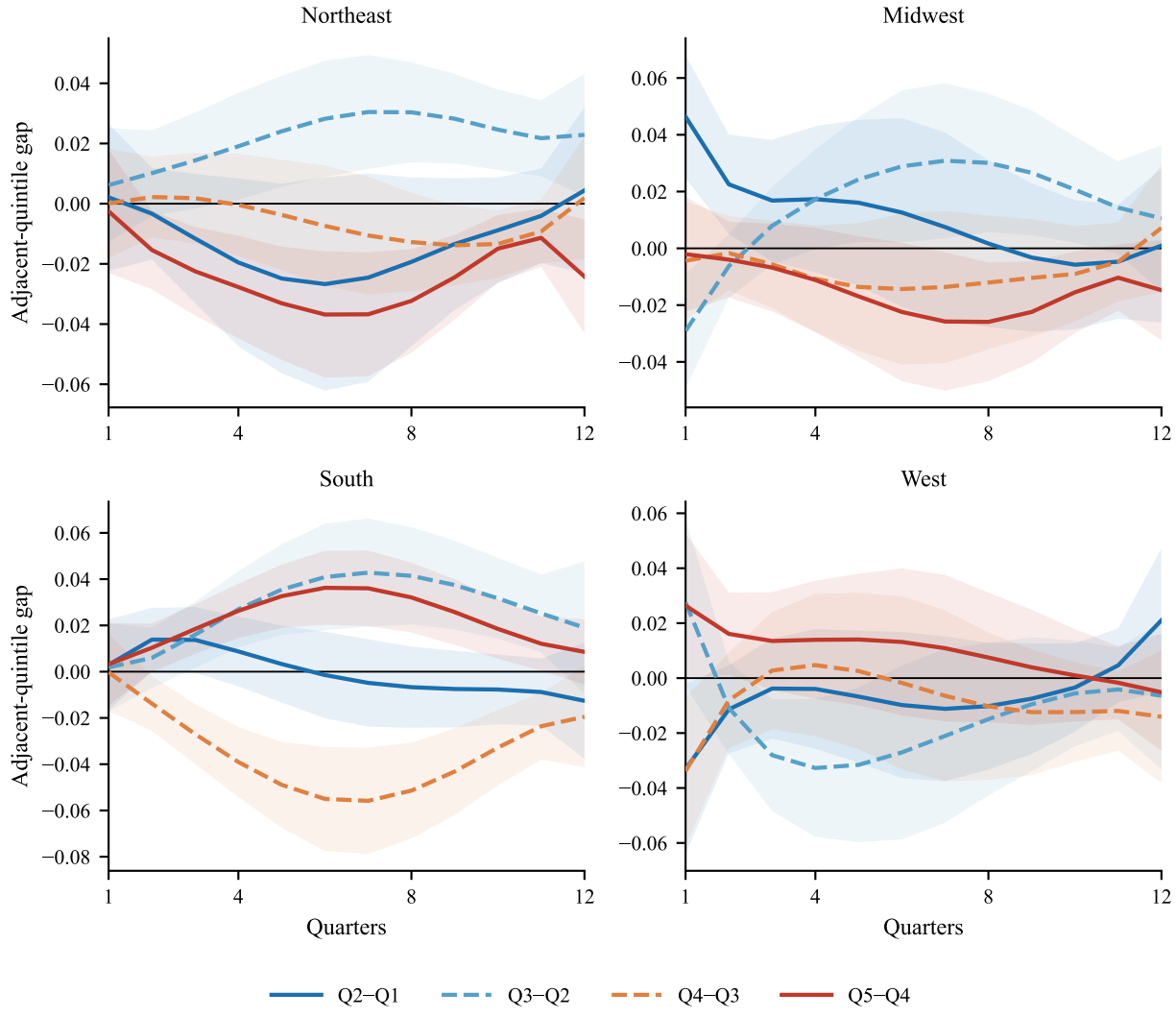


Figure D.6: Adjacent Income Gap: Log Non-Housing Consumption — FG Shock

Notes: Adjacent income gap; see Figure D.4 notes for definition. FG shock; see Figure 2 notes for general specification.

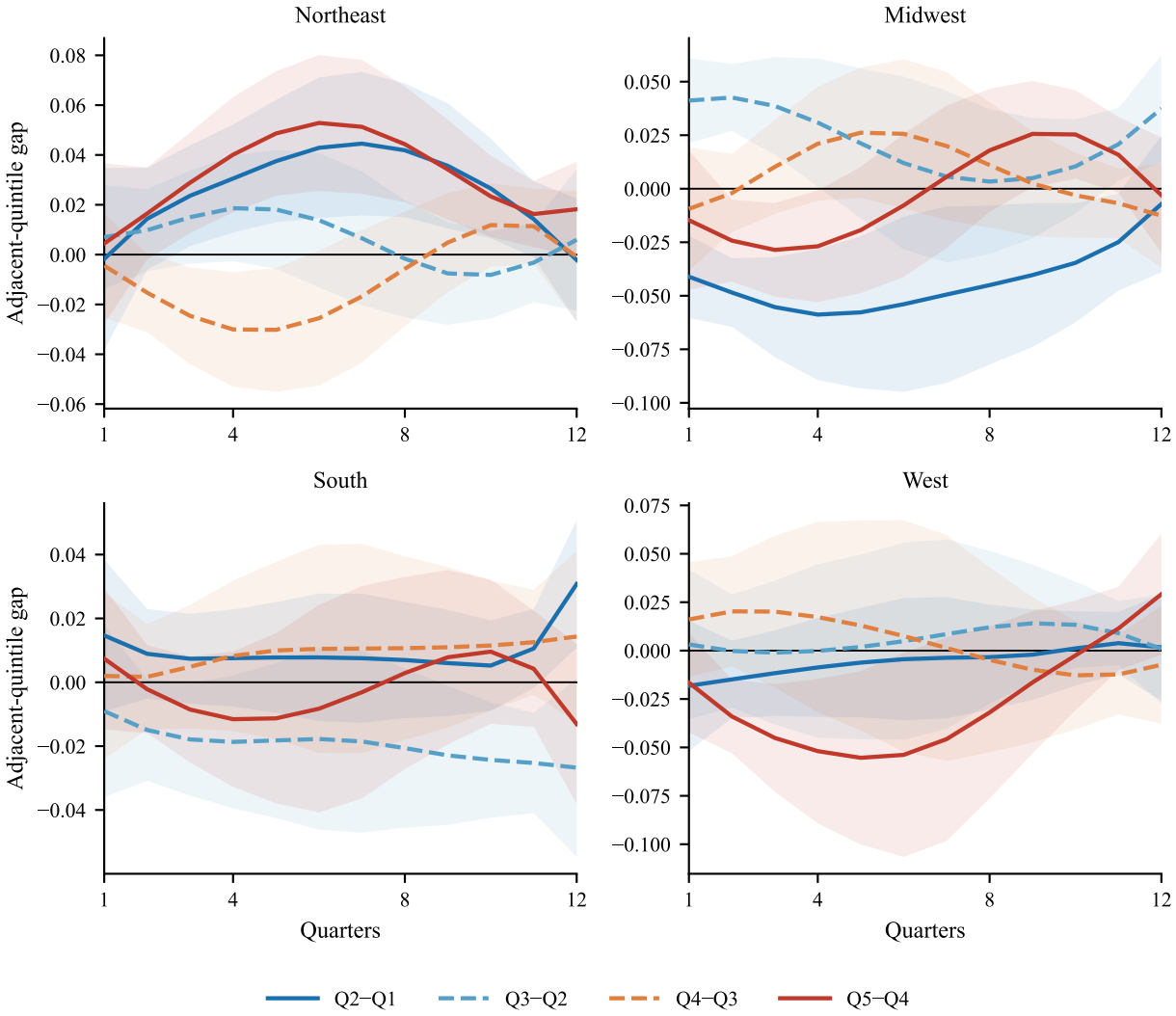


Figure D.7: Adjacent Income Gap: Total Consumption — LSAP Shock

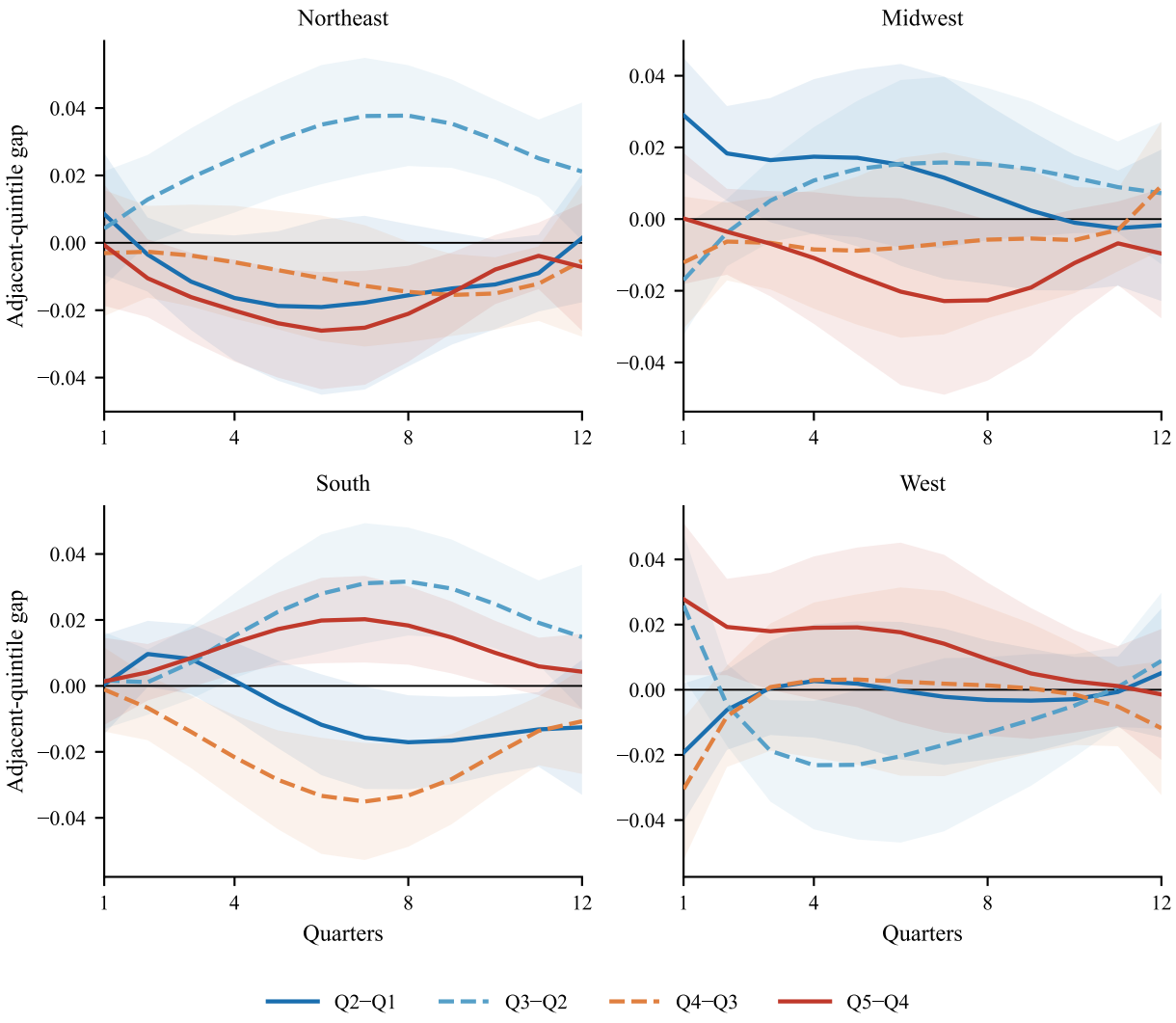


Figure D.8: Adjacent Income Gap: Total Consumption — FG Shock

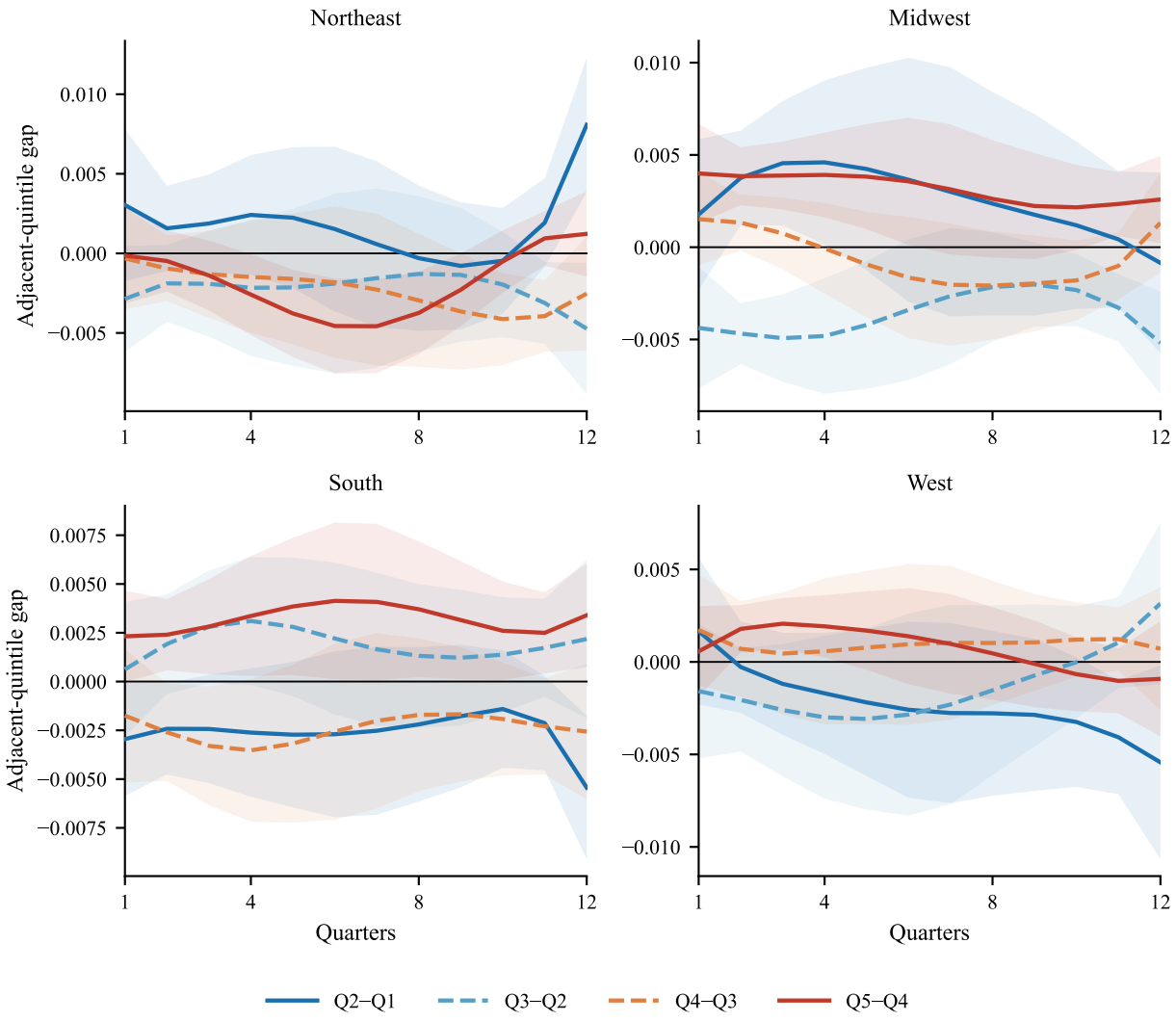


Figure D.9: Adjacent Income Gap: Food Share — LSAP Shock

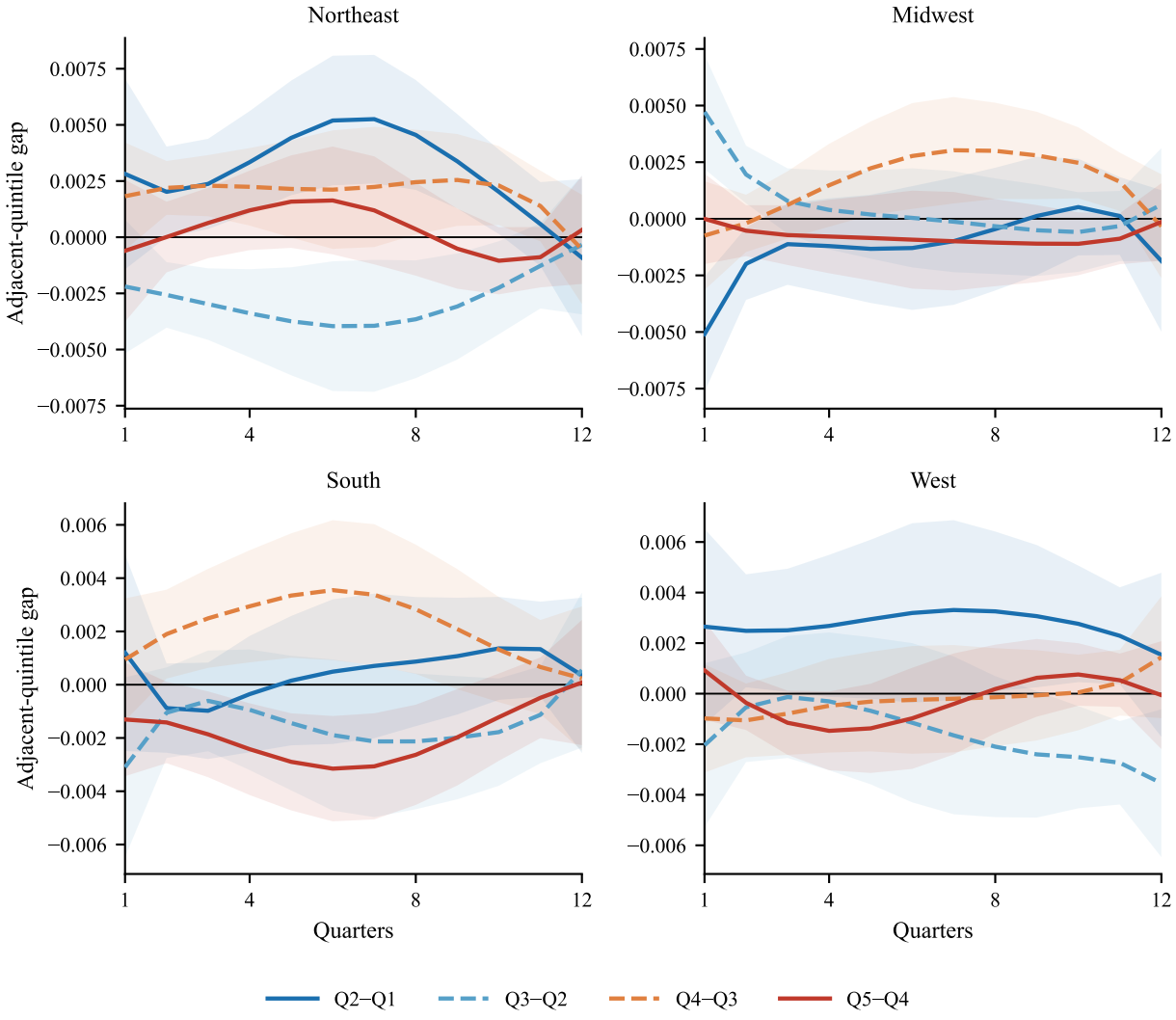


Figure D.10: Adjacent Income Gap: Food Share — FG Shock

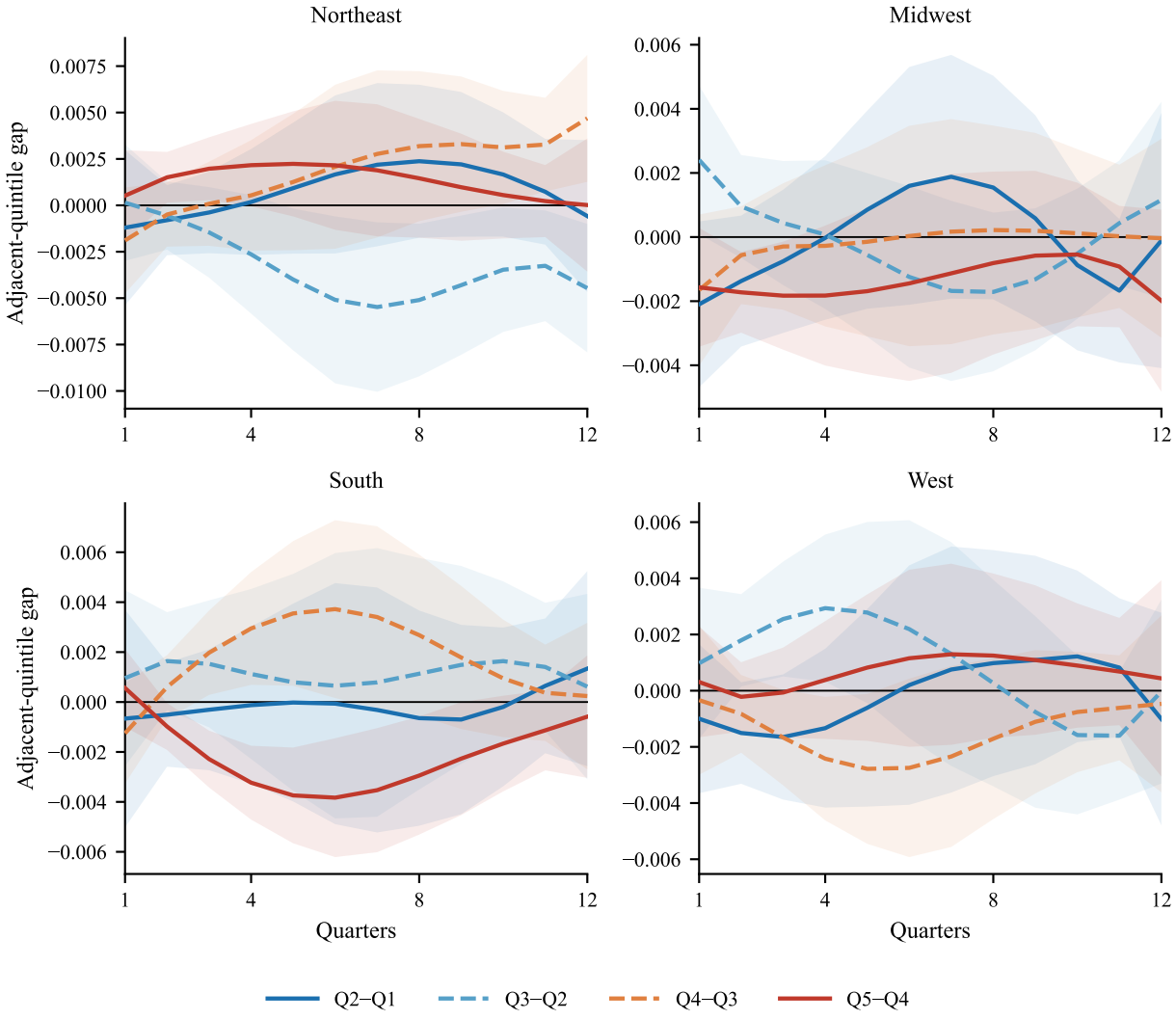


Figure D.11: Adjacent Income Gap: Food Share — FFR Shock

D.5 FFR Shock — Non-Housing Consumption

Figures D.12 and D.13 report the Smooth LP fan and adjacent-gap charts for the FFR shock referenced in Section 4.4; the corresponding Standard LP version is shown in Appendix C.

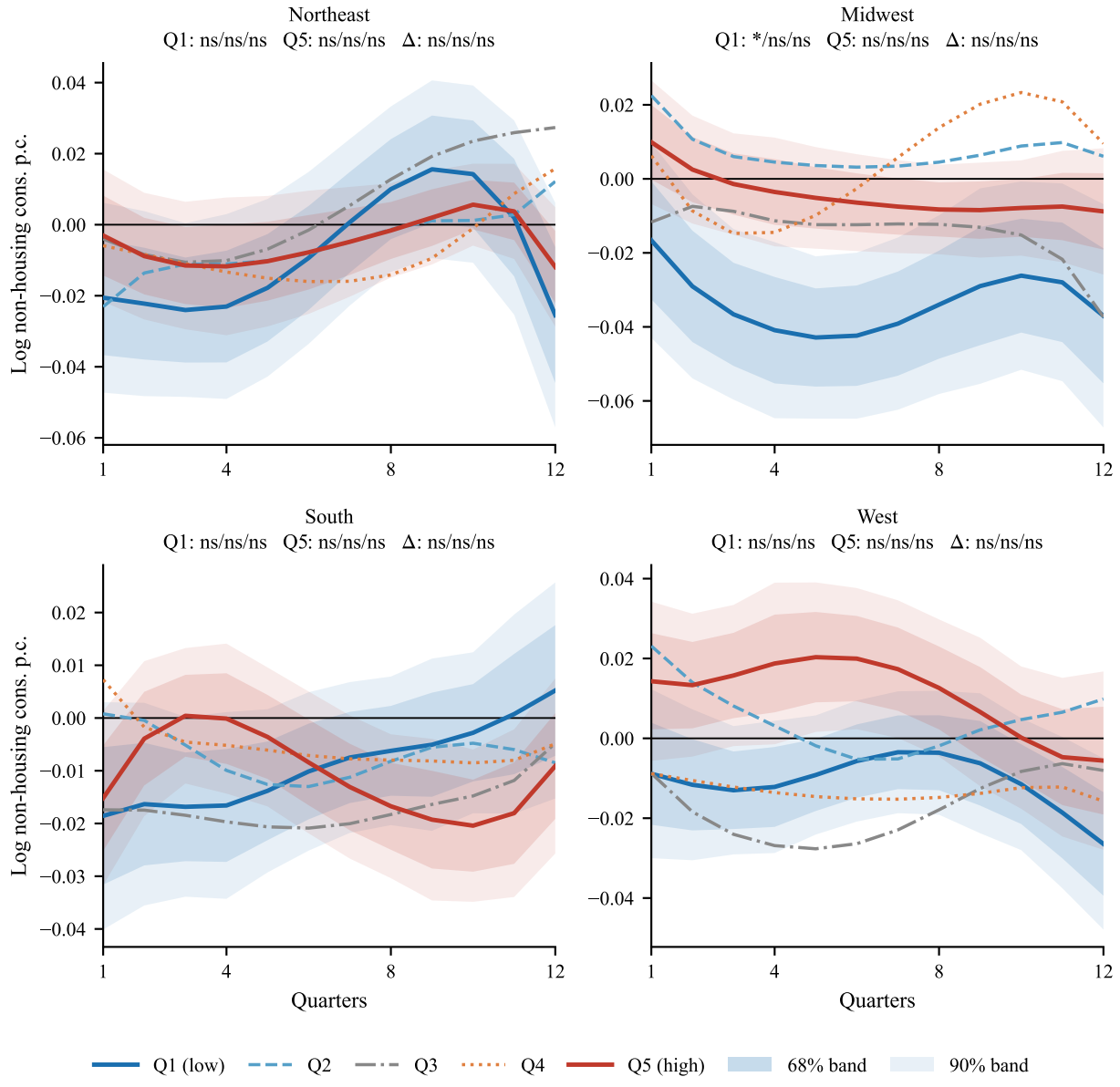


Figure D.12: Smooth LP IRF: Log Non-Housing Consumption — FFR Shock

The distributional effects of FFR shocks show complex regional heterogeneity, with the relative position of $Q1$ and $Q5$ differing from the LSAP results. The fan charts in Figure D.12 (smoothed local projections, used for display only and not for inference) show that non-housing consumption for $Q1$ declines in all four regions following a contractionary shock. Only in the South is the decline for $Q1$ slightly smaller than that for $Q5$. In the Midwest and West, the decline for $Q1$ is

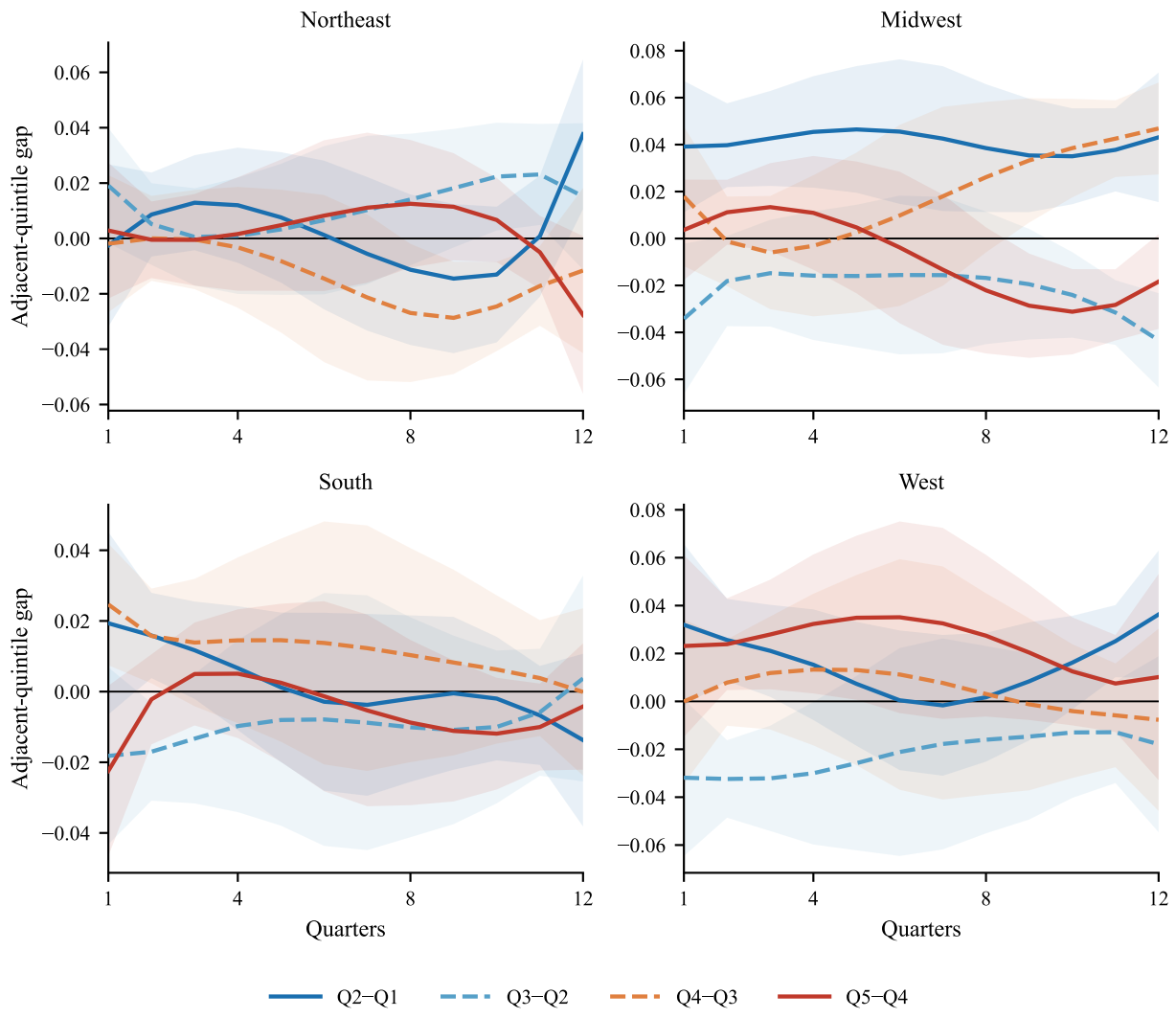


Figure D.13: Adjacent Income Gap: Non-Housing Consumption — FFR Shock

larger than that for $Q5$, consistent with the direction observed under LSAP, with the difference most pronounced in the Midwest; in the Northeast, the two responses are nearly identical. The specific point estimates for the South are provided in Table D.1.

The rise in total nominal consumption for $Q1$ in some regions requires interpretation alongside other consumption indicators. Under the FFR shock, the $Q1$–$Q5$ gap in the food expenditure ratio is insignificant in all four regions, so this measure cannot currently provide robust supplementary evidence on whether the rise in nominal consumption in the South (or any region) reflects a genuine improvement in welfare. At the same time, housing consumption for $Q1$ is positive in the Northeast, Midwest, and South, but negative in the West. This is consistent with rents rising with interest rates following a contractionary shock, while monthly payments on existing fixed-rate loans do not respond to short-term rates. This is an inherent limitation of nominal consumption measures, and

is the reason the main text uses non-housing consumption as the primary outcome variable.

The test for non-housing consumption does not reach conventional significance in any of the four regions, including the South. In terms of point estimates, the $Q1$–$Q5$ difference in the Midwest and West runs in the same direction as under LSAP, the Northeast is close to zero, and only the South runs in the opposite direction, with a small magnitude. This does not rule out the possibility that FFR shocks have distributional effects in more regions; the current sample may simply lack the statistical power to detect them, particularly during the 2009–2015 ZLB period when the FFR was near zero and the shock series lacked effective variation.

D.6 Food Share — All Three Shocks

The food expenditure ratio provides a complementary perspective based on consumption composition. Its primary function is to provide cross-validation for the findings from other measures, such as non-housing consumption and total consumption, particularly when a result appears to conflict with the main findings or requires additional explanation. It does not, in itself, constitute independent directional evidence. Engel's Law posits that as income levels rise, the proportion of household budgets allocated to food expenditures declines. Consequently, fluctuations in the food expenditure ratio in response to monetary policy shifts can be regarded as an indirect measure of alterations in expenditure composition. In contrast to nominal consumption levels, the food expenditure ratio is not directly influenced by aggregate price-level movements. This measure also has limitations. Griffith et al. (2016) find that during the Great Recession, households reduced food spending primarily by switching to cheaper items within food categories, purchasing more items on promotion, and changing retail outlets, while largely maintaining their nutritional intake. Changes in the food expenditure ratio therefore reflect both a budget-tightening component and a within-basket substitution component, with the magnitude of the latter not being fully independent of the unit food prices households actually pay. For the results in this paper, however, this channel biases the estimates in a conservative direction: downgrading behavior reduces food spending and hence lowers the ratio, implying that the rise in the food expenditure ratio for $Q1$ under contractionary shocks represents a net increase that remains after accounting for this effect.

The result for the Northeast is discussed in the main text (see Section 4.2); this section reports the evidence for the remaining three regions. The West ($p = 0.455$) runs in the same direction as the Northeast. The Midwest ($p = 0.587$) runs in the opposite direction, with $Q5$ rising more than $Q1$, and its direction changes repeatedly across the 12 horizons. The $Q1$–$Q5$ gap in the South ($p = 0.897$) is close to zero. None of the three is significant over the full horizon.

Under the FFR shock, the $Q1$–$Q5$ gap in the food expenditure ratio is insignificant in all four regions. Under the FG shock, the Northeast is significant only in the second-year window and

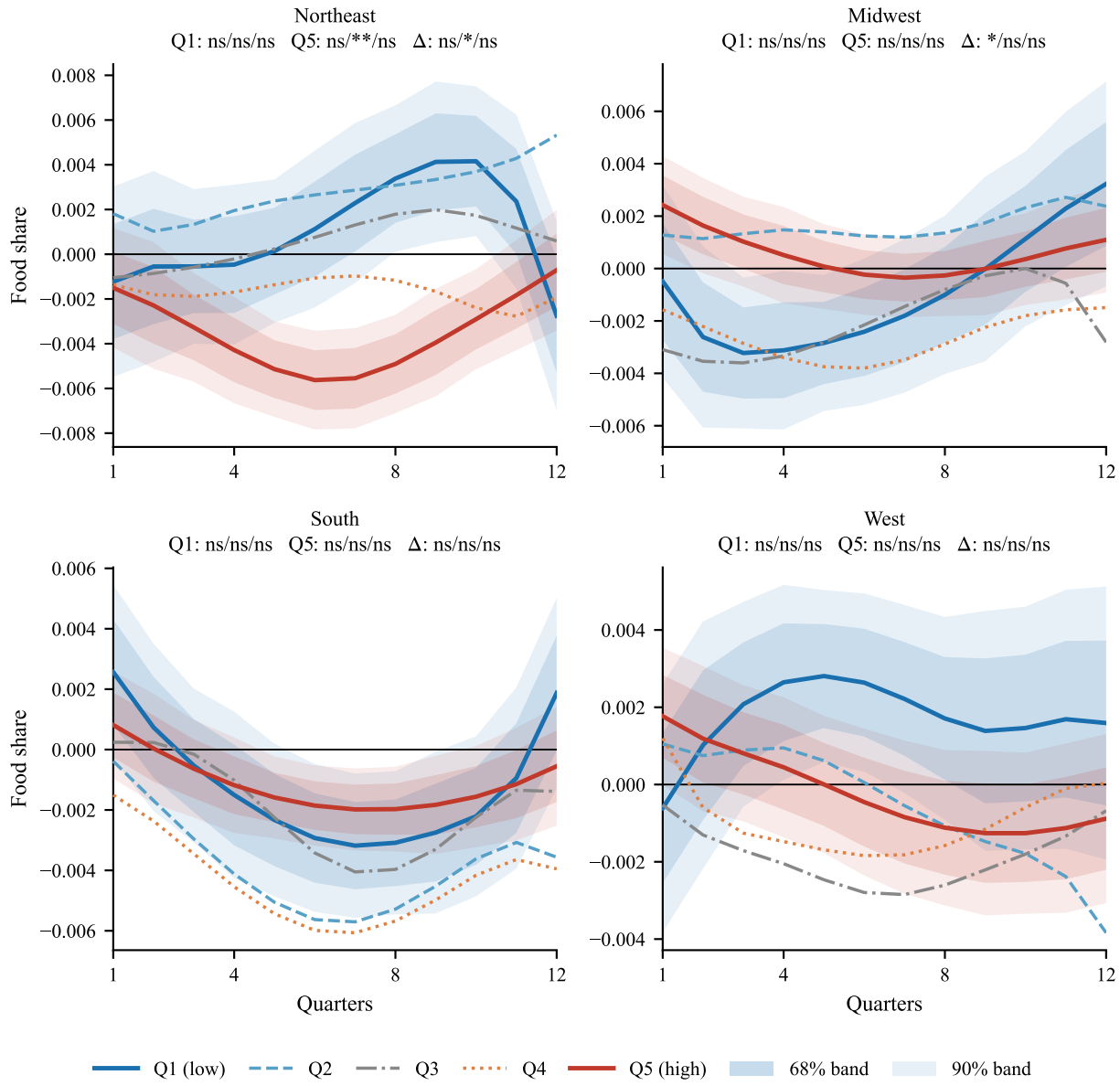


Figure D.14: IRF: Food Share — LSAP Shock

Notes: (1) Food share is food expenditure divided by total consumption expenditure. Unlike non-housing consumption, a progressive effect here implies $Q1$ *above* $Q5$: tightening worsens $Q1$'s consumption composition more (food share rises more). (2) See Figure 2 notes for general specification.

insignificant over the full horizon, whereas the Midwest is insignificant in every window.

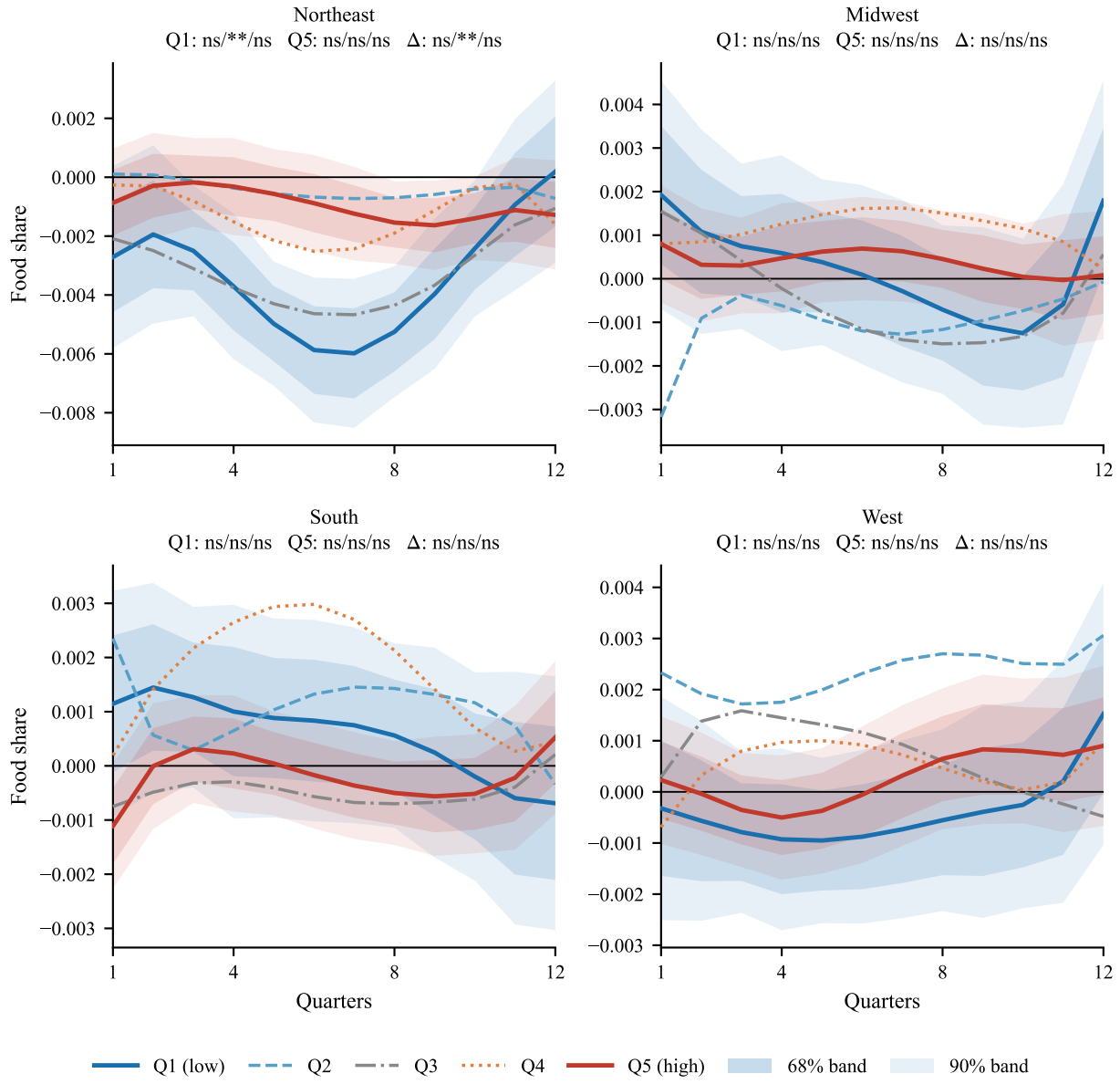


Figure D.15: Smooth LP IRF: Food Share — FG Shock

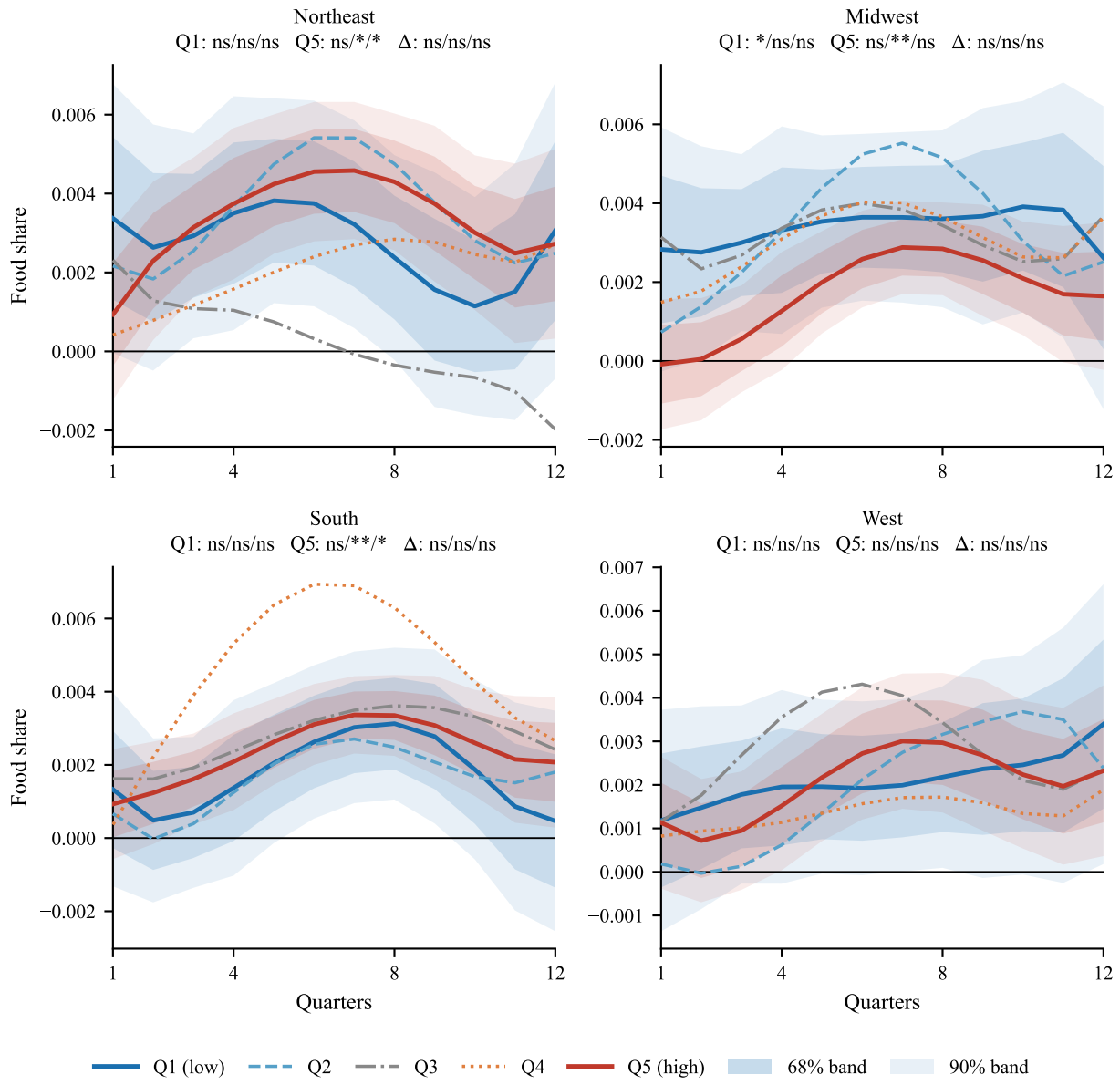


Figure D.16: Smooth LP IRF: Food Share — FFR Shock

D.7 The Q1–Q5 Gap: All Outcomes

Table D.2 directly compares the $Q1$–$Q5$ differences between pairs of instruments within the same region and consumption measure, covering all three instrument pairs.

Table D.1: The $Q1$–$Q5$ Gap by Region, Instrument, and Horizon Window

Outcome	Shock	Horizon	Northeast	Midwest	South	West
Non-housing	FFR	1–12	+0.31 (1.25)	+1.64 (1.09)	−0.12 (0.87)	+1.23 (1.20)
		1–4	+0.76 (1.42)	+2.22 (1.43)	+1.13 (0.90)	+1.81 (1.26)
		5–8	+0.24 (1.44)	+1.53 (1.40)	−0.52 (0.97)	+0.96 (1.34)
		9–12	−0.08 (1.51)	+1.16 (1.29)	−0.99 (1.09)	+0.92 (1.56)
	FG	1–12	−0.80 (1.09)	+0.18 (0.94)	+0.71 (0.75)	−0.98 (0.94)
		1–4	+0.04 (1.23)	+0.70 (1.19)	+1.35* (0.76)	−1.10 (0.99)
		5–8	−2.33* (1.33)	+0.13 (1.24)	+0.56 (0.90)	−0.95 (1.10)
		9–12	−0.10 (1.26)	−0.29 (1.03)	+0.23 (0.96)	−0.89 (1.21)
	LSAP	1–12	+3.29** (1.52)	−0.98 (1.43)	−1.06 (1.09)	−0.71 (1.43)
		1–4	+1.03 (1.85)	−2.46 (1.88)	−1.12 (1.23)	−1.74 (1.62)
		5–8	+5.02*** (1.75)	−0.95 (1.80)	−0.99 (1.26)	−1.88 (1.61)
		9–12	+3.83** (1.82)	+0.49 (1.65)	−1.07 (1.38)	+1.49 (1.81)
Total	FFR	1–12	−0.28 (0.85)	+1.04 (0.98)	−0.47 (0.68)	+1.25 (1.12)
		1–4	−0.52 (1.02)	+1.82 (1.26)	+0.73 (0.66)	+1.52 (1.26)
		5–8	−0.06 (0.97)	+0.95 (1.20)	−0.68 (0.78)	+1.20 (1.26)
		9–12	−0.24 (1.01)	+0.34 (1.15)	−1.47* (0.84)	+1.05 (1.43)
	FG	1–12	−0.30 (0.76)	+0.05 (0.81)	+0.14 (0.59)	−0.08 (0.94)
		1–4	+0.05 (0.90)	+0.34 (1.04)	+0.83 (0.56)	−0.03 (1.02)
		5–8	−1.02 (0.94)	−0.02 (1.05)	−0.21 (0.73)	+0.11 (1.08)
		9–12	+0.07 (0.85)	−0.17 (0.88)	−0.19 (0.75)	−0.32 (1.16)
	LSAP	1–12	+2.58** (1.08)	−0.87 (1.23)	−0.09 (0.83)	−0.95 (1.37)
		1–4	+1.35 (1.34)	−2.33 (1.61)	−0.22 (0.87)	−1.74 (1.59)
		5–8	+3.79*** (1.28)	−0.94 (1.53)	−0.29 (0.99)	−2.36 (1.51)
		9–12	+2.61** (1.25)	+0.64 (1.40)	+0.24 (1.06)	+1.24 (1.69)
Food share	FFR	1–12	+0.02 (0.12)	−0.11 (0.11)	+0.04 (0.09)	−0.02 (0.10)
		1–4	−0.04 (0.14)	−0.22 (0.14)	+0.04 (0.11)	−0.08 (0.11)
		5–8	+0.00 (0.13)	+0.00 (0.13)	+0.02 (0.10)	+0.08 (0.11)
		9–12	+0.11 (0.15)	−0.11 (0.14)	+0.06 (0.11)	−0.05 (0.13)
	FG	1–12	+0.10 (0.10)	−0.00 (0.10)	−0.04 (0.08)	+0.03 (0.08)
		1–4	+0.09 (0.11)	−0.05 (0.12)	−0.10 (0.09)	+0.03 (0.09)
		5–8	+0.26** (0.12)	+0.01 (0.12)	−0.06 (0.09)	+0.03 (0.09)
		9–12	−0.04 (0.12)	+0.03 (0.12)	+0.05 (0.09)	+0.04 (0.11)
	LSAP	1–12	−0.21 (0.15)	+0.08 (0.14)	−0.01 (0.11)	−0.10 (0.13)
		1–4	−0.05 (0.17)	+0.30* (0.18)	−0.09 (0.13)	+0.02 (0.16)
		5–8	−0.34* (0.17)	+0.10 (0.17)	+0.08 (0.12)	−0.10 (0.14)
		9–12	−0.25 (0.18)	−0.16 (0.18)	−0.03 (0.13)	−0.20 (0.16)
Housing	FFR	1–12	−1.30 (0.88)	−0.05 (1.18)	−1.00 (0.68)	+1.46 (1.49)
		1–4	−2.70*** (1.05)	+1.05 (1.36)	+0.10 (0.72)	+0.87 (1.66)
		5–8	−0.69 (1.11)	+0.17 (1.42)	−0.86 (0.86)	+1.71 (1.76)
		9–12	−0.52 (1.05)	−1.36 (1.35)	−2.23*** (0.89)	+1.79 (1.83)
	FG	1–12	+0.68 (0.78)	−0.20 (1.00)	−0.80 (0.58)	+1.65 (1.31)
		1–4	+0.03 (0.94)	−0.33 (1.13)	+0.02 (0.58)	+1.94 (1.36)
		5–8	+1.29 (0.99)	−0.30 (1.27)	−1.41* (0.75)	+2.27 (1.55)
		9–12	+0.73 (0.95)	+0.04 (1.14)	−1.03 (0.79)	+0.73 (1.64)
	LSAP	1–12	+0.94 (1.18)	−1.09 (1.43)	+1.63** (0.79)	−1.66 (1.83)
		1–4	+1.73 (1.50)	−2.23 (1.69)	+1.45 (0.90)	−1.87 (2.04)
		5–8	+1.44 (1.52)	−1.33 (1.77)	+0.92 (1.00)	−3.42 (2.12)
		9–12	−0.34 (1.36)	+0.29 (1.64)	+2.52** (1.09)	+0.30 (2.27)

Notes: each entry is the average of the gap $\delta^h \equiv \beta^h_{Q5} - \beta^h_{Q1}$ over the stated horizons, in percentage points per unit contractionary shock, with its bootstrap standard error in parentheses. Stars refer to the bootstrap test that the average is zero (residual moving block bootstrap, $B = 5000$, $L = 12$): $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. All three shock series are oriented so that a positive value is a monetary tightening. For the three consumption measures a positive value means that $Q1$ loses more under a contraction; for the food share the reading is reversed.

Table D.2: Instrument Contrasts of the $Q1$–$Q5$ Gap by Region and Horizon Window

Outcome	Contrast	Horizon	Northeast	Midwest	South	West
Non-housing	LSAP − FFR	1–12	+2.99 (2.30)	−2.61 (2.13)	−0.94 (1.63)	−1.94 (2.22)
		1–4	+0.27 (2.75)	−4.69 (2.81)	−2.25 (1.80)	−3.55 (2.42)
		5–8	+4.78* (2.64)	−2.48 (2.71)	−0.47 (1.84)	−2.84 (2.48)
		9–12	+3.91 (2.76)	−0.67 (2.47)	−0.09 (2.06)	+0.57 (2.82)
	LSAP − FG	1–12	+4.09* (2.09)	−1.16 (1.92)	−1.77 (1.47)	+0.27 (1.95)
		1–4	+0.99 (2.50)	−3.17 (2.49)	−2.47 (1.62)	−0.64 (2.10)
		5–8	+7.35*** (2.41)	−1.08 (2.45)	−1.55 (1.71)	−0.94 (2.20)
		9–12	+3.93 (2.49)	+0.78 (2.17)	−1.30 (1.86)	+2.38 (2.47)
	FG − FFR	1–12	−1.10 (1.58)	−1.46 (1.27)	+0.84 (1.15)	−2.21 (1.52)
		1–4	−0.73 (1.50)	−1.52 (1.41)	+0.22 (1.00)	−2.91** (1.33)
		5–8	−2.56 (1.89)	−1.40 (1.62)	+1.08 (1.35)	−1.90 (1.74)
		9–12	−0.02 (2.01)	−1.45 (1.57)	+1.21 (1.50)	−1.81 (2.05)
Total	LSAP − FFR	1–12	+2.86* (1.60)	−1.91 (1.85)	+0.38 (1.25)	−2.20 (2.09)
		1–4	+1.87 (1.98)	−4.14* (2.41)	−0.95 (1.28)	−3.25 (2.41)
		5–8	+3.85** (1.85)	−1.89 (2.29)	+0.39 (1.45)	−3.56 (2.32)
		9–12	+2.85 (1.86)	+0.30 (2.12)	+1.70 (1.57)	+0.20 (2.60)
	LSAP − FG	1–12	+2.88** (1.47)	−0.92 (1.67)	−0.24 (1.13)	−0.87 (1.90)
		1–4	+1.29 (1.82)	−2.67 (2.16)	−1.05 (1.14)	−1.71 (2.12)
		5–8	+4.81*** (1.75)	−0.91 (2.10)	−0.08 (1.36)	−2.47 (2.13)
		9–12	+2.54 (1.67)	+0.81 (1.85)	+0.43 (1.45)	+1.56 (2.35)
	FG − FFR	1–12	−0.02 (1.04)	−0.99 (1.14)	+0.62 (0.93)	−1.33 (1.42)
		1–4	+0.57 (1.06)	−1.47 (1.19)	+0.10 (0.79)	−1.55 (1.27)
		5–8	−0.96 (1.27)	−0.98 (1.40)	+0.48 (1.10)	−1.09 (1.61)
		9–12	+0.31 (1.31)	−0.51 (1.42)	+1.28 (1.20)	−1.36 (1.89)
Food share	LSAP − FFR	1–12	−0.24 (0.23)	+0.19 (0.21)	−0.06 (0.17)	−0.08 (0.19)
		1–4	−0.01 (0.26)	+0.53** (0.27)	−0.13 (0.20)	+0.10 (0.23)
		5–8	−0.34 (0.25)	+0.10 (0.24)	+0.06 (0.19)	−0.18 (0.20)
		9–12	−0.36 (0.28)	−0.05 (0.26)	−0.09 (0.19)	−0.15 (0.24)
	LSAP − FG	1–12	−0.32 (0.20)	+0.08 (0.19)	+0.03 (0.15)	−0.13 (0.17)
		1–4	−0.14 (0.23)	+0.35 (0.24)	+0.01 (0.18)	−0.02 (0.20)
		5–8	−0.59*** (0.23)	+0.09 (0.23)	+0.14 (0.17)	−0.13 (0.18)
		9–12	−0.22 (0.24)	−0.19 (0.24)	−0.08 (0.17)	−0.24 (0.22)
	FG − FFR	1–12	+0.08 (0.14)	+0.11 (0.14)	−0.08 (0.12)	+0.05 (0.12)
		1–4	+0.13 (0.14)	+0.18 (0.14)	−0.14 (0.11)	+0.12 (0.12)
		5–8	+0.26 (0.17)	+0.01 (0.16)	−0.09 (0.14)	−0.05 (0.14)
		9–12	−0.14 (0.19)	+0.14 (0.19)	−0.01 (0.15)	+0.09 (0.16)
Housing	LSAP − FFR	1–12	+2.25 (1.73)	−1.04 (2.16)	+2.63** (1.22)	−3.12 (2.75)
		1–4	+4.43** (2.16)	−3.28 (2.54)	+1.35 (1.37)	−2.74 (3.14)
		5–8	+2.13 (2.20)	−1.50 (2.66)	+1.78 (1.57)	−5.12 (3.22)
		9–12	+0.18 (1.99)	+1.66 (2.45)	+4.76*** (1.62)	−1.50 (3.41)
	LSAP − FG	1–12	+0.26 (1.60)	−0.89 (2.00)	+2.44** (1.12)	−3.31 (2.59)
		1–4	+1.70 (2.00)	−1.90 (2.31)	+1.42 (1.19)	−3.80 (2.78)
		5–8	+0.15 (2.01)	−1.02 (2.50)	+2.33 (1.44)	−5.69* (3.00)
		9–12	−1.06 (1.83)	+0.25 (2.27)	+3.55** (1.53)	−0.44 (3.24)
	FG − FFR	1–12	+1.98** (1.00)	−0.15 (1.45)	+0.19 (0.90)	+0.19 (1.92)
		1–4	+2.73** (1.10)	−1.38 (1.35)	−0.08 (0.85)	+1.06 (1.75)
		5–8	+1.98 (1.33)	−0.47 (1.75)	−0.55 (1.13)	+0.56 (2.27)
		9–12	+1.24 (1.23)	+1.41 (1.81)	+1.21 (1.17)	−1.06 (2.45)

Notes: each entry is the average over the stated horizons of the gap $\delta_A^h - \delta_B^h$, where $\delta_s^h \equiv \beta_{Q5,s}^h - \beta_{Q1,s}^h$ is the $Q1$–$Q5$ gap under instrument s and the row is labeled $A - B$, in percentage points per unit contractionary shock, with its bootstrap standard error in parentheses. Stars refer to the bootstrap test that the average is zero (residual moving block bootstrap, $B = 5000$, $L = 12$): $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Region, outcome, sample, and specification are held fixed, and only the instrument varies; the bootstrap uses one block index per draw for all regressions of an outcome.

D.8 National Aggregate

The national cells are constructed by aggregating the four regional cells using survey weights, with the specification, anchors, and bootstrap procedure of the main text. Table D.3 reports the $Q1$–$Q5$ differences under the three policy instruments, along with results based on national income ranks and alternative shock series for asset purchases.

Table D.3: National Aggregate: $Q1$–$Q5$ Gaps with the Four Regions Pooled

		$\bar{\delta}$ (pp)[a]	Boot. SE	p (1–12)	p (1–4)	p (5–8)	p (9–12)
Panel A: Baseline							
Non-housing consumption	FFR	+0.60	0.62	0.317	0.081	0.561	0.854
	FG	−0.17	0.51	0.734	0.679	0.433	0.593
	LSAP	−0.02	0.80	0.972	0.312	0.962	0.349
Total consumption	FFR	+0.27	0.53	0.598	0.234	0.634	0.760
	FG	−0.04	0.43	0.921	0.539	0.637	0.666
	LSAP	+0.12	0.67	0.848	0.436	0.922	0.172
Food share	FFR	+0.00	0.06	0.929	0.373	0.527	0.657
	FG	+0.02	0.05	0.698	0.776	0.548	0.511
	LSAP	−0.06	0.07	0.366	0.873	0.520	0.091
Panel B: Non-housing consumption under LSAP, alternative settings							
National income ranking		+0.49	0.79	0.539	0.772	0.668	0.164
LSAP factor set to zero before 2009		−0.06	1.23	0.956	0.783	0.567	0.491
Orthogonalized shocks		+0.69	0.92	0.460	0.755	0.450	0.123

[a] The gap $\beta^h_{Q5} - \beta^h_{Q1}$ averaged over $h = 1, \ldots, 12$, in percentage points per unit contractionary shock. 'Boot. SE' is its bootstrap standard deviation.

Notes: each national cell pools the four regional cells of an income quintile with their survey-weight totals; the food share is the weight-averaged cell food share. Specification and anchors are those of the main text. The p-values are from the bootstrap test that the average of the gap over the stated horizons is zero (residual moving block bootstrap, $B = 5000$, $L = 12$). Panel A uses the within-region income quintiles of the baseline. In Panel B, the first row assigns quintiles by the national income ranking (`INC_RANK`), and the other two rows use the shock series defined in the notes to Table F.2.

E Full Results: Housing Tenure × Income Quintile (Tenure Estimation)

This appendix reports the full set of results for the tenure-by-income estimation underlying the mechanism analysis in Section 5. Figure 4 in the main text presents the $Q1$–$Q5$ gap paths within each tenure group under LSAP using the standard local projections, as the smoothed local projections are not reliable for this estimation (see Appendix C). This appendix provides the corresponding fan charts, with LSAP estimated using the standard local projections and FG and FFR using the smoothed local projections, together with housing consumption and the food expenditure ratio under LSAP and the corresponding adjacent-income-gap charts. All tests reported in this appendix

are based on the standard local projections estimator and follow the inference conventions described in Section 3.3.

E.1 Tenure Composition by Income Quintile

Table E.1 reports the share of households in each income quintile falling into each of the three tenure categories, referenced in Section 5.2. Renters account for a substantially larger share of $Q1$ than of $Q5$ (60.4% vs. 11.3%), while mortgaged owners show the opposite pattern, rising from 11.5% of $Q1$ to 70.2% of $Q5$. Outright owners are most prevalent among $Q1$–$Q2$ households (28–33%) and least prevalent in $Q5$ (18.4%), a pattern consistent with this group disproportionately comprising older, retired homeowners whose current income is low relative to their accumulated housing wealth. This systematic association between tenure type and income is the reason the tenure decomposition in Section 5 must compare across tenure groups within a fixed income quintile, rather than across quintiles directly.

Table E.1: Housing Tenure Composition by Income Quintile

Tenure Type	$Q1$	$Q2$	$Q3$	$Q4$	$Q5$
Mortgaged	11.5%	21.6%	37.6%	56.2%	70.2%
Outright	28.1%	32.9%	25.5%	20.5%	18.4%
Renter	60.4%	45.5%	36.9%	23.3%	11.3%
Household-level observations (sum over sample period)					
Mortgaged	12,092	22,779	39,825	59,690	75,286
Outright	29,405	34,670	26,952	21,750	19,259
Renter	65,213	49,116	39,725	24,791	11,749

Notes: Each column reports the share of households within that income quintile (survey-weighted, summed over the full 1992Q1–2019Q4 CEX panel) falling into each of the three housing tenure categories; columns sum to 100%. The lower panel reports the corresponding unweighted household-level observation counts. Renters account for a substantially larger share of $Q1$ than of $Q5$ (60.4% vs. 11.3%). Mortgaged owners are most heavily concentrated in $Q5$ (70.2%) and least concentrated in $Q1$ (11.5%); outright owners are most prevalent in $Q1$–$Q2$ (28–33%) and least prevalent in $Q5$ (18.4%), consistent with this group disproportionately comprising older, retired homeowners with substantial housing wealth but comparatively low current income.

Table E.2 reports the tenure composition of each region and the share of housing in local household budgets. This table supports two claims in Section 5.3. First, the tenure composition is very similar across the four regions: the gap in the mortgagor share between the highest and lowest income quintiles is around 60 percentage points in each region, and the Northeast does not have the largest gap. Thus, differences across regions cannot be attributed to differences in tenure composition. Second, the Northeast has the highest share of housing in total expenditure among the four regions, particularly in the lowest income quintile.

Table E.2: Housing Tenure Composition and the Weight of Housing in the Budget, by Region

Region	Tenure composition			Housing share of expenditure
	Mortgaged	Outright	Renter	
Panel A. All households				
Northeast	36.3%	25.6%	38.1%	38.4%
Midwest	41.1%	26.8%	32.2%	34.7%
South	38.8%	27.7%	33.5%	34.9%
West	40.4%	18.9%	40.7%	37.7%
Panel B. Within $Q1$				
Northeast	7.9%	24.5%	67.7%	43.0%
Midwest	10.1%	27.5%	62.4%	39.4%
South	13.2%	33.2%	53.6%	40.1%
West	13.1%	23.3%	63.6%	42.3%
Panel C. Within $Q5$				
Northeast	66.7%	20.5%	12.8%	34.4%
Midwest	73.3%	19.9%	6.7%	30.7%
South	69.8%	19.1%	11.1%	31.2%
West	71.0%	13.9%	15.1%	34.1%
Memo: $Q5 - Q1$ mortgaged-share wedge (Panel C − Panel B)				
Northeast +58.8 pp, Midwest +63.2 pp, South +56.6 pp, West +57.9 pp				

Notes: Each panel covers one population: all households, those in $Q1$, and those in $Q5$. The first three columns report the share of that population falling into each of the three housing tenure categories and sum to 100%; the last column reports the survey-weighted mean of the household-level ratio of housing expenditure to total expenditure for the same population. All figures are survey-weighted and summed over the 1992Q1–2019Q4 CEX panel. Two features are referenced in Section 5.3. First, tenure composition is close to uniform across regions: the $Q5 - Q1$ mortgaged-share wedge ranges from 56.6 to 63.2 percentage points (South to Midwest), and the Northeast is not the extreme case, so differences in composition cannot account for the regional pattern in Section 4. Second, the Northeast has the largest housing share of total expenditure of the four regions, both overall (38.4%) and within $Q1$ (43.0%). Sample is the estimation sample of Section 2.

E.2 Detailed Transmission Discussion: Renters and Outright Owners

Among the three tenure groups, mortgagor households are the direct recipients of the cash flow channel. Mortgages in the United States are predominantly long-term and fixed-rate, so the monthly payment on an existing loan does not move with short-term rates, and monetary policy can affect the cash flow of these households only through the prepayment option. Asset purchases compress long-term rates and mortgage-backed securities spreads, which determine whether refinancing is worthwhile, but exercising the option requires paying closing costs and meeting income, equity, and credit requirements. Low-income borrowers have the highest ratio of monthly payments to disposable income and the thinnest liquidity buffers, making these requirements the hardest for them to meet. They are therefore squeezed most when a contractionary shock closes the refinancing channel. In Figure 4, the difference path for the mortgagor group is close to zero in the first year and widens window by window thereafter, consistent with the effects of refinancing taking time to

accumulate.

Neither renter households nor outright owner households lie on this channel. The housing cash flow of renters adjusts through rents as short-term rates move, while asset purchases do not act on rents directly. Accordingly, no income gradient appears within the renter group: non-housing consumption, total consumption, and housing consumption are insignificant in every window, and only the food expenditure ratio is significant at the 10% level in the first year ($p = 0.073$). It should be noted, however, that a gradient does appear within the renter group in the third year after excluding extreme shock quarters, so this comparison likewise depends on crisis-period shock variation (Appendix F). Outright owner households have neither monthly mortgage payments nor a refinancing margin. The only significant result for this group is housing consumption, reflecting changes in holding costs such as property taxes, insurance, and maintenance with interest rates rather than a response of consumption demand. This group contains a higher share of older households who have paid off their mortgages and rely more on fixed sources of income such as pensions, which is also consistent with the direction of the $Q1$ response for this group under forward guidance being opposite to that of the other two groups.

E.3 Non-Housing Consumption by Tenure Type — All Three Shocks

The three figures below present fan charts of non-housing consumption by tenure group under the three shocks, with the three panels in each figure showing mortgagor, outright owner, and renter households, respectively. The LSAP figure uses the standard local projections for the reason given in Appendix C, whereas the FFR and FG figures use the smoothed local projections.

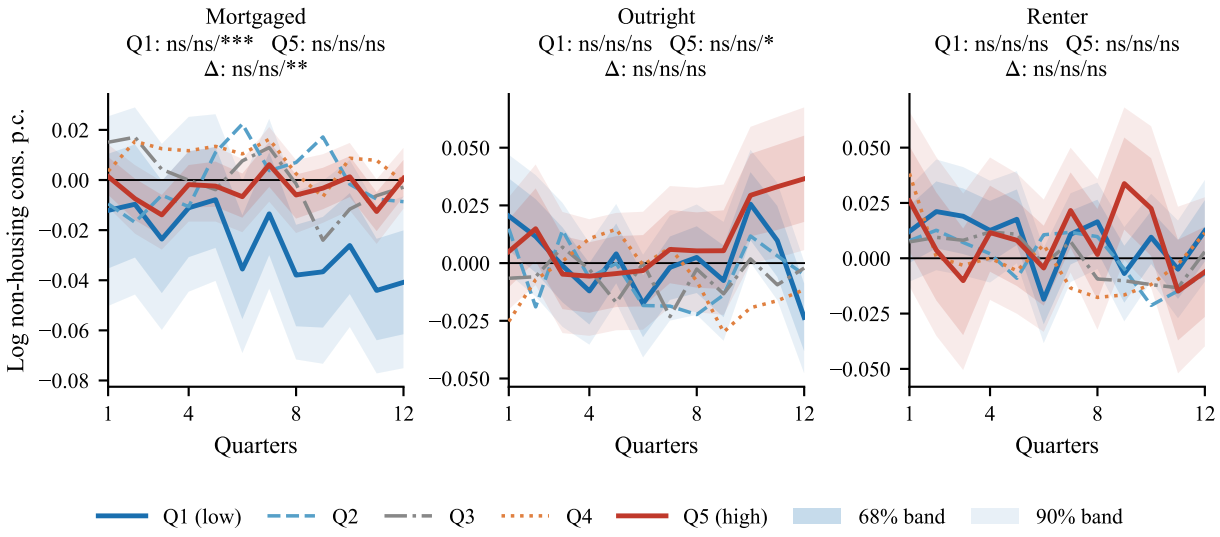


Figure E.1: IRF: Log Non-Housing Consumption per Capita — LSAP Shock, by Housing Tenure × Income Quintile

Notes: (1) Tenure estimation (housing tenure × income quintile); estimates are national-level. Three panels correspond to mortgaged owners, outright owners, and renters; lines within each panel are Q1–Q5 income quintiles. (2) This figure uses Standard LP estimates (not Smooth LP) because the B-spline smoothing introduces directional bias for the tenure estimation under LSAP shocks; see Appendix C. (3) Confidence bands and significance annotations come from the residual moving block bootstrap ($B = 5000$, $L = 12$); panel titles report the bootstrap tests of the window average (h = 1–4 / 5–8 / 9–12) for $Q1$, $Q5$ and Δ. See Figure 2 notes for the band definitions, significance symbols, and sample period.

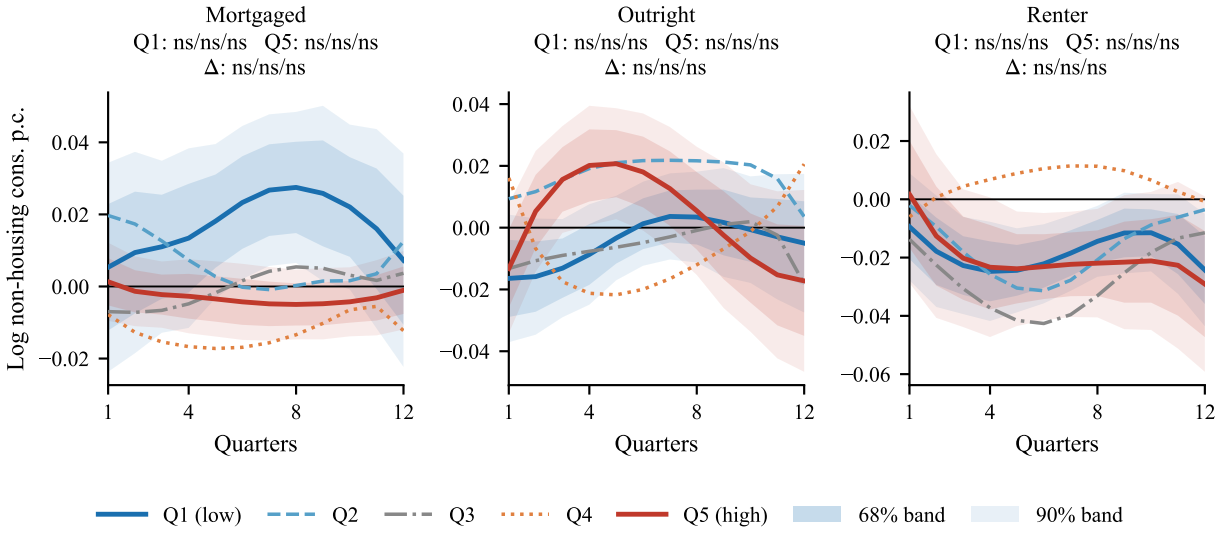


Figure E.2: Smooth LP IRF: Log Non-Housing Consumption — FFR Shock, by Tenure

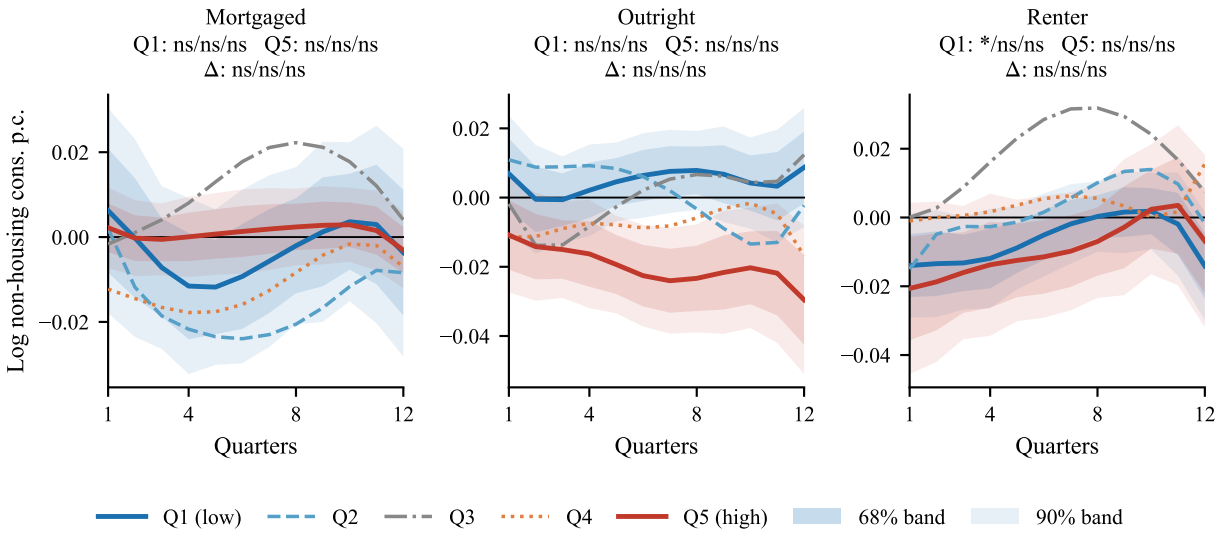


Figure E.3: Smooth LP IRF: Log Non-Housing Consumption — FG Shock, by Tenure

E.4 Housing Expenditure by Tenure Type

As in the regional estimation, housing consumption responds directly to interest rates: for renters, this operates mainly through rents, whereas for mortgagors, it operates mainly through refinancing activity rather than changes in existing monthly payments. Figure E.4 presents this result for LSAP as background to the non-housing consumption results discussed above.

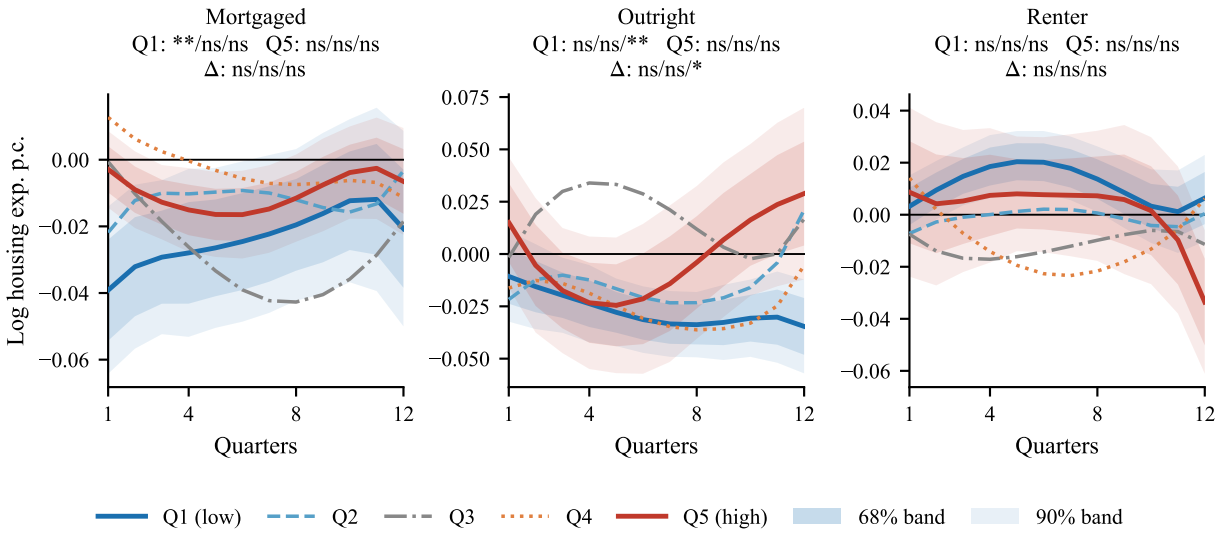


Figure E.4: Smooth LP IRF: Log Housing Expenditure — LSAP Shock, by Tenure

E.5 Food Share by Tenure Type

Figure E.5 reports the food expenditure ratio response to LSAP by tenure type, complementing the regional food-share evidence discussed in Section 4.2.

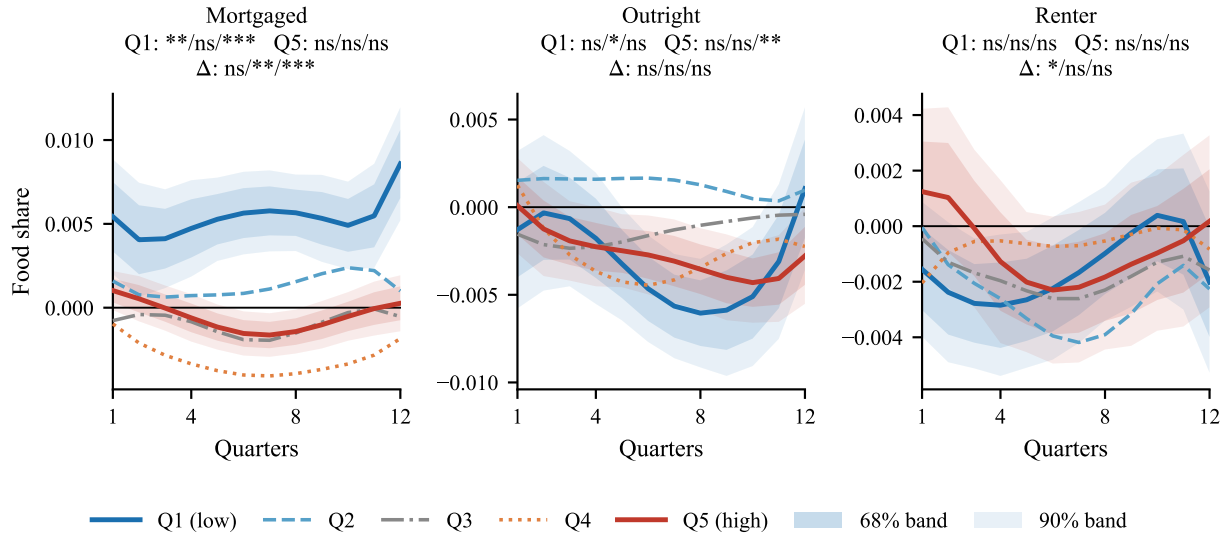


Figure E.5: Smooth LP IRF: Food Share — LSAP Shock, by Tenure

E.6 Adjacent Gap Figures — Tenure Estimation

This subsection reports adjacent income gap figures for the tenure estimation, paralleling the fan charts above. For non-housing consumption under LSAP, the main text uses Standard LP for display (Figure 4) because the Smooth LP fit is unreliable for this shock in the tenure estimation (Appendix C); the corresponding gap figure below uses the same Standard LP specification for consistency. The remaining gap figures use Smooth LP, matching the fan charts reported earlier in this appendix.

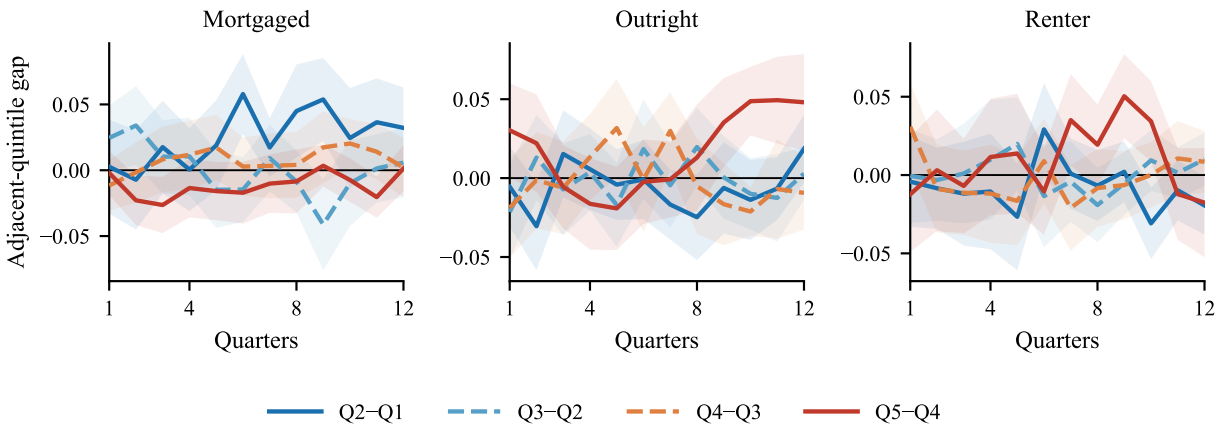


Figure E.6: Standard LP Adjacent Gap: Log Non-Housing Consumption — LSAP Shock, by Tenure

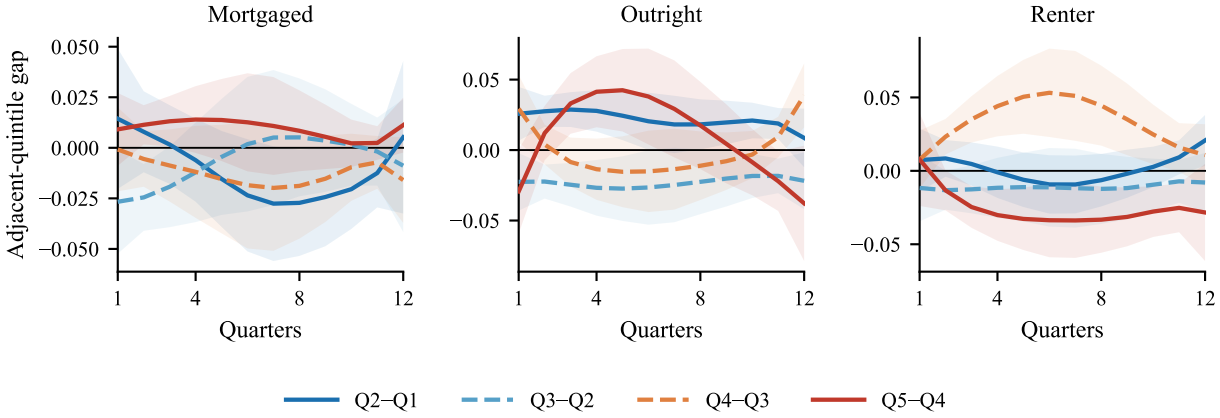


Figure E.7: Smooth LP Adjacent Gap: Log Non-Housing Consumption — FFR Shock, by Tenure

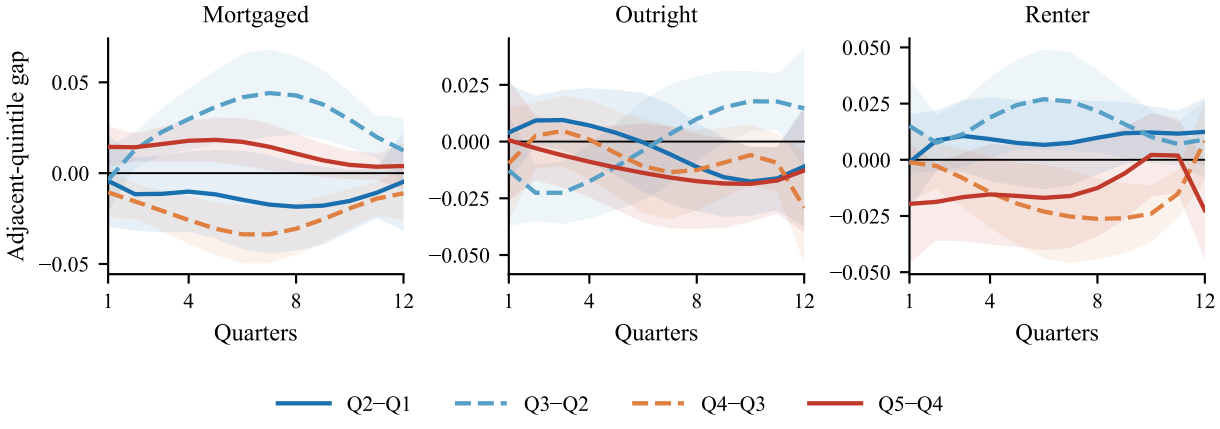


Figure E.8: Smooth LP Adjacent Gap: Log Non-Housing Consumption — FG Shock, by Tenure

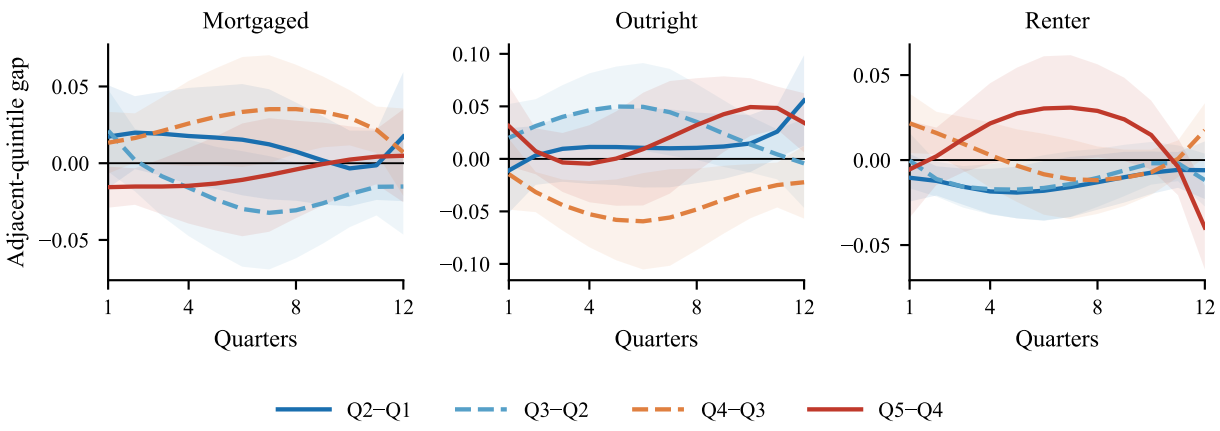


Figure E.9: Smooth LP Adjacent Gap: Log Housing Expenditure — LSAP Shock, by Tenure

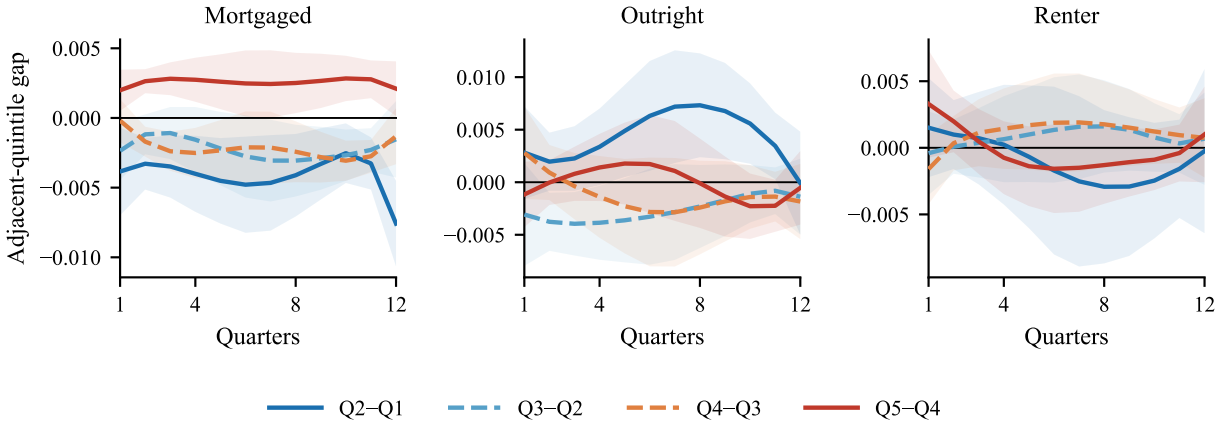


Figure E.10: Smooth LP Adjacent Gap: Food Share — LSAP Shock, by Tenure

E.7 Tables — Tenure Estimation

Table E.3 reports the $Q1$–$Q5$ gap within each tenure group underlying the mechanism discussion in Section 5.2, covering mortgagors, outright owners, and renters across the three shocks and each consumption variable. Table E.5 tests whether the renter-minus-mortgagor difference within the same income group is statistically distinguishable across pairs of instruments, providing a direct test of the reversal in direction described in Section 5.2.

Table E.3: Tenure Estimation: The $Q1$–$Q5$ Gap within Tenure Groups by Horizon Window

Outcome	Shock	Horizon	Mortgaged	Outright	Renter
Non-housing	FFR	1–12	−1.03 (1.02)	+0.36 (1.41)	−0.08 (0.90)
		1–4	−0.57 (1.30)	+1.57 (1.17)	+0.33 (1.21)
		5–8	−1.34 (1.22)	+0.11 (1.44)	+0.30 (0.93)
		9–12	−1.17 (1.21)	−0.60 (2.00)	−0.88 (1.20)
	FG	1–12	+0.14 (0.87)	−1.37 (1.01)	−0.10 (0.74)
		1–4	+0.21 (1.08)	−0.73 (0.91)	−0.42 (0.93)
		5–8	+0.04 (1.15)	−1.43 (1.08)	−0.33 (0.90)
		9–12	+0.18 (0.99)	−1.94 (1.42)	+0.46 (0.93)
	LSAP	1–12	+2.04 (1.32)	+0.96 (1.61)	+0.03 (1.19)
		1–4	+0.76 (1.81)	−0.21 (1.62)	−0.61 (1.61)
		5–8	+2.03 (1.59)	+0.53 (1.60)	+0.03 (1.21)
		9–12	+3.32** (1.51)	+2.55 (2.24)	+0.67 (1.54)
Total	FFR	1–12	−1.03 (0.76)	+0.10 (1.37)	−0.39 (0.81)
		1–4	−1.12 (1.02)	+0.92 (1.13)	−0.17 (1.09)
		5–8	−0.99 (0.95)	+0.41 (1.42)	−0.10 (0.78)
		9–12	−0.98 (0.89)	−1.03 (1.88)	−0.89 (1.07)
	FG	1–12	+0.46 (0.64)	−1.21 (0.98)	+0.12 (0.63)
		1–4	+0.36 (0.82)	−0.55 (0.87)	−0.12 (0.82)
		5–8	+0.66 (0.86)	−1.32 (1.07)	−0.07 (0.73)
		9–12	+0.37 (0.74)	−1.77 (1.34)	+0.54 (0.79)
	LSAP	1–12	+1.62 (0.99)	+1.37 (1.49)	+0.02 (1.04)
		1–4	+1.25 (1.39)	+0.26 (1.45)	−0.33 (1.47)
		5–8	+1.26 (1.26)	+0.63 (1.58)	−0.05 (1.00)
		9–12	+2.34** (1.16)	+3.23 (1.99)	+0.44 (1.35)
Food share	FFR	1–12	+0.24** (0.10)	−0.09 (0.13)	−0.20** (0.09)
		1–4	+0.20 (0.12)	−0.19 (0.16)	−0.22** (0.10)
		5–8	+0.17* (0.10)	−0.06 (0.13)	−0.15 (0.10)
		9–12	+0.36*** (0.12)	−0.02 (0.18)	−0.22* (0.12)
	FG	1–12	+0.09 (0.08)	+0.08 (0.10)	−0.11 (0.08)
		1–4	+0.06 (0.09)	+0.10 (0.11)	−0.13 (0.09)
		5–8	+0.12 (0.10)	+0.02 (0.11)	−0.07 (0.09)
		9–12	+0.10 (0.10)	+0.13 (0.13)	−0.13 (0.10)
	LSAP	1–12	−0.33*** (0.12)	−0.01 (0.16)	+0.06 (0.12)
		1–4	−0.26 (0.16)	−0.10 (0.20)	+0.25* (0.14)
		5–8	−0.28** (0.13)	+0.18 (0.15)	−0.02 (0.13)
		9–12	−0.44*** (0.16)	−0.10 (0.21)	−0.06 (0.16)
Housing	FFR	1–12	−1.04 (0.83)	−0.30 (1.55)	−1.09 (0.89)
		1–4	−1.82* (1.00)	−0.80 (1.51)	−1.48 (1.13)
		5–8	−0.49 (1.08)	+1.51 (1.71)	−1.15 (0.92)
		9–12	−0.81 (1.00)	−1.63 (2.02)	−0.64 (1.19)
	FG	1–12	+1.29* (0.68)	−0.49 (1.28)	+0.79 (0.67)
		1–4	+0.88 (0.82)	+0.29 (1.24)	+0.50 (0.84)
		5–8	+2.02** (0.88)	−0.62 (1.51)	+0.72 (0.76)
		9–12	+0.97 (0.85)	−1.14 (1.63)	+1.14 (0.88)
	LSAP	1–12	+0.89 (1.07)	+1.67 (1.86)	−0.57 (1.14)
		1–4	+2.00 (1.30)	+0.73 (1.94)	−0.23 (1.57)
		5–8	−0.07 (1.39)	+0.13 (2.19)	−0.72 (1.16)
		9–12	+0.74 (1.31)	+4.15* (2.29)	−0.76 (1.49)

Notes: each entry is the average of the gap $\delta^h \equiv \beta^h_{Q5} - \beta^h_{Q1}$ within the stated tenure group over the stated horizons, in percentage points per unit contractionary shock, with its bootstrap standard error in parentheses. Stars refer to the bootstrap test that the average is zero (residual moving block bootstrap, $B = 5000$, $L = 12$): $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. For the three consumption measures a positive value means that $Q1$ loses more under a contraction; for the food share the reading is reversed. The renter–mortgagor comparison within a fixed income quintile is reported in Tables 2 and E.4.

Table E.4: Renter versus Mortgagor Households within the Same Income Quintile: Other Outcomes

	FFR				FG				LSAP			
Quintile	h=1–12	1–4	5–8	9–12	h=1–12	1–4	5–8	9–12	h=1–12	1–4	5–8	9–12
Total consumption												
$Q1$	-1.75^{**}	-2.08^{**}	-1.72^{*}	-1.45^{*}	+0.01	−0.66	+0.51	+0.19	$+2.61^{***}$	$+2.95^{**}$	$+2.28^{*}$	$+2.61^{***}$
	(0.70)	(1.02)	(0.94)	(0.79)	(0.61)	(0.84)	(0.83)	(0.67)	(0.82)	(1.28)	(1.14)	(0.94)
$Q2$	−0.46	−0.57	−0.46	−0.34	+0.83	+0.61	+0.57	+1.31	+0.30	+1.49	−0.18	−0.41
	(0.83)	(0.83)	(1.00)	(1.07)	(0.76)	(0.72)	(0.94)	(0.98)	(1.06)	(1.12)	(1.29)	(1.34)
$Q3$	−0.94	−0.46	-1.35^{*}	−1.00	+0.13	−0.03	+0.32	+0.09	+0.21	−0.10	+0.12	+0.61
	(0.59)	(0.72)	(0.70)	(0.70)	(0.49)	(0.56)	(0.61)	(0.57)	(0.76)	(0.95)	(0.88)	(0.91)
$Q4$	+0.72	+0.37	+1.05	+0.73	+0.73	$+1.17^{**}$	+0.85	+0.17	−0.64	−0.14	-1.46^{*}	−0.31
	(0.46)	(0.61)	(0.64)	(0.56)	(0.47)	(0.57)	(0.64)	(0.53)	(0.63)	(0.86)	(0.83)	(0.72)
$Q5$	−1.10	−1.13	−0.83	−1.35	−0.33	−1.13	−0.22	+0.36	+1.02	+1.38	+0.97	+0.71
	(0.95)	(1.15)	(0.97)	(1.24)	(0.74)	(0.87)	(0.85)	(0.94)	(1.20)	(1.57)	(1.19)	(1.56)
Food share												
$Q1$	$+0.33^{***}$	$+0.30^{**}$	+0.18	$+0.51^{***}$	+0.16	$+0.20^{*}$	+0.13	+0.15	-0.40^{***}	-0.49^{***}	-0.28^{*}	-0.42^{**}
	(0.11)	(0.14)	(0.14)	(0.13)	(0.10)	(0.11)	(0.12)	(0.13)	(0.14)	(0.18)	(0.16)	(0.18)
$Q2$	+0.13	+0.06	+0.10	$+0.24^{*}$	+0.02	−0.04	+0.05	+0.04	−0.20	−0.18	−0.20	−0.23
	(0.11)	(0.13)	(0.13)	(0.13)	(0.09)	(0.10)	(0.11)	(0.11)	(0.13)	(0.17)	(0.15)	(0.16)
$Q3$	$+0.13^{*}$	+0.11	$+0.18^{*}$	+0.12	−0.04	+0.01	−0.06	−0.07	−0.05	−0.05	−0.00	−0.10
	(0.08)	(0.09)	(0.09)	(0.09)	(0.07)	(0.07)	(0.08)	(0.08)	(0.10)	(0.12)	(0.12)	(0.12)
$Q4$	+0.03	+0.07	+0.01	+0.03	−0.02	−0.03	−0.02	−0.01	$+0.12^{*}$	+0.07	+0.11	$+0.18^{**}$
	(0.05)	(0.06)	(0.06)	(0.07)	(0.05)	(0.05)	(0.06)	(0.06)	(0.07)	(0.08)	(0.08)	(0.09)
$Q5$	−0.11	−0.11	−0.14	−0.08	−0.04	+0.01	−0.07	−0.08	−0.01	+0.02	−0.02	−0.04
	(0.08)	(0.10)	(0.10)	(0.09)	(0.07)	(0.09)	(0.10)	(0.09)	(0.10)	(0.14)	(0.13)	(0.12)
Housing expenditure												
$Q1$	-1.40^{*}	-2.35^{**}	−1.10	−0.73	+0.68	−0.03	+1.36	+0.72	$+1.92^{**}$	$+3.31^{**}$	+1.43	+1.03
	(0.77)	(1.05)	(1.01)	(0.87)	(0.62)	(0.86)	(0.83)	(0.73)	(0.97)	(1.31)	(1.31)	(1.12)
$Q2$	+0.82	+1.03	+0.55	+0.89	+0.67	+1.21	−0.34	+1.15	+0.59	+0.84	+0.37	+0.57
	(0.94)	(1.03)	(1.10)	(1.22)	(0.86)	(0.85)	(1.03)	(1.11)	(1.19)	(1.38)	(1.36)	(1.55)
$Q3$	−0.04	+0.97	−0.12	−0.95	−0.40	−0.35	−0.17	−0.66	+0.77	−0.28	+1.19	+1.41
	(0.75)	(0.96)	(0.89)	(0.82)	(0.62)	(0.86)	(0.75)	(0.64)	(1.02)	(1.38)	(1.16)	(1.12)
$Q4$	+0.43	−0.16	+0.77	+0.68	+0.78	$+1.21^{**}$	$+1.44^{**}$	−0.32	−0.27	−0.48	−0.67	+0.34
	(0.62)	(0.67)	(0.75)	(0.85)	(0.56)	(0.57)	(0.71)	(0.75)	(0.81)	(0.91)	(0.99)	(1.06)
$Q5$	−1.45	-2.02^{*}	−1.76	−0.56	+0.18	−0.40	+0.06	+0.89	+0.46	+1.08	+0.78	−0.47
	(1.02)	(1.15)	(1.13)	(1.32)	(0.79)	(0.87)	(0.93)	(1.02)	(1.26)	(1.54)	(1.37)	(1.63)

Notes: each entry is the average of $d^h \equiv \beta^h_{\text{Renter}} - \beta^h_{\text{Mortgaged}}$ within the stated income quintile over the stated horizons, in percentage points per unit contractionary shock, with its bootstrap standard error in parentheses. Stars refer to the bootstrap test that the average is zero (residual moving block bootstrap, $B = 5000$, $L = 12$): $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. The table has the layout of Table 2 in the main text, which reports non-housing consumption.

Table E.5: Instrument Contrasts of the Renter–Mortgagor Difference within the Same Income Quintile: Log Non-Housing Consumption per Capita

	LSAP − FFR				LSAP − FG				FG − FFR			
Quintile	h=1–12	1–4	5–8	9–12	h=1–12	1–4	5–8	9–12	h=1–12	1–4	5–8	9–12
$Q1$	$+5.10^{***}$	$+4.50^{*}$	$+4.97^{**}$	$+5.84^{***}$	$+3.54^{**}$	$+3.70^{*}$	+2.91	$+4.02^{**}$	+1.56	+0.79	+2.06	+1.82
	(1.71)	(2.40)	(2.12)	(2.10)	(1.52)	(2.07)	(1.96)	(1.88)	(1.12)	(1.29)	(1.41)	(1.37)
$Q2$	+1.26	$+3.45^{*}$	+0.50	−0.16	−0.83	+1.67	−1.62	−2.54	+2.09	+1.78	+2.13	+2.38
	(1.83)	(1.89)	(2.33)	(2.25)	(1.70)	(1.75)	(2.14)	(2.13)	(1.30)	(1.16)	(1.61)	(1.69)
$Q3$	+1.38	+1.19	+1.80	+1.15	−0.26	+0.00	−0.63	−0.16	$+1.64^{*}$	+1.19	$+2.43^{**}$	+1.31
	(1.35)	(1.65)	(1.63)	(1.55)	(1.28)	(1.47)	(1.57)	(1.46)	(0.86)	(0.90)	(1.07)	(1.00)
$Q4$	−1.71	−0.65	-3.01^{**}	−1.46	−1.58	−1.13	-2.49^{*}	−1.13	−0.12	+0.48	−0.52	−0.33
	(1.03)	(1.45)	(1.44)	(1.17)	(1.02)	(1.38)	(1.39)	(1.13)	(0.69)	(0.78)	(0.95)	(0.80)
$Q5$	+2.15	+2.22	+1.33	+2.90	+1.78	+2.96	+1.28	+1.09	+0.37	−0.74	+0.04	+1.81
	(1.93)	(2.55)	(1.88)	(2.48)	(1.67)	(2.19)	(1.72)	(2.15)	(1.13)	(1.16)	(1.31)	(1.50)

Notes: $d_s^h \equiv \beta^h_{\text{Renter},s} - \beta^h_{\text{Mortgaged},s}$ is the renter–mortgagor difference within the stated income quintile under instrument s, as in Table 2. Each entry is the average of $d_A^h - d_B^h$ over the stated horizons for the contrast labeled $A - B$, in percentage points per unit contractionary shock, with its bootstrap standard error in parentheses. Stars refer to the bootstrap test that the average is zero (residual moving block bootstrap, $B = 5000$, $L = 12$): $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. A positive value means that, relative to renters, mortgagor households' consumption falls more under instrument A than under instrument B. The bootstrap uses one block index per draw for all six regressions of a quintile.

F Robustness Check Details

This appendix provides the full results for the robustness checks in Section 6 of the main text. Table 3 in the main text summarizes three quantities under each specification for the three results on which this paper makes its claims: the regional result (the Northeast, LSAP, non-housing consumption), the mechanism result (mortgagor households, LSAP, non-housing consumption), and the contrast between LSAP and FFR of the renter–mortgagor difference in the lowest income quintile. These are the magnitude of the gap or contrast, its bootstrap standard error, and the p-values over the full horizon and the three windows; Table F.2 reports the same quantities for the four main comparisons under every check. The CPI deflation does not apply to the food expenditure ratio, since the same price index cancels out in the numerator and denominator of the ratio, leaving the estimates identical before and after deflation.

F.1 Alternative Income Classification (INC_RANK)

Table F.2 reports the results after replacing the baseline within-region relative ranking with the Bureau of Labor Statistics' `INC_RANK` variable, which gives the national income percentile. The bootstrap p-value for the Northeast is 0.013, compared with 0.031 under the baseline. The point estimate of the difference rises from 3.29 percentage points in the baseline to 4.05 percentage points under the alternative classification. The timing of the effect is also preserved: the first-year window is insignificant under both classifications ($p = 0.33$ under the alternative and $p = 0.57$ under the baseline), while the second- and third-year windows are significant under both ($p = 0.009$ and $p = 0.015$ under the alternative, against $p = 0.005$ and $p = 0.033$ under the baseline). The core finding therefore depends neither on the income classification nor, in its timing, on which of the two is used.

F.2 Excluding NBER Recession Quarters

Table F.2 reports the results after excluding the 2001 recession (2001Q1–2001Q4) and the 2007–2009 recession (2007Q4–2009Q2), reducing the anchor periods available for estimation from 96 to 85. In this check the gap is no longer significant in any window: the bootstrap p-value for the Northeast rises from 0.031 to 0.380 over the full horizon.

The point estimate of the difference falls from 3.29 percentage points in the baseline to 2.29 percentage points, while the standard error rises by approximately 70% over the same comparison. By window, all three tests fall well short of the 10% threshold. Even when considering only the second-year window, in which the effect actually emerges, the reduced sample is far from sufficient to detect it. Excluding recession quarters does not merely reduce precision; it removes from the

sample the variation on which identification of this effect depends, since the excluded quarters are precisely those in which the unconventional instruments were used on a large scale.

F.3 Excluding Quarters with Extreme Shocks

Table F.2 reports the results after excluding the 15 quarters in which any shock series exceeds two standard deviations in absolute value, reducing the anchor periods available for estimation to 81. The bootstrap p-value for the Northeast rises from 0.031 to 0.124; the third-year window is 0.033 and the second-year window 0.058.

Unlike the exclusion of recession quarters, this exclusion does not shrink the point estimate. The $Q1$–$Q5$ gap per unit shock rises from 3.29 percentage points in the baseline to 4.39 percentage points. The rise in the p-value has a clear mechanical explanation: the excluded quarters are precisely those with the largest shock variance, concentrated mainly around 2001 and during the 2007–2009 financial crisis. Because observations with greater regressor variance provide more information for identifying the coefficient, removing these observations increases the standard errors by approximately 85%.

This check therefore does not show that the distributional effect varies with shock size; it shows that identifying the effect requires a sample that contains policy shocks of sufficient magnitude.

F.4 CPI-Deflated Consumption

Table F.2 reports the results after deflating non-housing consumption, total consumption, and housing expenditure to real values using CPIAUCSL (FRED, monthly data averaged to quarterly). If the regressions for the two quintiles used exactly the same controls, a common price deflator would cancel out in the income difference. The specification includes each cell's own lagged dependent variable, however, so deflation does not affect the estimates for the two quintiles identically. This check is therefore not mechanical; rather, it directly tests whether the distributional effect could be driven by price factors. The results bear this out: the bootstrap p-value for the Northeast under the LSAP shock moves from 0.0312 in nominal terms to 0.0316 in real terms, and total consumption from 0.0172 to 0.0178. It should be noted that the assumption of symmetric price movements across income groups is not uncontested. Cravino et al. (2020) find that price stickiness varies across the income distribution, implying that the effective price indices faced by different income groups respond differently to monetary policy. This paper uses only the national consumer price index and therefore cannot construct income-group-specific price indices. Consequently, the influence of this channel on the results cannot be ruled out.

F.5 Shock Series, Influential Quarters, and Specification

This section reports the full results for the last set of checks in Section 6. Table F.1 presents regressions of the three factors on six pre-announcement variables, and the orthogonalized shock series are the residuals from these regressions aggregated to quarterly frequency. Table F.2 gives the means and window-specific p-values for the four main comparisons under sixteen settings, and Table F.3 gives the remaining results cited in the main text.

Table F.1: Predictability of the Three Policy Factors from Pre-Announcement Information

	FFR	FG	LSAP	LSAP, 2009 on
Nonfarm payrolls surprise	+0.0015	+0.0003	−0.0002	+0.0007
	(2.69)	(0.42)	(−0.68)	(0.79)
Nonfarm payrolls, 12-month log change	+0.0027	+0.093	+0.0012	+0.040
	(0.10)	(1.91)	(0.03)	(0.51)
S&P 500, 13-week log change	+1.543	+0.596	+0.041	+6.083
	(1.53)	(0.46)	(0.05)	(1.48)
Treasury slope (10y − 3m), 13-week change	−0.189	+0.057	+0.249	−0.171
	(−1.42)	(0.38)	(2.10)	(−0.59)
Commodity prices, 13-week log change	+0.382	+2.216	+0.311	+0.958
	(0.49)	(2.07)	(0.76)	(0.80)
Implied skewness of the 10-year yield	+0.268	+0.475	−0.150	−0.030
	(1.59)	(2.01)	(−0.80)	(−0.14)
Announcements	241	241	241	84
R^2	0.107	0.087	0.046	0.185
Joint test, p-value	0.053	0.029	0.050	0.053

Notes: each column regresses one of the three factors of Swanson (2021), at the frequency of FOMC announcements, on six variables dated before the announcement; heteroskedasticity-robust t-statistics are in parentheses. The six variables are those of Bauer and Swanson (2023), taken as computed by the authors from the data file published by the Federal Reserve Bank of San Francisco: the surprise in the most recent nonfarm payrolls release, the log change in nonfarm payrolls over the preceding 12 months, the log change in the S&P 500 and in the Bloomberg commodity index and the change in the slope of the Treasury yield curve over the 13 weeks to the day before the announcement, and the implied skewness of the 10-year Treasury yield. The joint test is that the six coefficients are zero. The first three columns use the 241 announcements from July 1991 to June 2019, and the last column the 84 announcements from 2009 on. The orthogonalized shock series are the residuals of these regressions, summed within each quarter.

Table F.2: Main Results under Alternative Shock Series, Influential Quarters, Specifications, and Samples: Log Non-Housing Consumption per Capita

	$\bar{\delta}$ (pp)[a]	Boot. SE	p (1–12)	p (1–4)	p (5–8)	p (9–12)
Panel A: Northeast, $Q1$–$Q5$ gap, LSAP (regional estimation)						
Baseline	+3.29	1.52	0.031	0.574	0.005	0.033
LSAP factor set to zero before 2009	+4.58	2.26	0.049	0.329	0.013	0.100
LSAP factor set to zero from 2009 on (placebo)	+2.94	2.08	0.156	0.886	0.036	0.119
Shocks orthogonalized to pre-announcement information	+3.22	1.79	0.071	0.758	0.012	0.071
Orthogonalized, LSAP factor zero before 2009	+3.21	2.77	0.244	0.376	0.150	0.613
Excluding 2009Q1	+2.68	1.57	0.087	0.910	0.030	0.041
Excluding 2013Q3	+3.91	1.59	0.014	0.536	0.002	0.009
Excluding 2009Q1 and 2013Q3	+3.85	1.82	0.035	0.851	0.014	0.003
One lag of the shock ($p_s = 1$)	+3.33	1.53	0.030	0.566	0.006	0.033
No lags of the shock ($p_s = 0$)	+2.63	1.24	0.033	0.599	0.012	0.027
Lagged interest rates added	+3.21	1.51	0.034	0.440	0.005	0.087
Three shocks entered jointly	+3.99	1.77	0.024	0.293	0.012	0.021
National income ranking (`INC_RANK`)	+4.05	1.67	0.013	0.331	0.009	0.015
Excluding recession quarters	+2.29	2.60	0.380	0.879	0.320	0.273
Excluding extreme-shock quarters	+4.39	2.82	0.124	0.997	0.058	0.033
Consumption deflated by the CPI	+3.29	1.52	0.032	0.581	0.004	0.035
Panel B: Northeast gap minus the mean gap of the other three regions, LSAP						
Baseline	+4.21	1.63	0.010	0.134	0.002	0.071
LSAP factor set to zero before 2009	+5.86	2.39	0.021	0.135	0.002	0.146
LSAP factor set to zero from 2009 on (placebo)	+4.17	2.30	0.071	0.385	0.034	0.135
Shocks orthogonalized to pre-announcement information	+3.19	1.97	0.104	0.502	0.019	0.315
Orthogonalized, LSAP factor zero before 2009	+3.47	3.09	0.253	0.353	0.082	0.986
Excluding 2009Q1	+3.54	1.69	0.039	0.297	0.010	0.129
Excluding 2013Q3	+4.87	1.72	0.005	0.145	0.001	0.021
Excluding 2009Q1 and 2013Q3	+4.79	1.98	0.016	0.353	0.006	0.022
One lag of the shock ($p_s = 1$)	+4.25	1.65	0.010	0.136	0.002	0.070
No lags of the shock ($p_s = 0$)	+3.48	1.35	0.011	0.169	0.003	0.047
Lagged interest rates added	+3.75	1.67	0.027	0.107	0.003	0.268
Three shocks entered jointly	+4.40	1.90	0.021	0.143	0.011	0.068
National income ranking (`INC_RANK`)	+4.39	1.67	0.009	0.146	0.004	0.035
Excluding recession quarters	+3.64	2.81	0.198	0.550	0.056	0.419
Excluding extreme-shock quarters	+3.66	3.25	0.270	0.791	0.049	0.275
Consumption deflated by the CPI	+4.21	1.64	0.011	0.135	0.002	0.073

[a] The contrast averaged over $h = 1, \ldots, 12$, in percentage points per unit contractionary shock; 'Boot. SE' is its bootstrap standard deviation. Panel A: $\beta^h_{Q5} - \beta^h_{Q1}$. Panel B: the Panel A gap minus the average of the same gap in the Midwest, South, and West. Panel C: $\beta^h_{\text{Renter}} - \beta^h_{\text{Mortgaged}}$, as in Table 2. Panel D: the Panel C difference under LSAP minus the same difference under FFR.

Notes: each row changes one element of the baseline and re-estimates the contrast. The p-values are from the bootstrap test that the average of the contrast over the stated horizons is zero (residual moving block bootstrap, $B = 5000$, $L = 12$). Shock series: 'set to zero before 2009' replaces the LSAP factor by zero in every quarter before 2009Q1, and 'set to zero from 2009 on' keeps only its values before 2009Q1, a period without asset purchases; the orthogonalized series replace each factor by the residual from a regression across FOMC announcements on the six pre-announcement variables of Bauer and Swanson (2023) (Table F.1), summed to quarters, and the row that follows in addition sets the LSAP factor to zero before 2009 and takes its residual over the announcements from 2009 on. The other rows are defined in the notes to the second part of the table, on the next page.

Table F.2: Main Results under Alternative Shock Series, Influential Quarters, Specifications, and Samples: Log Non-Housing Consumption per Capita (continued)

	$\bar{\delta}$ (pp)[a]	Boot. SE	p (1–12)	p (1–4)	p (5–8)	p (9–12)
Panel C: Lowest income quintile, renter minus mortgagor, LSAP (tenure estimation)						
Baseline	+3.20	1.12	0.005	0.079	0.039	0.006
LSAP factor set to zero before 2009	+3.74	1.85	0.042	0.109	0.235	0.063
LSAP factor set to zero from 2009 on (placebo)	+2.68	1.56	0.082	0.473	0.175	0.044
Shocks orthogonalized to pre-announcement information	+4.13	1.28	0.001	0.031	0.012	0.005
Orthogonalized, LSAP factor zero before 2009	+3.99	2.17	0.064	0.352	0.249	0.021
Excluding 2009Q1	+3.24	1.19	0.006	0.110	0.041	0.009
Excluding 2013Q3	+2.62	1.13	0.022	0.157	0.092	0.023
Excluding 2009Q1 and 2013Q3	+2.05	1.27	0.105	0.356	0.185	0.156
One lag of the shock ($p_s = 1$)	+3.23	1.13	0.005	0.075	0.038	0.006
No lags of the shock ($p_s = 0$)	+2.31	0.97	0.017	0.117	0.059	0.026
Lagged interest rates added	+3.55	1.14	0.001	0.060	0.018	0.003
Three shocks entered jointly	+2.65	1.30	0.042	0.308	0.153	0.025
National income ranking (`INC_RANK`)	+2.81	0.97	0.002	0.055	0.029	0.018
Excluding recession quarters	+2.52	1.91	0.190	0.513	0.195	0.214
Excluding extreme-shock quarters	+0.31	2.15	0.888	0.682	0.532	0.415
Consumption deflated by the CPI	+3.19	1.11	0.004	0.075	0.039	0.007
Panel D: Lowest income quintile, renter minus mortgagor, LSAP minus FFR						
Baseline	+5.10	1.71	0.002	0.061	0.021	0.006
LSAP factor set to zero before 2009	+5.64	2.24	0.010	0.060	0.084	0.020
LSAP factor set to zero from 2009 on (placebo)	+4.59	2.13	0.028	0.287	0.068	0.030
Shocks orthogonalized to pre-announcement information	+5.47	1.80	0.001	0.027	0.009	0.029
Orthogonalized, LSAP factor zero before 2009	+5.33	2.54	0.031	0.197	0.111	0.034
Excluding 2009Q1	+5.15	1.71	0.003	0.068	0.015	0.007
Excluding 2013Q3	+4.59	1.69	0.006	0.094	0.036	0.012
Excluding 2009Q1 and 2013Q3	+4.00	1.75	0.021	0.168	0.051	0.045
One lag of the shock ($p_s = 1$)	+5.01	1.73	0.003	0.069	0.022	0.008
No lags of the shock ($p_s = 0$)	+3.99	1.56	0.007	0.097	0.025	0.024
Lagged interest rates added	+5.74	1.74	<0.001	0.044	0.008	0.002
Three shocks entered jointly	+3.53	1.41	0.012	0.146	0.048	0.019
National income ranking (`INC_RANK`)	+4.79	1.40	0.001	0.028	0.006	0.015
Excluding recession quarters	+2.95	2.63	0.268	0.515	0.494	0.181
Excluding extreme-shock quarters	+1.81	2.69	0.511	0.727	0.310	0.780
Consumption deflated by the CPI	+5.12	1.70	0.002	0.058	0.020	0.006

[a] See the first part of the table.

Notes: influential quarters: the stated anchor quarter is removed from every regression of the contrast, with leads and lags still taken from the complete series; 2009Q1 and 2013Q3 are the two quarters with the largest LSAP factor in absolute value. Specification: p_s is the number of lags of the shock in Equation (1) (baseline $p_s = 2$); 'lagged interest rates' adds the first lag of the federal funds rate and of the 10-year Treasury yield to the controls; 'three shocks entered jointly' adds the other two factors and two lags of each. Sample and measurement: the national income ranking forms the quintiles from the BLS variable `INC_RANK`, the household's percentile in the national income distribution ($B = 2000$); the two exclusions drop the NBER recession quarters (2001Q1–2001Q4 and 2007Q4–2009Q2) or the 15 quarters in which a shock exceeds two standard deviations in absolute value, and re-estimate on the remaining quarters; the last row deflates consumption by the consumer price index.

Table F.3: Further Results under Alternative Shock Series and Specifications

Panel A: Northeast, instrument contrasts of the $Q1$–$Q5$ gap, h = 5–8					
	LSAP − FG		LSAP − FFR		
	Non-housing	Total	Non-housing	Total	
Baseline	+7.35 (0.002)	+4.81 (0.004)	+4.78 (0.072)	+3.85 (0.034)	
LSAP zero before 2009	+9.04 (0.004)	+6.39 (0.003)	+6.48 (0.049)	+5.44 (0.017)	
Orthogonalized shocks	+6.78 (0.006)	+4.50 (0.012)	+2.64 (0.359)	+2.53 (0.209)	
Orthogonalized, LSAP zero before 2009	+6.36 (0.082)	+5.34 (0.034)	+2.23 (0.581)	+3.37 (0.219)	
Panel B: FFR, non-housing consumption, h = 1–12					
	$Q1$–$Q5$ gap		Renter minus mortgagor		
	Midwest	Northeast	$Q1$	$Q2$	$Q3$
Baseline	+1.64 (0.138)	+0.31 (0.802)	−1.90 (0.044)	−1.25 (0.208)	−1.32 (0.062)
Orthogonalized shocks	+2.79 (0.022)	+2.41 (0.111)	−1.34 (0.254)	−1.66 (0.167)	−1.25 (0.141)
No lags of the shock	+1.35 (0.212)	+0.34 (0.751)	−1.68 (0.073)	−1.18 (0.193)	−1.31 (0.047)
Lagged interest rates added	+1.43 (0.175)	+0.04 (0.970)	−2.19 (0.024)	−1.17 (0.225)	−1.19 (0.100)
Three shocks entered jointly	+1.88 (0.158)	+1.68 (0.238)	−0.88 (0.395)	−2.03 (0.076)	−1.60 (0.063)
Panel C: Mortgagors, $Q1$–$Q5$ gap, LSAP					
	Non-housing consumption		Food share		
	h = 1–12	h = 9–12	h = 1–12	h = 9–12	
Baseline	+2.04 (0.124)	+3.32 (0.030)	−0.33 (0.008)	−0.44 (0.006)	
LSAP zero before 2009	+2.23 (0.300)	+3.66 (0.137)	−0.27 (0.171)	−0.38 (0.128)	
LSAP zero from 2009 on (placebo)	+1.47 (0.422)	+3.03 (0.162)	−0.47 (0.006)	−0.60 (0.007)	
Orthogonalized shocks	+3.25 (0.030)	+4.21 (0.011)	−0.28 (0.059)	−0.31 (0.105)	
Orthogonalized, LSAP zero before 2009	+3.13 (0.195)	+5.63 (0.043)	−0.35 (0.101)	−0.52 (0.069)	
No lags of the shock	+1.38 (0.198)	+2.36 (0.069)	−0.23 (0.022)	−0.32 (0.021)	
Lagged interest rates added	+2.51 (0.047)	+3.68 (0.010)	−0.32 (0.012)	−0.41 (0.014)	
Three shocks entered jointly	+2.13 (0.174)	+3.79 (0.031)	−0.24 (0.087)	−0.29 (0.105)	

Notes: each cell reports the contrast averaged over the stated horizons, in percentage points per unit contractionary shock, with the p-value of the bootstrap test that the average is zero in parentheses (residual moving block bootstrap, B = 5000, L = 12). Panel A: the Northeast gap $\beta^h_{Q5} - \beta^h_{Q1}$ under LSAP minus the same gap under the other instrument, for non-housing and total consumption, as in Table D.2. Panel B: the gap $\beta^h_{Q5} - \beta^h_{Q1}$ under FFR in two regions, and $\beta^h_{\text{Renter}} - \beta^h_{\text{Mortgaged}}$ under FFR within the three lowest income quintiles, as in Table 2. Panel C: the gap $\beta^h_{Q5} - \beta^h_{Q1}$ among mortgagor households under LSAP, as in Table E.3. The rows are defined in the notes to Table F.2, with shorter labels; a panel lists the settings that bear on the result it reports.